

UNIVERSITY OF BABYLON · COLLEGE OF NURSING

PRACTICAL TRAINING GUIDE

Clinical Pediatric Nursing

A Comprehensive Practical Guide

for Ward and Laboratory Training

SKILLS LABORATORY · WARD CLINICAL
TRAINING

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English Text

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Third-year nursing students preparing for the laboratory examination (ministerial practical examination) and hospital ward clinical training

PREPARED BY

A working first edition intended for teaching and laboratory training. Clinical decisions must always follow the current official hospital protocols and the prescribing physician's orders.

UNIVERSITY OF BABYLON · COLLEGE OF NURSING

Academic Year -----

Clinical Pediatric Nursing: A Comprehensive Practical Guide

for Ward and Laboratory Training — Third Year, College of Nursing

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Edition and Disclaimer

This is a working first edition intended for teaching and laboratory training. Clinical decisions must always follow the current official hospital protocols and the prescribing physician's orders.

Chapter 1 — Communication and Interviewing with the Child and Family

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 1

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Define communication and therapeutic communication, and list the three essential relationship factors (trust, respect, empathy).
2. Distinguish verbal from nonverbal communication techniques for interacting with children and families.
3. Apply the general rules for communicating with children.
4. Adjust communication to the developmental thought stage of the child (infancy, early childhood, school age, adolescence).
5. Communicate effectively with parents during the health interview (introduction, privacy, confidentiality, encouragement techniques).
6. Use play and transition objects as communication tools.
7. Perform the laboratory and hospital steps of a rapport-building interview, using the chapter checklist.
8. Write a communication-focused nursing care plan for the child and family.

2. Placement Box

4. Introduction and Definitions

Communication is "the process of sharing information, thoughts, and ideas between two people using a common system of symbols, signs, or behaviors". Infants begin communicating soon after birth when they cry to convey discomfort or hunger.

Therapeutic communication is goal-directed, focused communication with a health purpose: "Client- and family-centered communication increases satisfaction with nursing care and aids in improving knowledge and health care skills". Three factors create and foster effective therapeutic communication: **trust, respect, and empathy**.

In pediatric nursing the nurse's relationship with the child is frequently *mediated by the parent*, especially for younger children; assessment therefore gathers input from the child (verbal and nonverbal), the parent, and the nurse's own observations, and for older children the parent becomes a collaborator.

5. The Communication Process and Factors

- **Sender and receiver:** both child/parent and nurse.
- **Message:** words (verbal) or body language (nonverbal).
- **Feedback:** show that the message was received (nodding, reflecting back).
- **Climate:** trust, respect, empathy.
- **Barriers:** medical jargon, interruptions, noise, adult-directed questions when the child can speak, ignoring the child, lack of privacy.

Safety / Red flag: most parents are laypersons — avoid medical jargon and abbreviations; when a medical term is unavoidable, define it in developmentally appropriate language.

6. Verbal Communication Techniques

Verbal communication is communication through the use of words, written or spoken. General guidelines for appropriate verbal communication include:

- Use **open-ended questions** that do not restrict the child's or parents' answers.

- **Redirect** the conversation to maintain focus.
- Use **reflection** to clarify the parents' feelings.
- **Paraphrase** the child's or parents' feelings to demonstrate empathy.
- **Acknowledge emotions.**
- Demonstrate **active listening** by using the child's or family's own words.
- Avoid **medical jargon**; define any necessary medical term.

The nurse uses a variety of verbal techniques to encourage communication; some are useful to pose questions or explore concerns in a less threatening manner; others can be presented as word games, which children often receive well.

7. Nonverbal Communication Techniques

Nonverbal communication, or body language, includes attending to others and active listening; when people feel they are being heard, trust and rapport are established.

Guidelines:

- Relax; maintain an **open posture** with arms uncrossed.
- Sit opposite the family and lean forward slightly.
- Maintain **eye contact**.
- **Nod** the head to demonstrate interest.
- Note the child's or parent's posture, eye contact, and facial expressions.

TAKE NOTE: people of all ages desire to be listened to without interruption. Active listening may uncover fears that questioning would not reveal. For many children and adults, talking about feelings is difficult, and verbal communication may be more stressful than supportive — then use several nonverbal techniques.

8. Communicating with Parents

8.1 Establishing a setting

- **Appropriate introduction:** introduce yourself and ask the name of every family member present and how they wish to be addressed; address adults by title ("Mr.", "Mrs.") unless they specify a preferred name; record the preferred name in the medical record. This conveys respect.
- **Include children in the interaction:** ask children their name and age; do not direct all questions to adults when children are old enough to speak for themselves.
- **Privacy and confidentiality:** the environment must allow privacy, with distractions (noise, interruptions, TV, mobile phones) kept to a minimum; provide play material for young children during the interview. Inform the family about the limits of confidentiality (e.g., suspected child abuse or suicidal teenager).

8.2 Encouraging parents to talk

- Whatever the parent sees as a problem should be a concern of the nurse.
- Careful phrasing with **broad, open-ended questions** (e.g., "What is Andy eating now?") provides more information than several single-answer questions (e.g., "Is Andy eating what the rest of the family eats?").
- Take the lead when the parent needs direction, based on an observation: "Joseph seems unhappy today" or "How do you feel when Joseph cries?".
- Ask about the parent's own coping: "What do you do to relax?" / "What help do you have with the children?".
- Give positive reinforcement and an opening for questions: "You handle Jamey very well. What kind of experience have you had with babies?".
- Often a nod or an "uh-huh" keeps parents talking.

8.3 Directing the focus

The nurse directs the focus of the interview while allowing maximum freedom of expression.

9. Communicating with Children: General Rules

Nursing Care Guidelines — Communicating With Children:

- Allow children time to feel comfortable.

- Avoid sudden or rapid advances, broad smiles, extended eye contact, and other gestures that may be seen as threatening.
- Talk to the parent if the child is initially shy.
- Communicate through **transition objects** (dolls, puppets, stuffed animals) before questioning a young child directly.
- Give older children the opportunity to talk without the parents present.
- Assume a position that is **at eye level** with the child.
- Speak in a quiet, unhurried, and confident voice.
- Speak clearly, be specific, and use simple words and short sentences.
- State directions and suggestions positively.
- Offer a choice only when one exists.
- Be honest with children.
- Allow children to express their concerns and fears.
- Use a variety of communication techniques.



Nurse sitting at eye level with the child



Nurse sitting at eye level with the child

10. Communication Related to the Development of Thought Processes

Thought processes progress from **sensorimotor** → **perceptual** → **concrete** → **abstract (formal operations)**. This frames how the nurse communicates at each age.

10.1 Infancy (sensorimotor)

- Infants use and understand mostly **nonverbal** communication; they smile and coo when content and cry when distressed (hunger, pain, restraint, loneliness).
- They become quiet when cuddled, rocked, or gently touched; they are comforted by a soft voice; loud harsh sounds and sudden movements are frightening.
- **Techniques:** speak softly, use touch, gentle handling; **peek-a-boo is an excellent means of initiating communication with infants.**

10.2 Early childhood (toddler–preschooler, egocentric)

- Children younger than 5 years are **egocentric** — focus communication on them, tell them what they can do and how they will feel.
- They are concrete and literal — they may attach literal meaning to phrases like "a little stick in the arm"; use simple, direct language.
- Allow them to touch and examine articles they will come in contact with (a stethoscope bell feels cold; palpating a neck might tickle).
- Toddlers communicate with their hands (push away, point, pull someone).

10.3 School-age (concrete)

- They rely less on what they see, more on what they know; they want explanations and reasons.
- They are interested in the functional aspect of procedures — show how squeezing the bulb makes the "arrow" move; let them operate the bulb.
- They have a heightened concern about **body integrity**, including possessions; encourage them to voice concerns to reduce anxiety.

10.4 Adolescence (formal operations; fluctuates)

- Adolescents fluctuate between child and adult thinking; they may seek the security of childhood when tensions rise.
- **Interview dilemma:** talk with the adolescent alone or with parents? Talking with the adolescent first identifies with the young person; talking with parents first may give insight into family relationships. Give both parties a chance to be included; clarify time limits to avoid appearing to take sides.
- Two views of a problem often exist (teenager vs parents); providing an open, unbiased atmosphere is therapeutic in itself.

11. Play as a Communication Tool

- **Play is the universal language of children** and one of the most important forms of communication.

- The form and complexity of play give clues about physical, intellectual, and social development.
- **Peek-a-boo** initiates communication in infants; **puppets are universally effective for communicating with children.**
- Nonverbal communication devices that employ pictures can help children with speech difficulties.

12. Nursing Care Plans (Communication Focus)

Care Plan 1 — Impaired Verbal Communication (child) / Therapeutic communication barrier

Assessment: child hides behind parent, refuses eye contact; parent answers for the child; child uses limited words.

NANDA diagnosis: *Risk for impaired parent–infant attachment / Anxiety (child or parent)* related to hospitalization and unknown environment.

Goals: The child will interact with the nurse using at least one communication technique within the encounter; the child will express fears verbally or nonverbally.

Interventions (with rationale):

1. Use a transition object (doll/puppet) and talk to the parent first — reduces threat, builds trust.
2. Position self at the child's eye level; use a quiet, unhurried voice — reduces perceived threat.
3. Offer choices only where a choice exists — respects autonomy, decreases power struggles.
4. Acknowledge emotions and paraphrase the child's feelings — demonstrates empathy.
5. Evaluate: child speaks, points, or nods; parent reports decreased anxiety.

Care Plan 2 — Deficient Knowledge (parent) / Anxiety (parent)

Assessment: parent uses medical terms incorrectly, repeats questions, appears anxious.

NANDA diagnosis: *Deficient knowledge related to the child's condition / Anxiety (parent).*

Goals: The parent will describe the condition and the planned care in own words; the parent will verbalize reduced anxiety.

Interventions (with rationale):

- 1. Use broad open-ended questions to uncover concerns ("What worries you most about your child?") — yields more information than yes/no questions.
- 2. Avoid jargon; define terms in plain language — parents are laypersons.
- 3. Provide privacy and undivided attention — the setting is almost as important as the interview.
- 4. Reflect concerns and validate feelings — therapeutic technique that promotes the nurse–family relationship.
- 5. Evaluate: parent restates key points; anxiety cues reduced.

13. Procedure — Laboratory Step: The Rapport-Building Interview (Checklist)

Purpose: to practise establishing therapeutic communication before clinical use. **Equipment:** quiet room, chair, pen, paper/observation sheet, a doll/puppet, practice scenario card.

#	Step	Done	Time
1	Introduce yourself (name, role).	☐	
2	Ask each family member's name and preferred address.	☐	
3	Ensure privacy; minimize interruptions.	☐	
4	Position yourself at eye level with the child; open posture.	☐	
5	Start with a broad open-ended question ("Tell me about...").	☐	
6	Use active listening (nods, "uh-huh", reflecting the parent's own words).	☐	
7	Use a transition object with the young child.	☐	
8	Avoid jargon; define any medical term used.	☐	

#	Step	Done	Time
9	Acknowledge emotions; paraphrase feelings.	☐	
10	Close by summarizing and stating the next step.	☐	



Nurse sitting at eye level with the child

14. Procedure — Hospital Step: Bedside Application

- Repeat checklist items 1–10 with a real family, in the ward.
- Document: preferred name, chief concern in the parent's words, communication observations, and the family's understanding.
- Never perform the interview a first time in the ward: the lab step comes first.
- If the child is old enough, give the older child the opportunity to talk without the parents present.

15. Worked Scenario Examples (Communication)

Scenario A — The literal child. A 4-year-old is told he will get "a little stick in the arm." He becomes terrified that a stick will be pushed into his arm. *Correction:* say "You will feel a small pinch like a mosquito bite, then it will stop" — simple, direct, honest.

Scenario B — The silent toddler. A 2-year-old won't speak to the nurse. *Correction:* talk to the parent first, then use a puppet/doll, and let the child handle the stethoscope before use.

Scenario C — The anxious parent. A mother answers in short phrases and wants to finish. *Correction:* ask a broad open-ended question ("What is Ali eating now?") and give positive reinforcement ("You handle him very well — what experience have you had with babies?").

Scenario D — The adolescent. A 13-year-old gives a different story from the parents. *Correction:* give both parties a chance, clarify the differing perceptions openly, and do not take sides.

16. Case Study (Communication)

History of a therapy session placement. Rania, aged 4 years, is admitted to the Medical ward with pneumonia. She hides behind her mother, refuses to look at the nurse, and cries when the nurse approaches. The mother is anxious and speaks quickly, mainly about "giving the injection."

Critical-thinking questions:

1. Why did the nurse talk to the mother first?
 2. Which transition object would you use, and at which point?
 3. How would you involve the child in her care without startling her?
 4. What are the priorities of the nurse's relationship before gathering the history?
-

17. Chapter MCQs (10)

Built from the ministerial question bank and the source books. Answers with rationale at the end.

Q1. Which action is most effective when interviewing a parent about the child's symptoms?

- a) Using pauses to give the parent time to think
- b) Closed yes/no questions only
- c) Speaking quickly to cover all topics
- d) Avoiding repetition to prevent confusion

Rationale (Ch key): open-ended questioning and allowing the parent time produces more detailed, reliable information than yes/no rapid sequences.

Q2. The nurse plans to use a transitional object to facilitate communication. Which item is the best choice?

- a) Puppet or stuffed animal
- b) Stethoscope (real)
- c) Blood-pressure cuff
- d) Medical chart

Q3. A preschooler is anxious about an injection; which nonverbal communication should the nurse prioritize?

- a) Standing upright to show authority
- b) Matching the facial expression with the message (calm seriousness)
- c) Smiling broadly to reassure him
- d) Avoiding eye contact to reduce intimidation

Q4. When communicating with an infant, the nurse should primarily:

- a) Use elaborate sentences
- b) Use the infant's own words to reflect
- c) Use nonverbal cues, soft voice, cuddling, and touch
- d) Speak loudly to get attention

Q5. A school-age child is interested in how a procedure works. The best approach is:

- a) Explain the function simply and let the child participate (e.g., operate the cuff bulb)
- b) Tell the child not to look
- c) Promise a reward
- d) Do the procedure silently and quickly

Q6. The nurse tells a hospitalized adolescent and the parents: "I have 10 minutes — I'd like to hear each of your views." This does all EXCEPT:

- a) Prevents the nurse from appearing to take sides
- b) Gives both parties a chance to be included
- c) Assumes the adolescent is an adult
- d) Clarifies two differing perceptions

Q7. Which statement is a *broad* open-ended question?

- a) "Is he eating the same food as the family?"
- b) "What is Andy eating now?"
- c) "He doesn't eat fruit, does he?"
- d) "Does he like fish?"

Q8. A child with limited speech can best be supported by:

- a) Nonverbal communication devices that employ pictures
- b) Avoiding communication
- c) Writing long notes
- d) Speaking only to the parents

Q9. The most important factors to create effective therapeutic communication are:

- a) Technology, speed, volume
- b) Trust, respect, empathy
- c) Uniforms, badges, equipment
- d) Punctuation, grammar, spelling

Q10. Reasons why play is used in pediatric communication include all EXCEPT:

- a) Play is a universal language of children
- b) Play reveals developmental progress
- c) Play distracts from the need for honesty
- d) Play lowers the child's perceived threat

Answers: Q1 a — Q2 a — Q3 b — Q4 c — Q5 a — Q6 c — Q7 b — Q8 a — Q9 b — Q10 c.

18. Key Points

- Therapeutic communication requires **trust, respect, and empathy**.
- Use **broad open-ended questions** with parents; redirect, reflect, paraphrase, acknowledge emotions, avoid jargon.
- Communicate with children at **eye level**; use transition objects; be honest; offer choices only when choices exist.
- Adjust to development: infants = nonverbal + soft voice/touch; toddlers/preschoolers = simple concrete language (literal meaning); school-age = explanations and participation; adolescents = balanced interview with child and parents.
- **Play is the universal language of children.**
- The lab interview checklist is the same tool used later at the bedside (Lab → Hospital).

19. Normal Values / Norms Table Relevant to This Chapter

Development of communication (language milestones) — norms:

Age	Expected communication behavior
0–3 months	Crying, cooing; responds to soft voice
4–6 months	Babbling, laughs, responds to peek-a-boo
1 year	First meaningful words ("mama", "dada")
2 years	2-word phrases; points and names objects
3 years	3–4 word sentences; talks about the present
4–5 years	Sentences; tells simple stories; asks "why?"
School age	Explains "how" and "why"; understands procedures when explained
Adolescence	Abstract reasoning; can participate in decisions

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 2

Chapter 2 — Health History Taking

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 2

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. State the parts of the pediatric health history (identifying information, CC, HPI, PMH, ROS, family history, psychosocial history).
2. Elicit the chief complaint with open-ended, neutral questions.
3. Take the history of present illness as a narrative from onset to present (onset, interval, present status, reason for seeking help now).
4. Take the birth history (mother's health during pregnancy, labor and delivery, infant condition after birth) and explain its relevance to parents.
5. Ask the past medical history (illnesses, injuries, surgeries, allergies, current medications, immunizations, growth and development).
6. Adapt the history depth to the child's age and to emergent vs well-child situations.
7. Perform the lab and hospital steps of the history-taking interview using the chapter checklist.

2. Placement Box

4. Introduction: The Pediatric Health History

The interview is a specific form of **goal-directed communication**. In the child, the health history is taken from the parent(s) or caregivers, with input from the child when old enough to speak for themselves. An organized approach is most effective.

The complete pediatric health history is outlined in **Box 4.4**, which includes:

1. **Identifying information** — name, address, telephone, birth date and place, race/ethnic group, sex, religion, date of interview, informant.
2. **Chief complaint (CC)** — the major specific reason for seeking health care.
3. **History of present illness (HPI)** — all details related to the CC.
4. **Past medical history (PMH)** — birth history, previous illnesses/injuries/surgeries, allergies, current medications, immunizations, growth and development, habits.
5. **Review of systems (ROS)** — each body system screened.
6. **Family history** — relevant hereditary or familial conditions.
7. **Psychosocial history** — family structure, habits, school, environment.
8. **Dietary/nutritional history** — detailed for the child.

5. The Chief Complaint (CC)

- The chief complaint is **the specific reason for the child's visit**; it may be the theme, with the present illness viewed as the description of the problem.
- Elicit the CC with **open-ended, neutral questions**: "What seems to be the matter?", "How may I help you?", "Why did you come here today?".
- **Do not** use labeling-type questions ("How are you sick?", "What is the problem?") — they limit information.
- The reason for the visit may not be an illness — it may be a developmental concern, school referral, or procedure.

6. History of Present Illness (HPI)

The HPI is a **narrative of the chief complaint from its onset through its progression to the present**. Four major components:

1. Details of **onset**
2. A complete **interval history**
3. The **present status**
4. The **reason for seeking help now**

The focus is on **all factors relevant to the main problem**, even if they have disappeared or changed.

Analyzing a symptom

- **Type** (character)
- **Location**
- **Severity** — best determined by how it affects the child's usual behavior; pain that prevents playing, interacting, sleeping, and eating is most often severe; use a rating scale such as the numeric or **Wong-Baker FACES Pain Rating Scale**
- **Duration** — onset, frequency, described in terms of activity/behavior
- **Influencing factors** — precipitating, relieving, temporal, positional, and associated events

7. Past Medical History (PMH)

7.1 Birth history

Includes all data concerning:

1. The **mother's health during pregnancy**
2. The **labor and delivery**
3. The **infant's condition immediately after birth**

Because prenatal influences significantly affect a child's physical and emotional development, a thorough birth history is essential. Explain its relevance when parents question it: *"I will be asking you some questions about your pregnancy and [child's name]'s birth. Your answers will give me a more complete picture of his overall health".*

7.2 Previous illnesses, injuries, and surgeries

- Begin with a general question ("What other illnesses has your child had?"), then specifically ask about colds, earaches, measles, rubella, chickenpox, mumps, pertussis, diphtheria, tuberculosis, scarlet fever, strep throat, recurrent ear infections, influenza, gastroesophageal reflux, tonsillitis, and allergic manifestations.
- Ask about injuries that required intervention (falls, poisoning, choking, concussion, fractures, burns), surgeries, procedures, and hospitalizations with dates and diagnoses.

7.3 Allergies

Ask about hay fever, asthma, unusual drug/food/latex reactions, and contact agents. Ask specifically and reliably about drug reactions : name of the drug, the reaction, route, time to reaction, recurrence on rechallenge, and whether it was documented.

Nursing Alert: information about allergic reactions is essential — failure to document a serious reaction places the child at risk if the agent is given.

7.4 Current medications

Ask about vitamins, antipyretics (especially aspirin), antibiotics, antihistamines, decongestants, supplements, herbs/essential oils, homeopathic medicines. Record name, dose, schedule, duration, reason. Ask parents to bring the containers or call the pharmacy (OTC drugs won't appear on prescriptions).

7.5 Immunizations

A record of all immunizations is essential; list the disease, number of injections, dose, date, and any reaction. Sources: provider record, school record, centralized immunization registry.

Screening for contraindications and precautions before every vaccine.

7.6 Growth and development

- Growth: measurements of height/length and head circumference at birth, patterns on the growth chart, deviations from previous percentiles, family concerns.

- Development: age of head control, sitting alone, walking, first words, bladder/bowel control, school grade and performance, best friend, interactions with peers and adults. Use specific, detailed questions ("sitting up" can mean different things).
-

8. Adapting the History to the Child's Age and Situation

- **Infants:** emphasize the birth history, detailed feeding history, immunizations, and growth and development.
 - **Older children and adolescents:** history taking should include the child directly; give older children the opportunity to talk without the parents present.
 - **Emergent situations:** a focused health history — chief complaint and essentials only.
-

9. Nursing Care Plan (History/Interview Focus)

Assessment: parent anxious, short answers; child plays in background.

NANDA diagnosis: *Risk for compromised family coping / Anxiety (parent)* related to the health concern.

Goals: The parent will provide a complete, accurate history; anxiety cues will decrease during the interview.

Interventions (with rationale):

1. Establish the setting (introduction, privacy, preferred names) before questions — builds trust and reliability.
 2. Begin each section with an open-ended statement ("Tell me about your child's birth") then use fact-finding questions to focus — yields the most complete data.
 3. Explain why birth-history questions are asked for an older child — increases cooperation.
 4. Use the medication containers and the allergy checklist — improves accuracy.
 5. Evaluate: history complete; parent reports feeling heard; no vital data omitted.
-

10. Case Study (History led to diagnosis)

Hiba, 6 weeks. The mother brought her to the clinic because "she vomits after every feed." CC confirmed with an open-ended question. HPI: projectile vomiting began at 4 weeks, worsening; the mother states the baby is hungry again soon after feeding. PMH birth history: full term, normal delivery, birth weight 3.2 kg.

- Which CC question was most appropriate? (A neutral open-ended: "What seems to be the matter?")
- Which part of the HPI is now incomplete? (Analyzing the symptom: type, severity, duration, influencing factors — here the feeding–vomiting relationship)
- Why does the nurse explain the birth-history questions? (To improve cooperation)

This history pattern (projectile, non-bilious, hungry after feeds) is the classic presentation re-trained in Chapter 19 (Pyloric stenosis).

11. Procedure — Laboratory Step: History-Taking Interview (Checklist)

Purpose: practise the full pediatric history before clinical use. **Equipment:** history form, chair, quiet room, pen.

#	Step	Done
1	Introduce self; identify informant; record demographics.	☐
2	Elicit CC with an open-ended neutral question.	☐
3	Take HPI narrative (onset, interval, present status, reason now).	☐
4	Analyze the main symptom (type, location, severity, duration, influencing factors).	☐
5	PMH: birth history (pregnancy, labor/delivery, neonatal) with explanation of relevance.	☐
6	PMH: past illnesses, injuries, surgeries.	☐
7	Allergies via the allergy checklist; document reactions.	☐

#	Step	Done
8	Current medications (name, dose, schedule, reason).	☐
9	Immunization record (disease, #, dose, dates, reactions).	☐
10	Growth and developmental milestones.	☐
11	Family history and psychosocial/dietary sections (open-ended start, fact-finding follow-up).	☐
12	Summarize the history back to the informant; thank them.	☐



Completed pediatric health history form

12. Procedure — Hospital Step

- Repeat the checklist at the bedside, using the ward's admission form.
- Always record the **informant** and the reliability of the information.

- Document allergies on the allergy/medication record (red alert).
 - Never perform an incomplete history "because the ward is busy" — the full outline protects the child.
-

13. Chapter MCQs (10)

Q1. The chief complaint is best elicited by:

- a) "How are you sick?"
- b) "What seems to be the matter?"
- c) "What is the problem?"
- d) "Is it his chest or his head?"

Q2. The history of present illness is:

- a) A list of the child's past diagnoses
- b) A narrative of the CC from onset to present
- c) The immunization record
- d) The family history

Q3. Which is NOT part of the birth history?

- a) Mother's health during pregnancy
- b) The labor and delivery
- c) The child's current school grade
- d) The infant's condition immediately after birth

Q4. When a parent asks why birth-history questions are needed for a school-age child, the nurse should:

- a) Skip those questions
- b) Explain that pregnancy/birth questions complete the health picture
- c) Ask the teacher instead
- d) Say it is hospital policy only

Q5. Pain severity in a child is best determined by:

- a) How loudly the child cries

- b) How it affects usual behavior (play, sleep, eating) + a rating scale
- c) The parent's guess
- d) The child's age

Q6. A reliable allergy history includes all EXCEPT:

- a) Name of the medication
- b) Description of the reaction
- c) Documented notification of providers
- d) The insurance number

Q7. An interval item in the vaccine record is:

- a) The name of the disease
- b) The number of injections and doses
- c) The date administered
- d) Occurrence of any reaction following immunization
- e) All of the above

Q8. In the medication history the nurse must ask specifically about:

- a) Vitamins, antipyretics (especially aspirin), and OTC drugs
- b) Only prescribed antibiotics
- c) Only the current IV fluids
- d) Nothing outside the discharge summary

Q9. For an emergent admission the health history should be:

- a) Complete with all 8 sections always
- b) Focused on the chief complaint and essentials
- c) Postponed to the next day
- d) Taken from the school record only

Q10. A developmental history question about milestones would ask:

- a) "At what age did he sit alone without support?"
- b) "How much does he weigh now?"
- c) "Which vaccine came last?"
- d) "Does his father work?"

Answers: Q1 b — Q2 b — Q3 c — Q4 b — Q5 b — Q6 d — Q7 e — Q8 a — Q9 b — Q10 a.

14. Key Points

- The pediatric history follows **Box 4.4** : identifying info, CC, HPI, PMH (birth/allergies/meds/immunizations/growth), ROS, family, psychosocial, dietary.
- Elicit the CC with **open-ended, neutral questions**; never labeling questions.
- The HPI is a **narrative** with four components: onset, interval history, present status, and reason for seeking help now.
- The **birth history** covers pregnancy, labor/delivery, and neonatal condition — explain its relevance when challenged.
- Document **allergies** reliably — failure to document puts the child at risk.
- Adapt depth: infants need birth/feeding/immunization/growth emphasis; emergencies get a focused history.

15. Normal Values / Forms Table

Section	Key recorded values
Demographics	Name, DOB, place of birth, sex, informant
Growth (from chart)	Weight-for-age %ile, length/height %ile, head circumference %ile
Vital signs (companion chapter)	T, P, R, BP (age-based, Appendix A)
Immunizations	Disease, # doses, dates, reactions
Allergies	Drug + reaction + time + documentation
Development	Milestones ages (head control, sitting, walking, first words)

Chapter 3 — Physical Examination of the Infant and Child

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 3

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Prepare the examination environment (lighting, warmth, privacy, play) and gather the required equipment.
2. Approach the child according to developmental age and stage (infant, toddler, preschooler, school-age, adolescent).
3. Apply the Nursing Care Guidelines for performing a pediatric physical examination (nonthreatening order, secure position, ABC in emergencies).
4. Use the four assessment techniques (inspection, palpation, percussion, auscultation) in the correct order for each system.
5. Sequence the examination head-to-toe with painful/intrusive areas last.
6. Read the child's readiness signals and apply the refusal techniques.
7. Perform the lab and hospital steps of a systematic head-to-toe examination using the checklist.

2. Placement Box

Ward: All wards. Each ministerial chapter re-trains a focused examination:

NICU — newborn examination (jaundice: skin/sclerae; RDS: respiratory);

Blood ward — lymph nodes, liver/spleen, joints;

Medical ward — chest auscultation (pneumonia), skin turgor/capillary refill (dehydration);

Surgical ward — abdominal examination (Hirschsprung, pyloric stenosis).

4. Preparing the Physical Examination

Being prepared and organized ensures efficiency; the approach depends on the child's developmental stage.

Environment and equipment:

- Room well lit and decorated with neutral colors; temperature comfortably warm.
- Strange, potentially frightening equipment kept **out of sight**; bring out one piece at a time.
- Toys/games available; washable toys as distractions.
- **Privacy**, especially for school-age children and adolescents.
- Provide time for play and becoming acquainted.

Gather the equipment: clean gloves, stethoscope, thermometer, sphygmomanometer, tape measure, reflex hammer, penlight, otoscope/ophthalmoscope, tongue depressor, cotton ball, infant or adult scale, stadiometer.

- Display a **confident, matter-of-fact** approach; frustration is sensed by child and parent.
- Keep the room warm; allow the child to remain dressed until the time of the examination.

5. Approaching the Child by Developmental Stage

General rule: approach the child according to developmental age; if several children are seen, begin with the most cooperative child to model the behavior.

Newborns and infants:

- If the infant is asleep, **auscultate the heart, lungs, and abdomen first** while the baby is quiet; count HR and RR before undressing.
- Completely undress down to the diaper; remove the diaper last (genitalia, anus, spine, hips).
- Examine 1–2 hours before a feeding; have the parent hold the child; parent stays the nurturer, not an assistant of painful procedures.
- Head-to-toe order; **most traumatic (ears, nose, mouth, throat) last**; delay the **Moro reflex** until the end (startle → crying).
- Warm hands and stethoscope; soft crooning voice, smile, eye contact; bright objects; pacifier if crying.
- **Stranger anxiety** is normal in older infants — keep the parent in view.

Toddlers:

- Remove clothing one item at a time; replace clothing after each area; gown not needed before school age.
- Use little touch at the beginning; introduce equipment slowly; let the child touch it, take a parent's temperature, put the cuff on a teddy bear.
- Sit on the caregiver's lap; knee-to-knee for the abdominal examination.
- **Praise** cooperation.
- Use short phrases telling what you'll do, not asking permission. Toddlers are egocentric — don't reference another child's behavior.

Preschoolers:

- May **fear body invasion/mutilation**; withdraw from intrusive steps; use simple explanations and reassurance.
- Let the child "help" (hold stethoscope/penlight); offer real choices; compliment cooperation.
- Games encourage deep breathing: hold up a finger or penlight and "blow it out".

School-age:

- Concrete thinking — avoid jargon and double-meaning words: "Let's see how warm you are" instead of "take your temperature".
- Respect body privacy: keep underpants on under the gown; replace clothing promptly.

- Commenting on findings is interesting to them — they like to learn how the body works.

Adolescents:

- Provide privacy while undressing; expose only the area examined; matter-of-fact comments about physical changes; allow questions without the caregiver present; no "dumb questions".
-

6. General Rules: Performing the Pediatric Examination

Nursing Care Guidelines — Performing Pediatric Physical Examination:

- **Observe readiness signals:** talking to the nurse, eye contact, accepting equipment, allowing touch, choosing the examining table over the parent's lap.
 - **If not ready:** talk to the parent while "ignoring" the child, making complimentary remarks; tell a funny story or do a magic trick; use a hand puppet "friend".
 - **If the child refuses:** assess the reason (an unduly afraid child may have had a traumatic experience); involve child and parent; avoid prolonged explanations; use a firm, direct approach; perform quickly; gentle restraint by an attendant; minimize stimulation; quiet calm confident voice.
 - **Begin nonthreateningly:** games (cranial-nerve tests, Simon Says); **paper-doll technique** — trace the child's outline and use it to demonstrate what will be examined.
 - **Involve the child:** choices (table vs lap), allow handling equipment, demonstrate on a doll or parent.
 - **Sequence:** usually head-to-toe with exceptions: alter for age (Table 4.2); examine **painful areas last; in an emergency examine ABC (airway, breathing, circulation) first.**
 - **Reassure** throughout (especially pubertal body concerns); discuss findings with the family at the end; **praise and reward** the child.
-

7. Assessment Techniques (IPPA)

The four techniques are used in system-specific order. For most systems the order is **inspection** → **palpation** → **percussion** → **auscultation**; for the **abdomen the order is inspection** → **auscultation** → **percussion** → **palpation** — **palpation last** because it may alter bowel sounds.

- **Inspection:** observation first (color, shape, movement, symmetry). Adequate lighting is necessary.
- **Palpation:** superficial then deep; light for most areas; to reduce tickling, place the child's hand over the palpating hand or have the child squeeze the parent's hand.
- **Percussion:** tone assessment (resonance/dullness) in chest and abdomen.
- **Auscultation:** use the appropriate side of the stethoscope (bell = low frequency e.g., heart sounds; diaphragm = high frequency e.g., breath sounds); in infants auscultate **before** palpation/percussion of the chest and do it when the infant is quiet.

8. Sequence of the Examination (Head-to-Toe Model)

The examination is recorded following the head-to-toe model even when the actual order is altered for the child.

1. **Vital functions** first in an emergency (ABC).
2. **Growth measurements** — weight, length/height, head circumference, arm circumference (see Chapter 4).
3. **General appearance** — alertness, posture, hygiene, interaction.
4. **Head and neck** — fontanelles, suture lines, eyes (red reflex), ears, nose, throat (last, most traumatic), neck lymph nodes.
5. **Chest** — inspection (shape, retractions), auscultation of breath and heart sounds (quiet first).
6. **Abdomen** — inspection → auscultation → percussion → palpation (superficial first).
7. **Genitalia, anus, spine, hips** — last; in infants after undressing fully.
8. **Neurologic** — reflexes (Moro delayed to the end), motor tone, cranial nerves.

Record observations, not only the abnormal — normal findings are baseline data for the ward notes.

9. Nursing Care Plan (Examination/Preparation Focus)

Assessment: preschooler refuses the examination, hides behind mother, responds to the stethoscope by crying.

NANDA diagnosis: *Anxiety (child) related to perceived body invasion/mutilation.*

Goals: The child will cooperate with a modified examination; verbal/nonverbal distress will decrease.

Interventions (with rationale):

1. Let the child "help" with the stethoscope/penlight and offer choices (lap vs table) — restores a sense of control.
2. Use the paper-doll technique to demonstrate steps — reduces fear of the unknown.
3. Explain each step in simple, concrete language ("I want to listen to you breathe") — matches concrete thinking.
4. Use a game to elicit deep breathing ("blow out the light") — gains cooperation without force.
5. Preserve privacy (underpants under the gown) and praise cooperation — supports body integrity and positive reinforcement.
6. Evaluate: the child allows auscultation/palpation; crying stops; parent reports reduced fear.

10. Procedure — Laboratory Step: Systematic Examination (Checklist)

Equipment: gloves, stethoscope, thermometer, sphygmomanometer, tape measure, penlight, otoscope, reflex hammer, scale, stadiometer.

#	Step	Done
1	Prepare the room (light, warmth, privacy, distractions hidden).	☐
2	Approach by developmental stage; observe readiness signals.	☐
3	Auscultate heart/lungs while infant is quiet (before handling).	☐

#	Step	Done
4	Measure weight, length/height, head circumference (Ch 4).	☐
5	Inspect general appearance; note color, posture, hygiene.	☐
6	Head/neck: fontanelles, eyes red reflex, ears, nose, throat (last).	☐
7	Chest: inspection then auscultation of breath and heart sounds.	☐
8	Abdomen: inspection → auscultation → percussion → palpation.	☐
9	Back/spine, genitalia, anus; hips (infants).	☐
10	Neurologic: reflexes (Moro last in infants), tone.	☐
11	Painful/intrusive areas last; emergency = ABC first.	☐
12	Reassure, praise, reward; record findings head-to-toe.	☐



Examination positions/order schematic

11. Procedure — Hospital Step

- Repeat the checklist at the bedside with the real child and family.
 - Verify the child's readiness; if refusal occurs, use the refusal techniques rather than forcing.
 - Record normal and abnormal findings; report abnormal findings to the physician.
 - Always document the parent's presence and the child's cooperation level.
-

12. Case Study (Examination)

Ali, 4 years. Admitted for pneumonia. On approach, he hides and refuses any touching. The nurse uses the paper-doll technique, a game ("let's see how warm you are"), and lets him hold the stethoscope on his teddy bear first. He then allows chest auscultation, where crackles are heard at the right base. Temperature 38.9°C.

Critical-thinking questions:

1. Why was chest auscultation deferred until rapport was established? (*Sequence flexibility — painful/intrusive last; readiness signals honored.*)
 2. How does "let's see how warm you are" help? (*Avoids medical jargon with double meaning; concrete language for the preschooler.*)
 3. Which examination order rule was applied when the nurse examined the chest before the throat? (*Most traumatic areas last.*)
 4. What should be documented besides the abnormal finding? (*Normal baseline findings, cooperation level, parent presence.*)
-

13. Chapter MCQs (10)

Q1. The best time to auscultate an infant's heart and lungs is:

- a) After undressing completely
- b) While the infant is quiet/sleeping, before other handling
- c) After feeding

- d) After eliciting the Moro reflex

Q2. Which area is examined LAST in the infant examination?

- a) The chest
- b) The throat (ENT—ears, nose, mouth, throat)
- c) The abdomen
- d) The hands

Q3. The Moro reflex in an infant is best elicited:

- a) At the beginning of the examination
- b) After the throat examination
- c) At the end of the examination
- d) Only during sleep

Q4. A toddler who refuses to cooperate should be approached by:

- a) Asking the child for permission for each step
- b) Using short phrases that tell what will happen
- c) Offering rewards for not crying
- d) Referencing a well-behaved sibling

Q5. For the abdomen, the correct technique order is:

- a) Palpation → percussion → auscultation
- b) Inspection → auscultation → percussion → palpation
- c) Auscultation → inspection
- d) Percussion only

Q6. In an emergency, examination begins with:

- a) Head-to-toe throughout
- b) Airway, breathing, circulation
- c) The painful area
- d) The throat

Q7. A preschooler with fear of body invasion needs:

- a) A firm, direct approach only
- b) Simple explanations, choices, and letting the child "help"
- c) Restraint by an attendant

- d) Avoiding the examination

Q8. To encourage deep breathing during lung auscultation in a preschooler, the nurse should:

- a) Count "1, 2, 3" loudly
- b) Use a game — "blow out the penlight"
- c) Ask the parent to demonstrate
- d) Tell the child to be brave

Q9. Which signal indicates the child is NOT ready to cooperate?

- a) Talking to the nurse
- b) Making eye contact
- c) Turning away and hiding behind the parent
- d) Accepting the offered equipment

Q10. During the adolescent examination the nurse should:

- a) Keep the caregiver in the room at all times
- b) Expose only the area being examined and provide privacy
- c) Comment on every body change loudly
- d) Skip the head-to-toe sequence

Answers: Q1 b — Q2 b — Q3 c — Q4 b — Q5 b — Q6 b — Q7 b — Q8 b — Q9 c — Q10 b.

14. Key Points

- Prepare the **environment** (light, warmth, privacy, equipment out of sight) and approach by **developmental stage**.
 - **ABC first in emergencies**; painful/intrusive areas last; Moro reflex last in infants.
 - Auscultate the quiet infant **first** (heart, lungs, abdomen).
 - Use **IPPA** with the abdomen order: inspection → auscultation → percussion → palpation.
 - Apply the readiness/refusal techniques and the paper-doll technique.
 - Record findings in the **head-to-toe model** regardless of exam order.
-

15. Normal Findings Table (Examination Basics)

Area	Normal finding (needs verification with the book tables)
Head circumference / length / weight	Age percentiles (Ch 4; Appendix A)
Fontanelles (anterior)	Flat, soft; closes ~12–18 months [bank Q11]
Breath sounds	Clear; crackles/wheezes = pneumonia (Ch 16)
Heart sounds	Regular; apical best heard < 2 years
Abdomen	Soft, non-tender; liver edge may be palpable < 2 years
Skin turgor	Immediate return (dehydration Ch 17)
Reflexes	Moro, grasp, rooting present in newborn; see Ch 6

(Full normal values: Appendix A; ward-specific norms in each ministerial chapter.)

END OF CHAPTER 3

Physical Examination of the Infant and Child

Next: **Chapter 4** — Anthropometric Measurements and Growth Assessment

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 4

Chapter 4 — Anthropometric Measurements and Growth Assessment

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 4

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Name the physical growth parameters: weight, recumbent length, height (stature), head circumference, arm circumference, and skinfold thickness.
 2. Measure recumbent length with a length board and two measurers, following the evidence-based technique.
 3. Measure standing height with a stadiometer (correct position, Frankfort plane, read at eye level).
 4. Measure weight correctly on an infant scale and a standing scale.
 5. Measure head circumference using a nonstretchable tape at the greatest frontooccipital circumference.
 6. Measure mid-arm (arm) circumference and identify the triceps skinfold sites.
 7. Plot measurements on the correct growth charts (WHO birth to 2 years; CDC 2 years and older) and interpret percentiles, including the BMI categories.
 8. Perform the lab and hospital steps of the anthropometric assessment using the checklist.
-

2. Placement Box

4. Growth Parameters

Measurement of physical growth is a **key element in evaluating health status**. Physical growth parameters include **weight, height (length), skinfold thickness, arm circumference, and head circumference**. Values are **plotted on percentile charts** and compared with the general population.

- **Height and head circumference reflect past nutrition; weight, skinfold thickness, and arm circumference reflect present (current) nutritional status.**
 - The most **reliable evaluation** compares growth measurements over time because growth is a continuous but uneven process.
-

5. Growth Charts and Interpretation

- Growth charts use a series of **percentile curves** to demonstrate the distribution of body measurements.
- **WHO growth standards** are used for infants and children **0 to 2 years** (reflects growth of predominantly breastfed children); **CDC growth charts** are used for children **2 years and older**.
- Children whose growth may be questionable:
- Height and weight percentiles **widely disparate** (e.g., height 10th and weight 90th with above-average skinfold thickness);
- **Failure to follow the expected growth velocity** in height and weight;
- A **sudden increase or decrease** in a previously steady pattern (crossing two major percentile lines, especially after 3 years);
- **Short stature in the absence of short parents.**

Nursing Alert:

BMI-for-age (sex-specific) categories:

- Underweight < 5th percentile
- Healthy weight 5th to < 85th percentile
- Overweight $\geq$ 85th and < 95th percentile
- Obese $\geq$ 95th percentile

Weight-for-length is used for infants/young children whose length (not height) is measured.

6. Measurement Techniques

6.1 Recumbent length

- Measure recumbent length in children **younger than 24–36 months** and in children who **cannot stand alone**.

- **Use a length board** (flat horizontal surface, stationary headboard, smoothly movable footboard at 90°, ruler in mm/ $\frac{1}{16}$ in). **Never use a tape measure alone** — inaccurate and unreliable.
- Prepare: remove clothing/shoes, loosen/remove diaper, remove hair ornaments on the crown.
- **Two measurers required:** assistant holds head in **midline with the crown against the headboard (Frankfort vertical plane)**; lead measurer extends the body (hand on both legs), ensures head against headboard, shoulders/hips not rotated, back not arched, legs not bent, then moves the footboard against the **heels of both feet with toes pointing upward**.
- Read to the **last completed millimeter**; measure **at least twice (ideally 3 times)** and take the **average**; record immediately.

6.2 Standing height (stature)

- Measure height in children **24–36 months and older who can stand alone**.
- **Use a stadiometer**; never wall charts or "floppy-arm" scale-mounted bars.
- Remove shoes and heavy outer clothing; undo hairstyles on the crown.
- Position: back against the vertical surface, **occiput, scapulae, buttocks, and heels in contact**; erect posture with lordosis minimized, knees fully extended, heels flat, head in the **horizontal Frankfort plane**.
- Move the headboard down to the crown (compressing hair); **read at eye level (avoid parallax error)** to the last completed millimeter; measure ≥ 2 times and average.
- **Diurnal variation:** height is less in the afternoon than in the morning due to gravity/spine compression — record the time of day.

6.3 Weight

- Use an appropriately sized **electronic or balance-beam scale**: nearest **10 g for infants, 100 g for children**.
- Balance the scale at 0 before weighing; infant scales tend to be more accurate than adult platform scales.
- Weigh **nude** when using birth-to-2-year charts; older children in underwear/gown/light clothing; always respect privacy.

- Child in the **center of the scale**; keep your hand slightly above the infant/young child to prevent falls.
- Weigh more than once and use the average; two nurses may weigh independently.

6.4 Head circumference

- Head circumference **reflects brain growth**; measure in children **up to 36 months** and in any child whose head size is questionable.
- Measure at the **greatest frontooccipital circumference** — slightly above the eyebrows and pinna of the ears, around the occipital prominence.
- Use a **paper or nonstretchable tape** (a cloth tape stretches and gives a falsely small measurement).
- Measure more than once (head shape and movement affect the reading) to the **last completed millimeter**.
- **Head circumference is larger than chest circumference in most newborns**; they approximate each other around **12 months**; later chest exceeds head by about **5–7 cm**.

6.5 Arm (midarm) circumference

- An **indirect measure of muscle mass**.
- Place the tape vertically along the posterior upper arm from the **acromial process to the olecranon process**; half of that length is the **midpoint** used for the circumference.
- WHO growth curves are available for triceps skinfold and arm circumference.

6.6 Skinfold thickness

- A **convenient measure of body fat** using special calipers (e.g., Lange calipers).
- Common sites: **triceps (most practical), subscapular, suprailiac, abdomen, upper thigh**; record the average of at least 2 measurements of one site.

6.7 Quality control

- Calibrate length boards/stadiometers regularly (ideally daily, at least monthly, and each time they are moved).

- Measure each child at least twice; measurements should agree within **0.5 cm** (ideally **0.3 cm**); use the mean.
-

7. Nursing Care Plan (Growth Assessment Focus)

Assessment: a 6-month-old plotted at the 3rd percentile for weight with a steady drop across two percentile lines; the mother reports normal feeding.

NANDA diagnosis: *Imbalanced nutrition: less than body requirements* (or *Risk for imbalanced nutrition*).

Goals: Growth trend will stabilize or improve on re-plotting after 1 month; the family will understand the measurement results.

Interventions (with rationale):

1. Re-measure with correct technique (length board, nude weight, calibrated scale) — rules out measurement error before alarming the family.
 2. Plot on the correct WHO/CDC chart by age — prevents misinterpretation (e.g., plotting length vs height on the wrong curve).
 3. Compare current to previous measurements (growth velocity) — growth is continuous but uneven; single values can mislead.
 4. Refer/further assess if BMI/weight-for-length is in an abnormal category.
 5. Evaluate: re-measurement confirms trend; the mother can explain the chart result.
-

8. Case Study (Anthropometrics)

Newborn, birth weight 3.1 kg, head circumference 34 cm. On the growth chart both plot near the 50th percentile. Chest circumference 33 cm.

- Which measurements reflect *past* nutrition? (Length and head circumference.)
- Why is head circumference measured here? (Reflects brain growth; newborn HC > chest circumference is normal; they approximate by ~12 months.)
- Why must the birth weight be accurate? (It sets the baseline percentile and is used for the first drug/fluid doses — mg/kg — in the NICU chapters.)

9. Procedure — Laboratory Step: Anthropometric Assessment (Checklist)

Equipment: calibrated infant scale, platform beam scale, length board, stadiometer, nonstretchable tape, calipers (skinfold), pen, growth chart.

#	Step	Done
1	Balance/calibrate the scale at 0; verify the length board/stadiometer calibration date.	☐
2	Weigh the infant nude (sitting/lying, center of scale, hand ready to prevent falls); nearest 10 g.	☐
3	Weigh older child in underwear (nearest 100 g); respect privacy.	☐
4	Measure recumbent length (length board, 2 measurers, Frankfort plane, heels against footboard).	☐
5	Measure standing height (stadiometer; occiput/scapulae/buttocks/heels; read at eye level).	☐
6	Measure head circumference (frontooccipital, nonstretchable tape, mm).	☐
7	Measure arm circumference (midpoint acromial to olecranon).	☐
8	Measure skinfold thickness with calipers (triceps site) — if required.	☐
9	Repeat measurements ≥ 2 times; average; record time of day.	☐
10	Plot on the correct chart (WHO < 2 y; CDC ≥ 2 y) and interpret percentiles/BMI.	☐
11	Document and report deviations to the physician.	☐



Length board technique (infant)



10. Procedure — Hospital Step

- Repeat the checklist at the bedside on admission and before each weight-based dose.
- Always record whether length or height was measured and the time of day.
- For children whose weight is affected by devices (IV armboard, prosthesis), note this when recording.
- Weigh on the ward scales and plot in the child's growth chart; report any measurement outside expected percentiles.

11. Chapter MCQs (10)

Q1. Which parameter reflects *past* undernutrition?

- a) Weight
- b) Skinfold thickness
- c) Height and head circumference
- d) Arm circumference

Q2. Recumbent length is measured in children younger than:

- a) 6 months
- b) 24–36 months who cannot stand alone
- c) 5 years always
- d) 12 months only

Q3. The correct instrument for measuring infant length is:

- a) A flexible cloth tape
- b) A paper tape measure
- c) A length board with two measurers

- d) A wall chart

Q4. When measuring standing height, the child's which of the following contact the vertical surface?

- a) Occiput, scapulae, buttocks, heels
- b) Only the heels
- c) Only the occiput
- d) Head and shoulders only

Q5. The reading for standing height should be taken:

- a) At the child's eye level
- b) At the examiner's eye level (avoid parallax)
- c) Looking downward
- d) Looking from the side

Q6. Head circumference should be measured in children up to:

- a) 12 months
- b) 36 months, or any child with questionable head size
- c) 5 years
- d) 2 years

Q7. The measurement tool that should be avoided for head circumference because it stretches is:

- a) A plastic tape
- b) A paper tape
- c) A cloth tape
- d) A steel tape

Q8. Head and chest circumferences approximate each other at about:

- a) 3 months
- b) 6 months
- c) 12 months
- d) 2 years

Q9. BMI $\geq$ 95th percentile for age and sex is classified as:

- a) Underweight

- b) Healthy weight
- c) Overweight
- d) Obese

Q10. Children whose growth should raise concern include those who:

- a) Follow their expected growth velocity
- b) Cross two major percentile lines downward after 3 years
- c) Are short with short parents
- d) Show stable growth over time

Answers: Q1 c — Q2 b — Q3 c — Q4 a — Q5 b — Q6 b — Q7 c — Q8 c — Q9 d — Q10 b.

12. Key Points

- Growth parameters: **weight, length/height, head circumference, arm circumference, skinfold thickness.**
- **WHO charts** 0–2 years; **CDC charts** ≥ 2 years.
- Measure recumbent length with a **length board (never a tape)**, height with a **stadiometer**, weight on a calibrated scale (10 g infants / 100 g children).
- Head circumference at the **frontooccipital circumference** with a **nonstretchable tape**, measured up to 36 months.
- Measure ≥ 2 –3 times and average; measurements should agree within 0.5 cm.
- Weight/height/length feed the **mg/kg drug doses and mL/kg fluids** used in every ward chapter.

13. Normal Values / Reference Table

Parameter	Instrument	Precision	Chart/Reference
Weight (infant)	infant scale	10 g	WHO < 2 y
Weight (child)	platform scale	100 g	CDC ≥ 2 y

Parameter	Instrument	Precision	Chart/Reference
Recumbent length	length board	mm / 1/16 in	WHO length curve
Standing height	stadiometer	mm / 1/16 in	CDC stature curve
Head circumference	nonstretchable tape	mm	head circumference curve (≤ 36 mo)
Arm circumference	paper/steel tape	mm	WHO arm-curve
Triceps skinfold	calipers	mm	WHO skinfold curves
BMI-for-age	calculated	—	CDC BMI curves

Newborn normals (context): birth weight ~2.5–4 kg; head circumference ~33–36 cm, larger than chest circumference at birth; they equalize at ~12 months; chest then exceeds by ~5–7 cm.

END OF CHAPTER 4

Anthropometric Measurements and Growth Assessment

Next: **Chapter 5** — Vital Signs in Infants and Children

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 5

Chapter 5 — Vital Signs in Infants and Children

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 5

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Take vital signs in the correct order for infants (respirations → pulse → temperature).

2. Measure temperature by the appropriate route (axillary, tympanic, temporal artery, rectal indications) with the correct nursing alerts.
 3. Assess the **apical pulse** for children younger than 2–10 years and palpate the radial pulse in older children, counting a full minute.
 4. Count the respiratory rate by the correct observation (abdominal in infants; thoracic after 1 year) for a full minute.
 5. Measure blood pressure with the correct cuff (bladder width $\geq$ 40% of arm circumference) and interpret cuff-size errors.
 6. State the age-based normal value bands for HR, RR, temperature, and BP (newborn table from the sources).
 7. Perform the lab and hospital steps of vital-sign measurement using the checklist.
-

2. Placement Box

4. Temperature

Definition and norms: core temperature reflects blood flow to the hypothalamus. For rectal temperatures in children, **37.0–37.5°C (98.6–99.5°F)** is an acceptable range where heat loss and production are balanced; for **neonates a core temperature of 36.5–37.6°C (97.7–99.7°F)** is desirable. In the neonate, temperature is monitored for thermoregulation adequacy, not just for fever.

Routes:

- **Axillary** — recommended for infants younger than 1 month; noninvasive, preferred when possible.
- **Tympanic membrane (ear)** — takes precedence for children **older than 2 years**; meta-analysis shows wide variability — do not base a fever-without-focus diagnosis on tympanic readings alone.
- **Temporal artery** — acceptable in all age-groups, correlates better with rectal than axillary/tympanic in studies.
- **Rectal** — closest to core temperature but invasive; take the temperature of the young infant supine with legs flexed, or prone/side-lying; insert past the anal

sphincter no more than **2.5 cm (1 inch)**; apply water-soluble jelly.

Nursing Alert: avoid the rectal route in the immunosuppressed child and in the child with diarrhea, a bleeding disorder, or a history of rectal surgery.

Nursing Alert: axillary temperature is lower than oral, but the difference varies too widely for a standard conversion — **do not add a degree** to the axillary reading. In neonates with fever, axillary alone should not be used for clinical decisions.

Order of measurement in infants (before disturbing): count respirations first, take the pulse next, then measure temperature; if the child is disturbed (crying), record the behavior along with the measurement.

5. Pulse (Heart Rate)

- Assess the heart rate while the child is **resting or sleeping**; the rate in infants is much faster than in adults and varies with anxiety, fear, or crying.
- **In children younger than 2 years, always assess the apical pulse** — the radial pulse is difficult to palpate accurately in this age group. **For children younger than 10 years, auscultate the apical pulse with the stethoscope for a full minute**; in older children palpate the radial pulse for a full minute.
- **Wait several seconds** after placing the stethoscope on the chest for the rate to slow from fear, then count **1 full minute**.
- Count 1 full minute in infants/young children because of possible irregularities.
- **Grade pulses** (Table 4.4): 0 not palpable; +1 weak, easily obliterated; +2 difficult to palpate, may be obliterated; **+3 easy to palpate (normal)**; +4 strong, bounding.
- Compare **radial and femoral pulses** at least once during infancy to detect coarctation of the aorta and other circulation impairment.

Drug-safety link: before digoxin or rate-limiting drugs, count the apical pulse for 1 full minute and withhold if the pulse is below the ordered rate (e.g., < 60 bpm adolescent, < 90 bpm infant).

6. Respiratory Rate

- Count respirations when the child is **resting or sitting quietly**; the rate changes with crying, feeding, and activity, and the most accurate rate is obtained **before disturbing the infant** (e.g., while the parent holds the child).
- **Count for a full minute** for accuracy.
- **Infants breathe primarily diaphragmatically — count abdominal movements; after 1 year of age count thoracic movements.**
- Infants normally display an **irregular breathing pattern with short pauses**; this may increase when ill.

7. Blood Pressure

When to measure: annually in children 3 years through adolescence and in children with renal/adrenal/cardiovascular disease, hypertension risk factors, emergency/ICU, and high-risk infants.

Method: auscultation is the gold standard under most circumstances; automated/oscillometric devices are acceptable for newborns and infants, where auscultation is difficult, and in intensive care. An elevated reading on an automated device should be **repeated using auscultation**; oscillometric readings (e.g., Dinamap) can run ~10 mm Hg higher than auscultation, so the two methods are not interchangeable.

Selection of cuff (the most important factor in an accurate BP):

- Cuff size refers to the **inner inflatable bladder**, not the cloth covering.
- Choose a cuff with a **bladder width at least 40% of the arm circumference** midway between the olecranon and acromion; this usually covers 80–100% of the arm circumference.
- **Too-narrow cuff → falsely high reading. Too-wide cuff → falsely low reading.**
- When the correct size is unavailable, use an **oversized** cuff rather than an undersized one; do not choose a cuff based on its name ("an infant cuff may be too small for some infants").

Positioning: measure with the cubital fossa at heart level, arm supported, stethoscope bell over the brachial artery proximal and medial to the cubital fossa **below the bottom edge of**

the cuff [p10292–10296].

Cuff bladder dimensions:

Cuff	Width (cm)	Length (cm)	Max arm circumference (cm)
Newborn	4	8	10
Infant	6	12	15
Child	9	18	22
Small adult	10	24	26
Adult	13	30	34

Interpretation: normal BP is based on **gender, age, and height percentiles** (tables with 50th, 90th, 95th, 99th percentiles) to avoid misclassifying tall/short children; lesions: BP < 90th percentile normal; 90th–95th prehypertension; ≥ 95th percentile possible hypertension (repeated twice, then staged).

8. Pulse Oximetry (SpO₂)

- Pulse oximetry (SaO₂) is reliable, noninvasive, and often routine in ill children; the pulse rate on the oximeter **should coincide with the apical pulse** to ensure the reading is accurate.
- Decide whether monitoring is continuous or intermittent (with vital signs).

9. Procedure — Laboratory Step: Vital Signs (Checklist)

Equipment: digital/infrared thermometer, stethoscope, sphygmomanometer, cuff set (multiple sizes), pulse oximeter, watch with seconds.

#	Step	Done
1	Order: count respirations first (before disturbing), then pulse, then temperature.	☐

#	Step	Done
2	RR: evaluate resting; count 1 full minute; abdominal movements in infants, thoracic after 1 year.	☐
3	Pulse: apical auscultation < 10 years (always < 2 years); radial palpation in older; count 1 full minute.	☐
4	Wait several seconds for rate to slow after stethoscope placement; note rhythm and grade.	☐
5	Temperature: choose route (axillary < 1 month; tympanic > 2 years; temporal artery any age; rectal with indications only).	☐
6	Rectal: jelly, insert ≤ 2.5 cm; avoid in immunosuppression/diarrhea/bleeding disorder/rectal surgery.	☐
7	BP: measure arm circumference; choose cuff with bladder $\geq 40\%$ of circumference (oversize if unsure).	☐
8	BP: cubital fossa at heart level; stethoscope bell over brachial below cuff edge.	☐
9	Pulse oximetry: on appropriate site; confirm oximeter pulse equals apical pulse.	☐
10	Record values with the child's behavior (crying/sleeping) and the method used; compare to age norms.	☐



Position for apical pulse in an infant



10. Procedure — Hospital Step

- Repeat the checklist at the bedside **before disturbing the child** where possible.
- Always document the method (apical/radial; axillary/tympanic/rectal) and the child's activity.
- Compare every recording to the normal band for that age and to previous readings — an isolated elevated BP may reflect stress, not hypertension.

11. Normal Values Table (Age-Based, from the Sources)

Newborn / infant norms (sources give the following):

Parameter	Newborn	Source
Temperature	36.5–37.5°C (core 36.5–37.6°C neonate)	—
Heart rate	110–160 beats/min (to 180 while crying)	—
Respiratory rate	30–60 breaths/min	—
Blood pressure	50–75 / 30–45 mm Hg	—

Child context values found in the sources:

- Heart rate 120–140 at rest with RR 30–60 breaths/min (infant).
- Axillary example: 98.9°F (37.2°C), pulse 80, RR within the normal range for a school-age child.

12. Nursing Care Plan (Vital Signs Focus)

Assessment: a crying 9-month-old in the ward; radial pulse is hard to feel; parent is anxious.

NANDA diagnosis: *Risk for unstable vital signs / Anxiety (parent) related to the child's illness.*

Goals: Vital signs will be measured accurately without false elevation; the parent will understand the readings.

Interventions (with rationale):

1. Count respirations first while the parent holds the child, before disturbing — most accurate rate.
 2. Take the apical pulse (child < 2 years), waiting for the rate to slow after stethoscope placement — avoids fear-related tachycardia.
 3. Choose the correct BP cuff (blade $\geq$ 40% of arm circumference); note that too small a cuff reads falsely high — prevents misdiagnosis.
 4. Reassure the parent by explaining the values against age norms and previous readings — anxiety reduction.
 5. Evaluate: values recorded with method + behavior; provider notified if abnormal.
-

13. Case Study (Vital Signs)

Newborn, 1 day old, in NICU. Axillary temperature 36.9°C, apical HR 148, RR 52 with abdominal movement, BP by oscillometry 62/40 (left arm). Spo₂ 96% on room air, oximeter pulse 146.

- Why was axillary temperature used? (*Infants < 1 month: axillary recommended; avoiding rectal risks.*)
 - Why the apical pulse? (*Child < 2 years — radial unreliable.*)
 - Why count RR for a full minute observing the abdomen? (*Infants breathe diaphragmatically and irregularly.*)
 - The BP was taken by oscillometric device; what should happen if it were elevated? (*Repeat by auscultation before interpreting — gold standard.*)
-

14. Chapter MCQs (10)

Q1. For an infant, the correct order for vital signs is:

- a) Temperature, pulse, respirations
- b) Respirations, pulse, temperature
- c) Pulse, respirations, temperature
- d) Blood pressure first

Q2. In a child younger than 2 years, the pulse is best obtained by:

- a) Radial palpation for 30 seconds
- b) Apical auscultation for a full minute
- c) Carotid palpation
- d) Femoral palpation

Q3. In infants, respirations are:

- a) Primarily thoracic
- b) Primarily diaphragmatic — count abdominal movements
- c) Identical to adults
- d) Counted for 15 seconds

Q4. The rectal route should be avoided in the child with:

- a) A feeding problem
- b) Diarrhea, a bleeding disorder, or immunosuppression
- c) Cough
- d) Constipation

Q5. Correct BP cuff size requires a bladder width of at least:

- a) 25% of the arm circumference
- b) 40% of the arm circumference
- c) 80% of the arm circumference
- d) The whole arm length

Q6. A BP cuff that is too small produces:

- a) A falsely low reading
- b) A falsely high reading
- c) No change
- d) Only diastolic error

Q7. When the correct cuff size is unavailable, the nurse should:

- a) Use the adult cuff always
- b) Use an oversized cuff rather than an undersized one
- c) Skip the measurement
- d) Use an infant cuff regardless of arm size

Q8. Tympanic (ear) thermometry is most reliable in children:

- a) Younger than 1 month
- b) Older than 2 years
- c) Only asleep
- d) With ear discharge

Q9. An elevated automated BP reading in a child should be:

- a) Documented as hypertension
- b) Repeated using auscultation
- c) Ignored
- d) Compared to adult norms

Q10. The apical pulse should be counted for 1 full minute before administering:

- a) Febrile antipyretics
- b) Rate-limiting drugs such as digoxin
- c) Antibiotics
- d) Oral rehydration

Answers: Q1 b — Q2 b — Q3 b — Q4 b — Q5 b — Q6 b — Q7 b — Q8 b — Q9 b — Q10 b.

15. Key Points

- Order for infants: **respirations** → **pulse** → **temperature** → **(BP)**.
- Apical pulse for **< 2 years always**; auscultate apically to 10 years, 1 full minute.
- Count **1 full minute**; infants are diaphragmatic and irregular.
- **BP cuff bladder ≥ 40% of arm circumference**; too small = falsely high, too large = falsely low; oversize rather than undersize.

- Auscultation is the **gold standard** BP method; repeat elevated automated readings by auscultation.
- Avoid **rectal temperature** in immunosuppression/diarrhea/bleeding disorder/rectal surgery.
- Newborn norms: T 36.5–37.5°C, HR 110–160 (180 crying), RR 30–60, BP 50–75/30–45.

END OF CHAPTER 5

Vital Signs in Infants and Children

Next: **Chapter 6** — Neurologic Assessment and Infant Reflexes

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 6

Chapter 6 — Neurologic Assessment and Infant Reflexes

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 6

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Perform the basic neurologic screen: level of consciousness, pupil responses, posture, motor tone, and fontanelle assessment.
2. Elicit the newborn reflexes correctly (sucking, rooting, grasp, Babinski, Moro, tonic neck, dance/step) using the described stimulation.
3. State the normal time of appearance and disappearance of each primitive reflex.
4. Recognize abnormal findings: asymmetric or absent Moro, persistent primitive reflexes (obligatory tonic neck, hyperactive Moro), and a positive Babinski persisting after 1 year.
5. Complete the lab and hospital steps of the neurologic/reflex assessment using the checklist.

2. Placement Box

4. The Basic Neurologic Screen

A useful screen before reflex testing:

- **Level of consciousness (LOC):** alert, lethargic, obtunded, comatose; observation periods according to age.
- **Pupils:** equal, round, reactive to light; note asymmetric or one dilated unresponsive pupil (emergency).
- **Posture and movement:** spontaneous activity (none in deep sleep), symmetric movement of all limbs; flaccidity or spasticity.
- **Fontanelles:** anterior fontanelle flat, soft, open (closes ~12–18 months); a **sunken fontanelle** suggests dehydration [Ch 17]; a tense/bulging fontanelle suggests increased intracranial pressure.
- **Cranial nerves:** tested at age-appropriate level (e.g., visual red reflex, pupillary response, sucking, facial symmetry).

Safety / Red flag: asymmetric pupils in a child with altered consciousness, a tense bulging fontanelle, or seizures are emergencies — report immediately.

5. Assessment of Reflexes in the Newborn

Reflex	How to elicit	Expected response	When it disappears
Blinking/corneal	Bright light or approach to cornea	Blink	Persists lifelong
Pupillary	Bright light to eye	Pupil constricts	Persists lifelong

Reflex	How to elicit	Expected response	When it disappears
Doll's eye	Slow turning of head	Eyes lag behind head	Disappears as fixation develops (persistence = damage)
Sneeze	Nasal irritation	Sneeze	Persists lifelong
Glabellar	Tap on glabella	Eyes close tightly	Persists
Sucking	Nipple or nonlatex finger in mouth	Strong, vigorous suck	Persists throughout infancy
Gag	Posterior pharynx stimulation	Gag	Persists lifelong
Rooting	Stroke the cheek	Head turns to that side and sucks	~3–4 months (may persist to 12 months)
Extrusion	Touch the tongue	Tongue forced outward	By 4 months
Grasp (palmar)	Touch palm near base of digits	Finger flexion	Lessens after 3 months (voluntary grasp)
Grasp (plantar)	Touch sole near base of toes	Toe flexion	Lessens by 8 months
Babinski	Stroke outer sole from heel upward across the ball	Toes hyperextend, hallux dorsiflexes	After 1 year
Ankle clonus	Brisk dorsiflexion of foot (knee partially flexed)	One or two oscillating beats	Eventually no beats
Moro	Sudden jarring/change in equilibrium	Extension + abduction of extremities, fanning of fingers (index + thumb form C), then flexion + adduction; may cry	After 3–4 months (strongest first 2 months)

Reflex	How to elicit	Expected response	When it disappears
Startle	Sudden loud noise	Arm abduction with flexed elbows, hands clenched	By 4 months
Perez	Prone; thumb pressed along spine sacrum → neck	Crying, flexion, pelvis/head elevation, lordosis	By 4–6 months
Tonic neck	Turn head to one side	Arm and leg extend on that side; opposite side flexes	By 3–4 months
Trunk incurvation (Galant)	Stroke the back alongside the spine	Hips move toward the stimulated side	By 4 weeks
Dance/step	Hold infant, sole of foot touches a hard surface	Reciprocal flexion/extension simulating walking	After 3–4 weeks
Crawl	Place on abdomen	Crawling movements of arms and legs	~6 weeks
Placing	Hold upright; briskly place dorsal foot against a hard object	Leg lifts as if stepping	Varies

6. Eliciting the Key Reflexes (Technique Notes)

- **Sucking and rooting:** place a nipple or nonlatex gloved finger in the mouth (vigorous suck expected — weak suck should be reported); root by **stroking the cheek** and noting the head turning to the stimulated side and sucking.
- **Moro:** sudden jarring or change in equilibrium (e.g., mild head drop with support); the strongest response is seen in the first 2 months; **an asymmetric or partial Moro reflex is abnormal** (see Section 7).
- **Tonic neck:** after 2 months becomes obligatory for the "fencing" posture; **an obligatory tonic neck reflex at any age is abnormal.**
- **Grasp:** firm but gentle pressure on the palm/soles produces flexion.

- **Babinski:** stroke the **outer sole upward from the heel and across the ball of the foot**; toes hyperextend and the hallux dorsiflexes then fans in infants — normal; a **positive Babinski is significant when it persists after 1 year**, when it reflects pyramidal tract dysfunction.
 - **Dance/step:** hold the infant so the sole of the foot touches a hard surface — reciprocal flexion/extension simulates walking.
-

7. Clinical Significance and Abnormal Findings

- **Asymmetric or partial Moro** suggests brachial plexus injury, clavicle fracture, or CNS dysfunction; it should be reported.
 - **Persistent primitive reflexes** are the earliest clue to **cerebral palsy**: obligatory tonic neck reflex at any age, and non-obligatory or hyperactivity of the Moro, plantar, and palmar grasp reflexes beyond the normal age.
 - **Asymmetric tonic neck reflex** (evoked by flexing or extending the head) persisting abnormally has diagnostic significance.
 - **Absence of tonic neck reflex** and asymmetric posturing (except the tonic neck reflex) are reportable findings in the newborn.
 - **Exaggerated/hyperactive Moro** (e.g., in drug-exposed newborns) and inability to coordinate swallowing should be noted.
 - In a child, a **fluctuating Babinski reflex is often observed after seizures**.
-

8. Nursing Care Plan (Neurologic Assessment Focus)

Assessment: a 1-day-old newborn with a weak suck, no rooting, and asymmetric Moro response on the right arm.

NANDA diagnosis: *Risk for injury / Neurologic status: risk for altered intracranial* (assessment monitoring) related to suspected neurologic dysfunction.

Goals: Continue monitoring and report changes promptly; the parents will understand the referral.

Interventions (with rationale):

- 1. Elicit reflexes consistently in a quiet state and by the same method — reproducible baseline.
- 2. Document precisely: "Moro: asymmetric — right arm absent" — prompts investigation (nerve vs CNS).
- 3. Re-check before any major procedure and notify the physician of asymmetry/persistence — early diagnosis.
- 4. Support parental understanding without alarming — explain that a weak suck may be transient but warrants watching.
- 5. Evaluate: interval neurologic re-assessment completed; findings documented and reported.

9. Procedure — Laboratory Step: Neurologic and Reflex Screen (Checklist)

Equipment: penlight, nipple or nonlatex glove, firm surface, reflex hammer, quick crisp surface for step reflex.

#	Step	Done
1	Observe LOC, posture, and spontaneous movement before handling.	☐
2	Assess pupils equal/round/reactive to light.	☐
3	Palpate the anterior fontanelle (flat/soft/open) and note size.	☐
4	Elicit sucking (nipple/finger — expect vigorous suck).	☐
5	Elicit rooting (stroke cheek — head turns to side).	☐
6	Elicit palmar and plantar grasp (finger flexion; toe flexion).	☐
7	Elicit Babinski (stroke outer sole heel → ball; hallux dorsiflexion + fanning).	☐
8	Elicit Moro last (sudden jarring; extension/abduction then flexion/adduction); note symmetry.	☐
9	Elicit tonic neck (head turn — fencing posture) and dance/step (sole on surface).	☐

#	Step	Done
10	Document reflexes described objectively + compare to expected age timing.	☐
11	Report asymmetry, absence, or persistence beyond normal age immediately.	☐



Eliciting the grasp and Babinski reflexes



Eliciting the grasp and Babinski reflexes

10. Procedure — Hospital Step

- Repeat the checklist on the real newborn **in a quiet state**, with warm hands, before feeding if possible.
- Elicit the **Moro reflex at the end** to avoid the startle disturbing the rest of the examination.
- Document using the ward's admission sheet: "Describe reflexes observed: Moro, sucking, Babinski, plantar, and other age-appropriate reflex responses".
- Report asymmetric Moro, weak/absent suck, or persistent primitive reflexes immediately.

11. Case Study (Reflex Assessment)

A 2-day-old full-term newborn, examined before discharge. Sucking and rooting present and strong; palmar and plantar grasp present; Babinski positive. At the end of the examination the nurse raises the head slightly and allows it to drop — the newborn extends and abducts both arms, fingers fan, with the index and thumb forming a C, then adducts and flexes; both sides respond equally and the baby cries briefly.

- Why was the Moro reflex left for the end? (*Startle may make the infant cry and disturb the rest of the exam.*)
- Is the response normal? (*Yes — symmetric Moro with extinction after flexion; expected up to 3–4 months, strongest in the first 2 months.*)
- Which finding would have been abnormal? (*Asymmetric response — e.g., absent arm movement — suggesting brachial plexus/CNS injury.*)
- The positive Babinski is expected at this age; when would it be significant? (*After 1 year of age.*)

12. Chapter MCQs (10)

Q1. The rooting reflex is elicited by:

- a) Placing a finger in the mouth
- b) Stroking the cheek alongside the mouth
- c) Sudden jarring
- d) Stroking the sole from heel to ball

Q2. The Moro reflex normally disappears by:

- a) 2 weeks
- b) 3–4 months
- c) 6 months
- d) 1 year

Q3. The Babinski reflex (dorsiflexion of the hallux with fanning of the toes) is significant as a pyramidal sign when it:

- a) Is present at birth
- b) Persists after 1 year of age
- c) Appears only during sleep

- d) Is unilateral for one week

Q4. Palmar grasp lessens at about:

- a) 3 months of age
- b) 8 months
- c) 1 year
- d) 2 years

Q5. An asymmetric Moro reflex should make the nurse suspect:

- a) Benign fussiness
- b) Brachial plexus or CNS dysfunction (report)
- c) Feeding intolerance
- d) Normal findings

Q6. The persistence of the tonic neck reflex beyond the normal age is described as:

- a) A startle response
- b) An obligatory tonic neck reflex — abnormal
- c) A normal variant to 2 years
- d) A feeding reflex

Q7. The tonic neck (fencing) position is:

- a) Extension of arm and leg on the same side as the turned head; flexion of the other side
- b) Flexion on both sides
- c) Symmetric
- d) Arms extended both sides

Q8. Which reflex simulates walking when the sole touches a hard surface?

- a) Dance or step reflex
- b) Crawl reflex
- c) Perez reflex
- d) Placing reflex

Q9. A weak sucking reflex should be:

- a) Ignored in newborns
- b) Reported — it may signal neurologic difficulty

- c) Treated with a pacifier
- d) Noted only after 6 months

Q10. Persistent primitive reflexes beyond the normal age are an early clue to:

- a) Normal tone
- b) Cerebral palsy
- c) Colic
- d) Jaundice only

Answers: Q1 b — Q2 b — Q3 b — Q4 a — Q5 b — Q6 b — Q7 a — Q8 a — Q9 b — Q10 b.

13. Key Points

- Use the newborn reflex table for technique, expected response, and disappearance age.
- Elicit the **Moro reflex last** in the examination.
- An **asymmetric/partial Moro** and weak/absent suck should be reported.
- **Persistent primitive reflexes** (obligatory tonic neck, hyperactive Moro beyond normal age) are early clues to cerebral palsy.
- Babinski is normal in infancy; a positive Babinski after 1 year is a pyramidal sign.

14. Normal Values / Timeline Table

Reflex	Appears	Usual disappearance
Sucking	At birth	Persists through infancy
Rooting	At birth	3–4 months (to 12 months)
Palmar grasp	At birth	Lessens after 3 months
Plantar grasp	At birth	Lessens by 8 months
Moro	At birth	3–4 months (strongest first 2 months)

Reflex	Appears	Usual disappearance
Startle	At birth	By 4 months
Tonic neck	At birth	3–4 months
Dance/step	At birth	After 3–4 weeks
Babinski (positive)	At birth	After 1 year

Neurofunction values: newborn HR 100–180 shortly after birth, then 120–140 at rest; respirations 30–60 irregular and abdominal.

END OF CHAPTER 6

Neurologic Assessment and Infant Reflexes

Next: **Chapter 7** — Fluid, Electrolyte, and Calorie Calculations

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 7

Chapter 7 — Fluid, Electrolyte, and Calorie Calculations

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 7

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain why fluid requirements per kilogram are higher in infants than in older children.
2. Calculate daily maintenance fluid and calorie requirements using the Holliday–Segar method.

3. Classify dehydration (mild, moderate, severe) and prescribe the correct first-line therapy (oral rehydration vs IV).
4. Calculate ORS volumes for replacement therapy and IV infusion flow rates (drops/min).
5. Monitor IV therapy, intake/output, urine output, and daily weight in the hospitalized child.

2. Placement Box

4. Why Children Differ in Fluid Balance

- Infants have a **proportionately greater amount of body water** than adults, so their **fluid intake requirement per kilogram is greater**, and they are at higher risk of fluid loss.
- Kidney immaturity limits water and electrolyte conservation; insensible losses are relatively larger in infants (larger body surface area per kg).
- Calorie needs for growth are high, so very young children need **100 kcal/kg/day** for the first 10 kg — matching the 100 mL/kg/day figure of the Holliday–Segar method.

5. Maintenance Fluid and Calorie Requirements (Holliday–Segar Method)

Parenteral maintenance fluid for a hospitalized child is planned by this method:

Body weight	Fluid per kg per day	Example
First 10 kg	100 mL/kg/day	$10 \text{ kg} \times 100 = 1000 \text{ mL/day}$
Second 10 kg	50 mL/kg/day	$10 \text{ kg} \times 50 = 500 \text{ mL/day}$
Each kg above 20 kg	20 mL/kg/day	$(\text{weight} - 20) \times 20 \text{ mL/day}$

Total for the day = sum of the three columns. The same 100/50/20 pattern is used for maintenance **calories** (kcal/day).

Worked example: A **25-kg** child needs:

- First 10 kg → 1000 mL/day
- Second 10 kg → 500 mL/day
- Remaining 5 kg → $5 \times 20 = 100$ mL/day
- **Total = 1600 mL/day** ✓

Daily fluid requirements by age (practical guide): newborn/infant about **140–160 mL/kg/day**; child 100 mL/kg/day for the first 10 kg then as above.

5.1 Worked Calculations (Holiday–Segar)

1. **8-kg infant:** $8 \times 100 = 800$ mL/day (≈ 33 mL/hr).
2. **12-kg toddler:** $(10 \times 100) + (2 \times 50) = 1000 + 100 = 1100$ mL/day (≈ 46 mL/hr).
3. **40-kg school child:** $1000 + 500 + (20 \times 20) = 1000 + 500 + 400 = 1900$ mL/day (≈ 79 mL/hr) — plus calories similarly.

6. Dehydration: Classification and First-line Therapy

Use the AAP dehydration classification and therapy plan:

Degree	Weight loss / signs	Rehydration therapy	Replacement of stool losses	Maintenance
Mild (5–6%)	Increased thirst; slightly dry buccal mucous membranes	ORS 50 mL/kg within 4 hours	ORS 10 mL/kg (infants) or 150–250 mL per diarrheal stool (older children)	Continue breastfeeding/formula; usual diet for solids
Moderate (7–9%)	Loss of skin turgor; dry buccal mucous membranes; sunken eyes;	ORS 100 mL/kg within 4 hours	Same as above	Same as above

Degree	Weight loss / signs	Rehydration therapy	Replacement of stool losses	Maintenance
	sunken fontanel			
Severe (>9%)	Moderate signs plus rapid thready pulse, cyanosis, rapid breathing, lethargy, coma	IV fluids (lactated Ringer), 40 mL/kg until pulse and level of consciousness normalize; then ORS 50–100 mL/kg	Same as above	Same as above

- ORT is the treatment of choice for most cases of dehydration; it is more effective, safer, less painful, and less costly than IV therapy.
- **Vomiting is not a contraindication** to ORS — give small frequent amounts (spoon or small syringe, 5–10 mL every 1–2 minutes).
- ORS should contain about **75 mmol/L sodium** and **13.5 g/L glucose** (standard solutions: Pedialyte, Infalyte, Ricelyte). **Do not use** juice, carbonated drinks, gelatin — they are high in carbohydrate and low in electrolytes.
- Homemade ORS: 1 L water + 8 tsp sugar + 1 tsp salt (use only if no commercial ORS).
- **For severe dehydration and shock**, IV fluids (normal saline or lactated Ringer) are initiated; the classic initial bolus is **20 mL/kg**, reassessing after each bolus.
- **NURSING ALERT:** clear fluids (juices, soft drinks, gelatin) **do not help** diarrhea; a BRAT diet is contraindicated.

7. Intravenous Fluid Therapy in the Hospital

Patients usually requiring IV fluids: severe dehydration, uncontrollable vomiting, inability to drink (extreme fatigue, coma), and severe gastric distention.

- The IV solution is usually a saline solution with **5% dextrose (D5)**; sodium bicarbonate may be added because acidosis accompanies severe dehydration.

- Rapid replacement is **contraindicated in hypertonic dehydration** because of the risk of water intoxication.
- Frequency of assessment: monitor the correct concentration, flow rate, infusion site, and intake/output; weigh the child daily — **accurate weight is one of the best indicators of fluid status.**

7.1 Drop-Rate Calculations (Worked Examples)

Formula: **Drops/min = (Total mL × Drop factor) ÷ Minutes**

- **Macro drip sets** deliver 15–20 gtts/mL; **microdrip** delivers **60 gtts/mL** (so, drop rate in gtts/min = mL/hr).
- **Example (macro 20):** 500 mL over 8 hours = $500 \times 20 \div 480 = \approx 21$ drops/min.
- **Example (microdrip 60):** 800 mL over 24 hours = 33 mL/hr → **33 drops/min.**
- Teach: 1000 mL over 24 h with a microdrip = $1000/24 \approx 42$ mL/hr ≈ 42 gtts/min.

Recheck the electric pump by **drop counting for 1 minute** and verify against the written order; investigate any discrepancy.

8. Monitoring Fluid Status: Urine Output and I/O

- Minimal acceptable urine output: **0.5–1 mL/kg/hr** in children; oliguria is urine output **< 1 mL/kg/hr** and anuria = no urine.
- Maintain strict **intake/output record** and weigh daily; monitor capillary refill, sensorium, skin turgor, oral mucosa, and presence of tears every 4–8 hours.
- The **most reliable indicators** of volume status are **capillary refill, alterations in sensorium, and urinary output.**

9. Electrolytes and Calories at the Bedside

- **Sodium (Na):** main cation of the extracellular fluid, 135–145 mmol/L. Low-sodium ORS/fluids are used in maintenance to avoid hyponatremia.

- **Potassium (K):** 3.5–5.5 mmol/L; **only add K to IV fluids when urine output is adequate; never give K by IV push.**
 - **Calories:** 1 g carbohydrate $\approx$ 4 kcal; glucose is 3.4 kcal/g. Use the 100/50/20 method for maintenance calories of parenteral support.
-

10. Nursing Care Plan (Dehydration — Risk for Deficient Fluid Volume)

Assessment: an 8-month-old with 2 days of watery diarrhea, 6 stools in 8 hours, decreased urine output (2 wet diapers/day), sunken eyes, dry mucous membranes, slightly decreased skin turgor, weight unchanged from 2 weeks prior.

NANDA diagnosis: *Deficient fluid volume* related to excessive fluid losses (diarrhea) as evidenced by decreased urine output, dry mucous membranes, and sunken eyes.

Assessment classification: mild dehydration (weight stable, tears present).

Goals: Rehydrate with ORS; restore urine output to $\geq$ 0.5–1 mL/kg/hr; maintain daily weight.

Interventions (with rationale):

1. Give **ORS 50 mL/kg within 4 hours** in 5–10 mL amounts every 2–5 min — replaces deficit slowly and reduces vomiting.
 2. Continue breastfeeding during and after rehydration — supports energy and fluid.
 3. Replace stool losses: 10 mL/kg ORS per diarrheal stool — prevents recurrence of dehydration.
 4. Record I/O, count wet diapers, weigh daily — objective evaluation of rehydration.
 5. Avoid juices, gelatin, and caffeinated drinks — they worsen the diarrhea.
 6. Evaluate: hydration restored (tears, moist mucosa, urine output $\geq$ 0.5 mL/kg/hr), plan for early refeeding.
-

11. Procedure — Laboratory Step: IV Infusion Rate Setting and Fluid Balance Chart (Checklist)

Equipment: IV catheter/site, prescribed solution, macro or micro infusion set, infusion pump or watch, fluid balance chart, scales.

#	Step	Done
1	Verify the written order: solution, additives, and rate (mL/hr).	☐
2	Calculate maintenance rate by Holliday–Segar (100/50/20).	☐
3	Convert mL/hr → drops/min using the set's drop factor (60 micro / 15–20 macro).	☐
4	Regulate the pump or count drops for 1 full minute; confirm the calculated value.	☐
5	Verify the site (patency, no infiltration) and the child's identity/wristband.	☐
6	Record starting volume, time, amount infused, and discard/end time on the fluid chart.	☐
7	Do not add K to IV fluid unless urine output is adequate; never IV-push K.	☐
8	Reassess at least hourly: site, rate, and running volume.	☐
9	Measure urine output and enter on the chart (mL/kg/hr).	☐
10	Weigh the child daily at the same time and enter on the chart.	☐



Obtaining the digit span rhythm in IV rate setting



12. Procedure — Hospital Step

- Obtain an **accurate admission weight** (undressed, same scale) before ordering fluids.
- For severe dehydration, assist with the **20 mL/kg** LR/NS bolus per protocol and reassess after each bolus.
- Teach the mother ORS technique (spoon/syringe, 5–10 mL, every 1–2 min), stool-loss replacement, and to bring the child back if urine output falls or lethargy develops.
- Document intake (including ORS) and output (diarrhea/vomiting/urine) carefully; report urine output $< 0.5\text{--}1$ mL/kg/hr.

13. Case Studies

A) Maintenance fluid (Holliday–Segar): A 12-kg toddler is NPO for surgery. Ordered IV: D5 $\frac{1}{2}$ NS plus 20 mEq/L KCl. Calculate the 24-h rate. *Answer: $(10 \times 100) + (2 \times 50) = 1100$ mL/day = 46 mL/hr.*

B) ORS for moderate dehydration: A 10-kg child with sunken eyes, dry mucosa, loss of skin turgor (moderate). *Answer: ORS 100 mL/kg $\div 4$ h = 1000 mL over 4 hours; then maintenance therapy and stool-loss replacement.*

C) IV drip rate: Order 500 mL D5 $\frac{1}{2}$ NS over 8 h with a **microdrip**. *Answer: 62.5 mL/hr $\rightarrow \approx 63$ gtts/min (microdrip = mL/hr).*

14. Chapter MCQs (10)

Q1. Using the Holliday–Segar method, the maintenance fluid for an 8-kg infant is:

- a) 400 mL/day

- b) 800 mL/day
- c) 1000 mL/day
- d) 1600 mL/day

Q2. A 25-kg child needs daily maintenance fluid of:

- a) 1000 mL
- b) 1200 mL
- c) 1600 mL
- d) 2000 mL

Q3. The ORS volume for **moderate** dehydration within 4 hours is:

- a) 20 mL/kg
- b) 50 mL/kg
- c) 100 mL/kg
- d) 40 mL/kg

Q4. Severe dehydration with shock is initially managed with:

- a) ORS 100 mL/kg
- b) IV lactated Ringer 40 mL/kg until pulse and consciousness normalize
- c) Plain water
- d) Juice

Q5. Minimal acceptable urine output in children is:

- a) 0.5–1 mL/kg/hr
- b) 2–3 mL/kg/hr
- c) 5 mL/kg/hr
- d) 10 mL/hr total

Q6. Vomiting during oral rehydration is:

- a) A contraindication to ORS
- b) Managed by giving small frequent amounts (5–10 mL every few minutes)
- c) Managed by giving fruit juice instead
- d) Managed by clear fluids only

Q7. The IV "microdrip" set delivers:

- a) 10 gtts/mL

- b) 15 gtts/mL
- c) 20 gtts/mL
- d) 60 gtts/mL

Q8. A 500 mL IV over 8 hours with a 20-gtts/mL macro set runs at approximately:

- a) 62 gtts/min
- b) 21 gtts/min
- c) 30 gtts/min
- d) 10 gtts/min

Q9. The initial bulk fluid bolus for severe dehydration in an infant is:

- a) 5 mL/kg
- b) 20 mL/kg
- c) 100 mL/kg
- d) 150 mL

Q10. Which statement about potassium is correct?

- a) Give by IV push for rapid correction
- b) Add to IV fluids only when urine output is adequate; never give by IV push
- c) Free water is the best replacement
- d) Potassium is not needed in diarrhea

Answers: Q1 b — Q2 c — Q3 c — Q4 b — Q5 a — Q6 b — Q7 d — Q8 b — Q9 b — Q10 b.

15. Key Points

- Use the **Holliday–Segar 100/50/20** method for maintenance fluid and calories.
- Mild (5–6%) → ORS 50 mL/kg/4 h; moderate (7–9%) → ORS 100 mL/kg/4 h; severe (>9%) → IV LR 40 mL/kg until pulse/LOC normalize, then ORS.
- ORS: 75 mmol/L Na, 13.5 g/L glucose; give small frequent amounts even if vomiting; replace each stool loss.
- IV bolus 20 mL/kg NS or LR for shock, reassessing after each bolus.
- **Never give K by IV push; only add K when urine output is adequate.**
- Monitor urine output ≥ 0.5 –1 mL/kg/hr and weigh daily.

16. Normal Values Table

Parameter	Normal value
Sodium (Na ⁺)	135–145 mmol/L
Potassium (K ⁺)	3.5–5.5 mmol/L
Urine output (child)	0.5–1 mL/kg/hr (minimum); 1–2 mL/kg/hr ideal in hospitalized
Daily maintenance water (Holliday–Segar)	100/50/20 mL/kg/day
Daily maintenance calories (Holliday–Segar)	100/50/20 kcal/kg/day
Infant fluid requirement	140–160 mL/kg/day

END OF CHAPTER 7

Fluid, Electrolyte, and Calorie Calculations

Next: **Chapter 8** — Medication Administration and Pediatric Dose Calculations

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 8

Chapter 8 — Medication Administration and Pediatric Dose Calculations

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 8

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain why pediatric dosing is weight-based and why it differs from adult dosing.
2. Calculate a safe mg/kg dose, verify it against the recommended range, and convert it to mL of the available preparation.
3. State when body surface area (BSA) dosing is used and calculate BSA with the Mosteller formula.
4. Apply the safety "rights," two-identifier verification, and high-alert drug double-checking.
5. Administer oral and intramuscular medication using atraumatic, aspiration-safe techniques.

2. Placement Box

4. Why Pediatric Dosing Is Different

- Neonates and preterm infants are especially vulnerable: **immature liver enzyme systems, lower plasma protein concentrations**, and **immature kidney function**.
- Beyond the newborn period, children **metabolize many drugs more rapidly than adults**, so they may need **larger doses per kilogram** and/or more frequent administration — important in pain control.
- Doses are therefore expressed **in mg/kg per dose or per day**, and some drugs (e.g., chemotherapy) are dosed by **BSA in mg/m²**.
- BSA vs weight: an infant who is shorter and weighs less has **relatively more BSA** than expected from the weight.

5. Weight-Based Dose Calculation

General formula:

- **Dose per day** = weight (kg) × recommended mg/kg/day

- **Dose per single administration** = weight (kg) × recommended mg/kg/dose (or divide the daily dose by the number of doses/day)
- **Volume to give (mL)** = ordered dose (mg) ÷ concentration (mg/mL)

5.1 Worked Examples

Example 1 (classic — bank Q14 style): Child weighs **22 lb**; recommend **20–40 mg/kg/day** divided every 8 h.

- Convert: $22 \text{ lb} \div 2.2 = \mathbf{10 \text{ kg}}$.
- Daily range: $10 \times 20 = 200 \text{ mg}$ to $10 \times 40 = \mathbf{400 \text{ mg/day}}$ ✓ (safe range 200–400 mg/day).
- Per dose (3 doses/day): 67–133 mg per dose.

Example 2 (paracetamol): 8-kg infant; 10–15 mg/kg/dose q4–6h, max 5 doses/day.

- 8×12.5 (midrange) = 100 mg/dose.
- Available syrup 160 mg/5 mL (32 mg/mL) → $100 \div 32 \approx \mathbf{3.1 \text{ mL}}$ per dose (give with an oral syringe in mL only).

Example 3 (antibiotic per day vs per dose): Amoxicillin 40 mg/kg/day divided **q8h** for a 12-kg child.

- $12 \times 40 = 480 \text{ mg/day} \rightarrow 160 \text{ mg/dose q8h}$.
- Suspension 250 mg/5 mL (50 mg/mL) → $160 \div 50 = \mathbf{3.2 \text{ mL}}$ q8h.

Teaching: always verify the calculated dose **falls within the safe range** before giving; if it does not, or if the preparation/route is doubtful, **check with the prescriber before proceeding** — the nurse is legally liable for the drug they administer.

6. Body Surface Area (BSA) Dosing

- Used for drugs where metabolic rate matters (e.g., chemotherapy); expressed as **mg/m²**.
- **Mosteller formula:** $\text{BSA (m}^2\text{)} = \sqrt{[(\text{height cm} \times \text{weight kg}) \div 3600]}$.
- The **West nomogram** is a graphic alternative.

- **Worked example:** height 90 cm, weight 15 kg $\rightarrow \sqrt{[(90 \times 15) \div 3600]} = \sqrt{(0.375)} = 0.61 \text{ m}^2$. A drug ordered at $50 \text{ mg/m}^2 = 50 \times 0.61 = \approx 30.6 \text{ mg}$.

7. Medication Safety Before Giving

- **Identify with two identifiers** (e.g., name and medical record number or birth date); the parent can verify identity for an infant or nonverbal child.
- Apply the **rights** of medication administration (right patient, drug, dose, route, time, documentation, reason) and independently **recheck dose calculations**.
- **High-alert medications** (e.g., antiarrhythmics, anticoagulants, chemotherapy, insulin, epinephrine, opioids, sedation agents) must be **double-checked by a second nurse** before administration.
- A **misplaced decimal point can cause a 10-fold or greater dosing error**.
- "Look-alike, sound-alike" drug names (e.g., DOBUTamine/DOPamine, predniSONE/prednisoLONE) use **Tall Man lettering** to emphasize differences.
- If part of an oral dose is **lost by spitting/vomiting, do not automatically repeat** — consult the provider; repeat dosing may cause overdose.

8. Oral Administration

- The **oral route is preferred** for ease of administration.
- **Measurement:** the most accurate device is the **plastic disposable oral syringe**; the AAP recommends dosing liquid medications **in milliliters only, never in teaspoons**. Cups (esp. <5 mL) and household spoons are inaccurate; droppers vary with viscosity.
- **Infant technique:** hold the infant **semireclining**; place the tip of the syringe in the pocket between the lower teeth and cheek; give **small increments**, allowing swallowing between each; never aim straight at the back of the throat (aspiration risk).
- **NURSING TIP:** blowing a small puff of air in the face of an infant up to 11 months old frequently elicits a swallow reflex.

- **Do not** add medicine to the infant's formula bottle (child may refuse the formula, and the full dose may not be taken).
 - **Crushing:** only crush when a liquid is unavailable; **do not crush** enteric-coated or sustained/slow-release products (overdose risk); **split only scored tablets**; a half dose can be made by dissolving a tablet in 5 mL of water and giving 2.5 mL.
 - **Atraumatic tips:** mix with a small amount (≈ 5 mL) of sweet substance; avoid essential foods; give a "chaser"; pinch the nose + straw works by blocking smell; give choices to the child to increase cooperation.
-

9. Intramuscular (IM) Administration

Volumes (single site): maximum ~ 1 mL in small children/older infants; **small infants ≤ 0.5 mL**; larger children approach adult volumes as muscle mass grows.

Recommended site (CDC/WHO/AAP): vastus lateralis for infants:

- Landmarks: greater trochanter and knee; divide the distance into thirds and inject into the **middle third**; needle inserted **perpendicular to the knee** (22–25 gauge, $\frac{5}{8}$ –1 inch).
- Ventrogluteal is suitable for children and infants **from 2 months**.

Technique details:

- Use a **tuberculin syringe (0.01-mL graduations)** for volumes < 1 mL and low-dose syringes for < 0.1 mL.
- Do **not** use an air bubble flush when giving < 1 mL — it adds extra drug; dead space in the hub is not part of the calibrated dose.
- Use the **Z-track method** for irritating drugs (e.g., iron dextran, tetanus toxoid) to prevent tracking into subcutaneous tissue.
- Needle gauge: smallest that safely delivers (25–30 gauge = least discomfort); larger gauges for viscous drugs.

Ear drops (child ≥ 3 years): pull the pinna **upward and backward** (bank Q15).

10. Nursing Care Plan (Medication Safety)

Assessment: an 18-month-old about to receive the first dose of an order written "80 mg PO q8h"; weight 8 kg; drug range 20–40 mg/kg/day.

Risk analysis: $\text{dose/day} = 80 \times 3 = 240 \text{ mg/day}$; $8 \text{ kg} \times 40 \text{ mg/kg} = 320 \text{ mg/day max}$ — dose is within range ($240 < 320$). Safe to give at $\sim 30 \text{ mg/kg/day}$.

NANDA diagnosis: *Risk for injury* related to possible medication error.

Interventions (with rationale):

1. Verify two identifiers and the order with the parent present — prevents wrong-patient error.
2. Recalculate using the weight and published range — the nurse legally verifies every dose.
3. Double-check if the drug is high-alert — reduces catastrophic error.
4. Measure with an oral syringe in mL — accurate small-dose delivery.
5. Give semireclining, small increments, suck-swallow pacing — prevents aspiration.
6. Evaluate: documented dose, time, and response; no error occurred.

11. Procedure — Laboratory Step: Safe Medication Administration (Checklist)

#	Step	Done
1	Compare the order with the medication record (drug, dose, route, time).	☐
2	Verify the child with two identifiers (+ parent verification for infants).	☐
3	Obtain the current weight in kg on the day of the dose.	☐
4	Calculate mg/kg/day and mg/kg/dose; confirm within the safe range.	☐
5	Compute the mL volume using the preparation's concentration.	☐
6	Double-check with a second nurse for high-alert drugs; recheck all math.	☐
7	Select the correct device: oral syringe (oral) / correct syringe & site (IM).	☐

#	Step	Done
8	Administer: semireclining infant, side-of-cheek pocket, small increments.	☐
9	Observe for swallowing/aspiration; document dose, time, and response.	☐
10	If vomiting/loss of dose occurred → consult the provider before repeating.	☐



Administering oral medication with an oral syringe



Administering oral medication with an oral syringe

12. Procedure — Hospital Step

- Give IM immunizations (CH09) at the **vastus lateralis** in infants; change sites for repeated doses; record site, lot, and route.
- For analgesic orders after surgery, use weight-based dose, reassess pain ≥ 30 –60 min after administration.
- Document every dose in the medication administration record; verify the next dose time to avoid double dosing.

13. Case Study (Dose Verification)

Ordered: paracetamol 90 mg PO q6h for an 11-kg child with fever (band: 10–15 mg/kg/dose, max 5 doses/day).

1. Safe range per dose: $11 \times 10 = 110$ mg to $11 \times 15 = 165$ mg. The ordered **90 mg is BELOW the range** — is it dangerous? No, but it is sub-therapeutic; the nurse should verify with the prescriber whether the dose was intended (perhaps written for a different weight).
2. What if the dose had been 160 mg/dose q6h? Day total = 640 mg; 75 mg/kg/day = 825 mg max — acceptable, but as a single 14.5 mg/kg dose within band.
3. Available syrup 120 mg/5 mL, ordered 90 mg $\rightarrow 90 \div 24 = \mathbf{3.75 \text{ mL}}$.

Teaching point: always compute the per-kg dose and compare with the published range before giving; document the calculation.

14. Chapter MCQs (10)

Q1. The child who weighs 22 lb needs a drug dosed 20–40 mg/kg/day. The safe daily range is:

- a) 300–600 mg
- b) 400–800 mg
- c) 500–1000 mg
- d) 200–400 mg (22 lb = 10 kg $\rightarrow$ 200–400 mg/day)

Q2. When giving ear drops to a 4-year-old child, the pinna is pulled:

- a) Downward and backward
- b) Upward and backward
- c) Forward and downward
- d) Forward and upward

Q3. The most accurate device for measuring a small liquid pediatric dose is:

- a) Household teaspoon
- b) Medicine cup
- c) Plastic disposable oral syringe
- d) Eye dropper

Q4. Paracetamol syrup 160 mg/5 mL; ordered dose 100 mg. The volume to give is:

- a) 2 mL

- b) 3.1 mL
- c) 5 mL
- d) 1 mL

Q5. Which tablet must not be crushed?

- a) Scored acetaminophen tablet
- b) Enteric-coated or slow-release products
- c) Chewable tablet
- d) Orally dissolving tablet

Q6. The recommended IM site for immunizations in infants is:

- a) Deltoid
- b) Vastus lateralis (anterolateral thigh)
- c) Dorsogluteal
- d) Abdominal

Q7. Maximum IM volume in one site for small children/older infants is about:

- a) 0.2 mL
- b) 5 mL
- c) 1 mL (small infants $\leq$ 0.5 mL)
- d) 2.5 mL

Q8. High-alert medications (e.g., insulin, opioids) should be:

- a) Given without documentation
- b) Double-checked by a second nurse before administration
- c) Crushed before giving
- d) Given only by mouth

Q9. If a child loses part of an oral dose by vomiting, the nurse should:

- a) Immediately repeat the full dose
- b) Consult the provider before repeating the dose
- c) Double the next dose
- d) Give it with milk

Q10. The Mosteller formula for BSA uses:

- a) $\sqrt{[(\text{height cm} \times \text{weight kg}) \div 3600]}$

- b) Height ÷ weight × 100
- c) Weight in lb × 2.2
- d) Age × 5

Answers: Q1 d — Q2 b — Q3 c — Q4 b — Q5 b — Q6 b — Q7 c — Q8 b — Q9 b — Q10 a.

15. Key Points

- Pediatric doses are weight-based (mg/kg); some drugs use BSA (mg/m²).
- Convert lb → kg (÷2.2), compute mg/kg/day and mg/kg/dose, verify against the safe range, then convert to mL using the concentration.
- Use **two identifiers**; double-check **high-alert drugs** with a second nurse; a decimal error = 10× dosed.
- Oral liquids: dose **in mL only** with an **oral syringe**; semireclining position, side-of-cheek pocket.
- IM: vastus lateralis in infants (middle third), ≤1 mL per site (0.5 mL in small infants), no air flush <1 mL, Z-track for irritants.

16. Common Pediatric Dose Examples Table (for practice only — always verify)

Drug	Recommended dose
Paracetamol (acetaminophen)	10–15 mg/kg/dose q4–6h; max 5 doses/24 h or 75 mg/kg/day (max 4000 mg)
Ibuprofen (children >6 months)	5–10 mg/kg/dose q6–8h; max 30 mg/kg/day or 3200 mg/day
Ondansetron	0.1–0.15 mg/kg IV/PO q4h; max 8 mg/dose
Diphenhydramine	1 mg/kg IV/PO q4–6h prn; max 25 mg/dose
Epinephrine (acute)	0.01 mg/kg up to max 0.5 mg

Drug	Recommended dose
Morphine IV	0.1 mg/kg q1–2h

Doses shown are reference examples for practice; confirm with the current ward formulary and drug reference before administration.

END OF CHAPTER 8

Medication Administration and Pediatric Dose Calculations

Next: **Chapter 9** — Immunization and the Cold Chain

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 9

Chapter 9 — Immunization and the Cold Chain

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 9

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Distinguish live-attenuated vaccines from inactivated/recombinant vaccines and apply the route and needle/site rules for each.
2. Describe the recommended (EPI-type) childhood immunization schedule and apply the current national schedule.
3. List the vaccine contraindications and precautions (severe febrile illness, severe immunocompromise, anaphylaxis to vaccine/component) and correct common misconceptions.

4. Manage the cold chain: storage at +2 to +8 °C (center shelf, protected from light/freezing), expiry and vaccine-vial-monitor checks, and "do not use" rule when temperature control is doubtful.
 5. Administer IM/SC vaccines (vastus lateralis in infants), provide atraumatic care, document, and manage vaccine reactions including anaphylaxis.
-

2. Placement Box

4. Principles of Immunization

- Vaccines for routine immunizations are **among the safest and most reliable drugs** available; minor side effects occur, and serious reactions are rare.
 - Vaccines may contain **inactive components** (preservatives, stabilizers, adjuvants, antibiotics such as neomycin, and culture-medium proteins such as egg) — the child may react to these rather than to the vaccine itself.
 - **Types:**
 - **Live-attenuated** (e.g., MMR, varicella, rotavirus, BCG): multiplication of the virus may be enhanced in the immunocompromised → generally **not** given to severely immunocompromised persons.
 - **Inactivated/recombinant** (e.g., DTaP, HepB, IPV, Hib, PCV, influenza): cannot replicate; side effects typically local and occur within hours–days.
 - HepB is a **recombinant** vaccine made by splicing a gene into yeast cultures; a severe yeast hypersensitivity would preclude its use without specialist review.
-

5. Immunization Schedule — A Representative EPI-Type Chart

IMPORTANT: The table below is a **representative WHO/EPI-type schedule** for practice. Before any vaccination, always apply **the current official schedule of the Ministry of Health (national EPI schedule)** posted at the ward/PHC unit.

Age	Vaccine (representative national EPI-style)	Route
Birth	BCG*; HepB (birth dose before hospital discharge); OPV-0	Intradermal / IM / oral
2 months (6 wk)	Pentavalent (DTP–HepB–Hib)-1; OPV-1; PCV-1; Rotavirus-1	IM / oral / IM / oral
4 months (10 wk)	Pentavalent-2; OPV-2; PCV-2; Rotavirus-2	IM / oral / IM / oral
6 months (14 wk)	Pentavalent-3; OPV-3/ IPV; PCV (per schedule)	IM / oral
9 months	Measles–Rubella (MR)/MCV-1	SC
15–18 months	MR/MCV-2; DTP booster; OPV booster	SC / IM / oral

*BCG is injected **intradermally** and is **very painful**. HepB birth dose is given **before hospital discharge** for every newborn when the mother's HBsAg status is negative (and with HBIG when the mother is positive).

6. The Cold Chain and Vaccine Storage

- **Proper vaccine storage is critical to vaccine efficacy. If you suspect a vaccine was not maintained at the proper storage temperature, do not use it** — an ineffective vaccine does not prevent disease.
- **Standard cold-chain temperature: +2 to +8 °C** (as per national cold-chain policy). Protect vaccines from **freezing** and from **light**.
- **Placement:** store vaccine bottles on a **center shelf of the refrigerator**, not in the door, where frequent temperature changes on opening can alter potency.
- Protect light-sensitive vials by wrapping in aluminum foil.
- Keep the refrigerator temperature log (twice-daily recording) and ensure **no vaccine is used after its expiration date**.
- **Follow the manufacturer's directions** for storage and reconstitution for each vaccine (diluent, mixing, and time limits after reconstitution).

7. Contraindications and Precautions

Contraindications (do not give):

- **Severe febrile illness** — the general contraindication for all immunizations.
- **Severe/unknown immunocompromise** — for **live** vaccines (MMR, varicella, rotavirus, BCG).
- **Pregnancy** — for MMR; breastfeeding is **not** a contraindication for any vaccine.
- **Recent passive immunity**: MMR and varicella postponed **≥3 months** after immunoglobulin/blood transfusions (except washed red cells).
- **Anaphylaxis** to a previous vaccine or a component (e.g., anaphylaxis to neomycin, gelatin) — proper contraindication.

NOT contraindications (common misconceptions):

- Minor illnesses such as the **common cold** are not a contraindication.
- **Egg sensitivity** is not a contraindication for MMR or inactivated influenza.
- Family history of seizures/SIDS, penicillin allergy, and neomycin contact dermatitis are not contraindications.
- A **precaution** is a condition that might increase the risk; give the vaccine only if benefit outweighs risk (e.g., a moderate acute illness with fever).

8. Administration Technique

Routes by vaccine type : IM — DTaP/DT/Tdap, HepA/HepB, Hib, PCV, HPV, MCV4, influenza (RIV/IIV); SC — MMR, IPV, varicella, MPSV4; intranasal — influenza LAIV.

Needle and site selection :

Age	Route	Needle	Length	Site
Birth–28 days	IM	25 g	5⁄8 in	Anterolateral thigh (vastus lateralis)
1–12 months	IM	23–25 g	1 in	Anterolateral thigh
1–2 years	IM	23–25 g	1–1¼ in	Anterolateral thigh or deltoid

Age	Route	Needle	Length	Site
3–18 years	IM	22–25 g	1–1¼ in; ⅝–1 in	Anterolateral thigh or deltoid
1–12 months	SC	23–25 g	⅝ in	Fatty tissue over anterolateral thigh
> 12 months	SC	23–25 g	⅝ in	Fatty tissue over thigh or triceps

- The **dorsogluteal site is not recommended** for children (sciatic nerve risk).
- **Aspiration for blood is no longer recommended** for IM injections — large vessels are absent at recommended sites, and aspiration causes more pain.
- Liquid **adjuvant** vaccines (e.g., DTaP) need **excellent IM technique**; subcutaneous/intracutaneous injection causes local irritation or abscess.
- **Rotate sites** with repeated doses.

Atraumatic care :

- EMLA (1 h) or LMX4 (30 min) topical anesthetic under an occlusive dressing before injection.
- **Beneficial positioning** — parent holds the infant; older children sit upright.
- **Breastfeeding** during/just before injections reduces pain.
- Distraction ("blow, blow, blow"), calm nurses, neutral words ("here I go"), and **no site massage**.

Emergency preparedness : Before any vaccine, have the anaphylaxis kit ready:

- EpiPen Jr **0.15 mg** (0.3 mL of 1:2000) IM for a child **15–30 kg**
- EpiPen **0.3 mg** (0.3 mL of 1:1000) IM for a child **≥30 kg**.

9. Nursing Care Plan (Vaccination Session)

Assessment: 2-month-old infant due for Pentavalent-1, OPV-1, PCV-1, Rotavirus-1. No fever, no known allergies, thriving.

NANDA diagnosis: *Risk for injury* (vaccine reaction) / *Deficient knowledge* (parents) related to immunization schedule and side effects.

Goals: Safe administration; parents informed; normal side effects managed; serious reactions detected immediately.

Interventions (with rationale):

- 1. Verify schedule, vaccine, expiry, and cold-chain record — prevents using ineffective vaccine.
- 2. Take an allergy/history screen for contraindications — prevents adverse reaction.
- 3. Provide the **vaccine information statement (VIS)** and answer questions before giving — informed consent.
- 4. Prepare the emergency tray (EpiPen Jr) before injection — immediate anaphylaxis management.
- 5. Give IM vaccines at **vastus lateralis**, 25 g ⁵/₈–1 in, no aspiration; offer breastfeeding during — reduces pain and reactions.
- 6. Observe ≥20–30 min after the session; document site, lot, route, and dose — documentation and prompt response.
- 7. Evaluate: no serious reaction; parent can state the next due date and expected mild side effects.

10. Procedure — Laboratory Step: Vaccination and Cold-Chain Check (Checklist)

#	Step	Done
1	Check refrigerator temperature log and vaccine batch/expiry before the session.	☐
2	Confirm the child is due per the current national schedule; screen contraindications.	☐
3	Provide the VIS and obtain consent.	☐
4	Wash hands; prepare the vaccine and correct syringe/needle per Table 31.5.	☐
5	Select the site: vastus lateralis (infants) / thigh or deltoid (older).	☐
6	Apply topical anesthetic (EMLA/LMX4) 30–60 min before if available.	☐
7	Inject IM without aspiration; apply gentle pressure; do not massage.	☐

#	Step	Done
8	Give OPV/rotavirus orally per package directions before injections.	☐
9	Observe 20–30 min; document site, route, dose, lot, and expiry.	☐
10	Teach post-vaccine care (acetaminophen if needed, fever, local tenderness; return if severe).	☐



Vastus lateralis vaccination position



Vastus lateralis vaccination position

11. Procedure — Hospital Step

- Record the **HepB birth dose before discharge** for every newborn (mother HBsAg negative).
- During the session, keep the **emergency tray (EpiPen Jr, EpiPen)** within reach.
- Document each vaccine in the child's immunization card and the ward record: date, vaccine, dose, route, site, lot number, and expiry; update the next-visit date.
- Report suspected cold-chain breaks (temperature outside +2 to +8 °C, freeze, or VVM change) to the supervising nurse; **quarantine, do not use**.

12. Case Study (Cold Chain Breach)

Scenario: A student nurse opens the vaccination refrigerator and finds the temperature log shows 9.5 °C for the morning reading, and a vial of MMR reconstituted 3 hours earlier is still in the tray.

1. What should be done with the MMR vial? *(Discard/quarantine — beyond its reconstitution time limit per the package insert; do not use.)*
 2. How should the suspected cold-chain breach be handled? *(Report to the supervising nurse/pharmacist; quarantine stock; document; do not use vaccines suspected of exposure.)*
 3. Which placement rule was violated by storing in the door? *(Vaccines must be on the center shelf; the door exposes them to temperature changes on opening.)*
 4. Would MMR cause harm to a severely immunocompromised child? *(Yes — a live vaccine; enhanced viral multiplication can cause severe vaccine-induced illness.)*
-

13. Chapter MCQs (10)

Q1. The recommended cold-chain storage temperature for vaccines is:

- a) Below 0 °C (freezer)
- b) +2 to +8 °C
- c) 15–25 °C
- d) 8–15 °C

Q2. In the vaccine refrigerator, vials should be stored:

- a) In the door shelves
- b) On a center shelf, protected from light
- c) Next to food for defrosting
- d) In the freezer compartment

Q3. The IM site of choice for vaccinations in an infant is:

- a) Dorsogluteal
- b) Vastus lateralis (anterolateral thigh)
- c) Deltoid in all infants
- d) Subcutaneous abdomen

Q4. Which vaccine is given intradermally and is very painful?

- a) OPV
- b) BCG
- c) Pentavalent
- d) HepB

Q5. A general contraindication for all immunizations is:

- a) The common cold
- b) Severe febrile illness
- c) Penicillin allergy
- d) Family history of seizures

Q6. Live virus vaccines (MMR, varicella) should not be given to:

- a) Children with egg allergy
- b) Severely immunocompromised children
- c) Breastfed infants
- d) Children with a cold

Q7. Which of the following is NOT a contraindication to MMR?

- a) Anaphylaxis to neomycin or gelatin
- b) Egg sensitivity
- c) Pregnancy
- d) Severe immunocompromise

Q8. Aspiration for blood during IM vaccine injection is:

- a) Recommended routinely
- b) No longer recommended (avoids pain and is unnecessary)
- c) Required for all infants
- d) Only for the deltoid site

Q9. The birth dose of hepatitis B vaccine should be given:

- a) At 9 months
- b) Before hospital discharge
- c) Only if the mother is HBsAg positive
- d) At 2 months

Q10. EpiPen Jr (0.15 mg IM) is used for anaphylaxis in a child weighing:

- a) 15–30 kg
- b) <5 kg
- c) >30 kg
- d) Any infant <1 year

Answers: Q1 b — Q2 b — Q3 b — Q4 b — Q5 b — Q6 b — Q7 b — Q8 b — Q9 b — Q10 a.

14. Key Points

- Vaccines are safe but need correct **route, site, needle**, and **cold chain** to be effective.
- Cold chain **+2 to +8 °C**, center shelf, protect from light/freezing, check expiry, follow manufacturer's reconstitution directions.
- **Do not use** any vaccine suspected of improper temperature exposure.
- Contraindications: severe febrile illness, severe/unknown immunocompromise (live vaccines), pregnancy (MMR), anaphylaxis to vaccine/component; **minor illness, egg allergy, and breastfeeding are NOT contraindications.**
- IM: vastus lateralis in infants; no dorsogluteal; **no aspiration**; atraumatic care (EMLA, positioning, breastfeeding, distraction).
- Emergency: EpiPen Jr 0.15 mg (15–30 kg) / EpiPen 0.3 mg (≥30 kg) IM.

15. Reference Table: Common Vaccine Routes

Route	Vaccines
Intramuscular	DTaP/DT/Tdap, HepA, HepB, Hib, PCV, HPV, MCV4, influenza IIV/RIV
Subcutaneous	MMR, IPV, varicella, MPSV4
Intranasal	Influenza (LAIV)
Intradermal	BCG*

*BCG intradermal, very painful.

END OF CHAPTER 9

Immunization and the Cold Chain

Next: **Chapter 10** — Core Ward Skills: Hand Hygiene, Atraumatic Care, Isolation, and Documentation

PART I - FOUNDATIONS (SKILLS LAB BASICS)

CHAPTER 10

Chapter 10 — Core Ward Skills: Hand Hygiene, Atraumatic Care, Isolation, and Documentation

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part I — Foundations of Pediatric Practical Care, Chapter 10

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Perform correct hand hygiene (soap and water / alcohol-based) — the single most important infection-control practice.
 2. Apply Standard Precautions and identify when Airborne, Droplet, and Contact Precautions are added.
 3. Prepare a child and family for isolation and minimize the psychological stress of isolation.
 4. Apply the principles of atraumatic care (prevent/minimize physical and psychological distress; minimize parent–child separation; promote family-centered care).
 5. Document nursing assessment, interventions, weights, and intake/output correctly on the ward record and give a structured handover.
-

2. Placement Box

4. Hand Hygiene and Standard Precautions

- Each day about **1 in 31 patients** acquires at least one hospital-acquired infection (HAI); HAIs are preventable with meticulous hand washing, cleaning, and disposal.
- **Hand hygiene is the single most important practice to reduce transmission** of infectious diseases in the health care setting; includes washing with **soap and water** and use of **alcohol-based hand disinfectants**.
- **Standard Precautions** synthesize universal precautions (blood/body fluids) and body-substance isolation:
- Use **barrier PPE** — gloves, goggles, gowns, and masks — for contact with blood, all body fluids/secretions/excretions (except sweat), nonintact skin, and mucous membranes.
- Standard Precautions apply to **all patients** because unrecognized sources of infection exist.
- **Respiratory Hygiene/Cough Etiquette** and **safe injection practices** are part of Standard Precautions: cover mouth/nose when coughing, mask coughing persons, 3 feet of space, tissues; use safety-engineered sharps.
- **Glove use:** gloves do not provide complete protection — **wash hands thoroughly after removing gloves**; disposable-diaper plastic lining is a sufficient barrier, so gloves are adequate for routine diaper changes (add gown for explosive/liquid stools).
- **Sharps:** dispose of all needles (uncapped, unbroken) in **rigid, puncture-resistant containers near the point of use**, placed out of children's reach; prefer needleless systems.
- **MRSA transmission is mainly patient-to-patient via the hands of health care providers; hand washing is the most critical infection-control practice.**

5. Transmission-Based Precautions

These are **in addition to** Standard Precautions:

Precautions	Route	Requirements	Examples
Airborne	Airborne droplet nuclei ($\leq 5 \mu\text{m}$) suspended in air	Airborne infection isolation room (AIIR) + special ventilation; N-95/respirator	Measles, varicella (incl. disseminated zoster), tuberculosis
Droplet	Large droplets ($> 5 \mu\text{m}$) travel ≤ 3 feet, from cough/sneeze/talk	Surgical mask; close for both child and personnel; no special air handling	Hib (meningitis, pneumonia, epiglottitis), meningococcus, diphtheria, pertussis, streptococcal pharyngitis, adenovirus, influenza , mumps, rubella
Contact	Direct (skin-to-skin) or indirect (contaminated objects/environment)	Gloves + gown; dedicated equipment	MRSA and other multidrug-resistant organisms; enteric infections with low infectious dose (C. difficile, E. coli O157:H7, Shigella, hepatitis A, rotavirus) in diapered/incontinent patients

Respiratory-distress note: infants and children with suspected airborne illness (e.g., measles, TB, varicella) must be placed in the AIIR and remain there.

6. Preparing the Child for Isolation (Pediatric Approach)

Admission to an isolation room **increases all the stressors** of hospitalization: further separation, loss of control, sensory deprivation, and strangers.

- **Preschoolers** cannot understand cause-and-effect; they **view isolation as punishment**. Use a simple story: "You need to be in this special room to help you get better; the infection made you sick and you could not help that".

- **Prepare the child for what they will see, hear, and feel:** show the mask, gloves, and gown and let the child "**dress up**" in them; playing with PPE lessens fear of staff entering.
 - Staff should **introduce themselves and let the child see their faces before donning masks**, so the child associates them with familiar people.
 - Provide appropriate play activities when the child's condition improves; keep explanations developmentally matched.
-

7. Principles of Atraumatic Care

Atraumatic care is therapeutic care that **minimizes or eliminates the psychological and physical distress** experienced by children and their families in the health care system (premise: "do no harm").

Box 30.1 — Principles :

- Prevent or minimize **physical stressors**: pain, discomfort, immobility, sleep deprivation, inability to eat/drink, changes in elimination.
 - **Avoid or reduce intrusive/painful procedures** (multiple punctures, injections, urethral catheterization).
 - Reduce other physical distress: noise, smells, shivering, nausea, sleeplessness, restraints, skin trauma.
 - **Control pain** by frequent assessment plus pharmacological and nonpharmacological interventions.
 - **Prevent or minimize parent–child separation.**
 - **Promote family-centered care** — treat the family as the patient; use core primary nursing.
 - Follow research on whether parents and children prefer to be together.
 - **Promote a sense of control** (tell the child what will happen, give choices, use distraction).
-

8. Documentation and Handover

- Document objectively and immediately: admission assessment, weight and head circumference, vital signs, I/O (including ORS), medication doses (route, site, lot), isolation precautions, and "reflexes observed, fontanelle status" where relevant [Ch 2, 6, 7].
- **Weight daily** — one of the best indicators of fluid status in children.
- Use the structured **SBAR** handover: Situation, Background, Assessment, Recommendation — especially during ward shift changes and when transferring an isolated child.
- **Legal note:** the nurse is accountable for what they document and administer; record the exact time of medications and observations, and report any discrepancy with the written order.

9. Nursing Care Plan (Risk for Infection — Hand Environment)

Assessment: 2-year-old on the medical ward with rotavirus gastroenteritis and loose explosive stools; nursing staff rotate frequently.

NANDA diagnosis: *Risk for infection* related to exposure to a multidrug-resistant organism / inadequate hand hygiene.

Goals: Prevent cross-infection on the ward; the child's course is uncomplicated; no staff/patient transmission.

Interventions (with rationale):

1. Apply **hand hygiene before and after every contact** — single most important practice.
2. Use **Contact Precautions (gloves + gown)** for diapered patient with rotavirus — interrupts fecal–oral route.
3. Hand hygiene **after glove removal** because gloves fail to provide complete protection.
4. Contain stool: gloves for routine changes, gown for explosive stools; dispose of diapers properly; clean environment.
5. Prepare the preschooler for isolation with play ("dress up"), face-to-face introductions, and simple explanations — decreases fear misconceived as punishment.

6. Evaluate: no secondary cases on the ward; family understands the isolation routine.

10. Procedure — Laboratory Step: Hand Hygiene and PPE (Checklist)

#	Step	Done
1	Remove jewelry/watches; nails short, no artificial nails.	☐
2	Wet hands; apply soap; rub all surfaces (palms, backs, fingers, thumbs, fingertips, wrists) ≥ 20 s.	☐
3	Rinse under running water; dry with a single-use towel; turn off the tap with the towel.	☐
4	OR: apply alcohol-based rub to dry hands; rub until dry (≈ 20 – 30 s).	☐
5	Perform hand hygiene at the WHO 5 moments (before/after patient contact; before aseptic task; after body-fluid exposure; after touching environment).	☐
6	Don gown; tie at neck/waist; don mask; then goggles; gloves last — cover gown cuff.	☐
7	Perform the task.	☐
8	Doff gloves first (inside-out), then gown (peel from inside), then goggles, then mask; avoid touching front surfaces.	☐
9	Perform hand hygiene immediately after doffing.	☐
10	Dispose of waste/PPE per ward policy.	☐



Correct hand-washing technique



11. Procedure — Hospital Step

- Enter isolation rooms only with the correct PPE displayed on the door sign; dedicate equipment where required (stethoscope, thermometer).
- At handover, clearly state the precautions for each child (e.g., "rotavirus — contact; measles — airborne, AIIR").
- Reassure and re-explain isolation to the child each shift using play and age-appropriate language.

12. Case Study (Contact Precautions in an Isolation Room)

Scenario: A 3-year-old is admitted with measles-like symptoms and placed in the AIIR. A second child on the ward has rotavirus diarrhea.

- Which precautions does the measles-suspected child need? (*Airborne + standard; AIIR with special ventilation; N-95 respirator for staff.*)
- Which precautions does the rotavirus child need? (*Contact + standard; gloves and gown.*)
- The preschooler with rotavirus keeps crying when nurses enter in gowns and masks. How would you prepare him? (*Let him "dress up" in gloves/gown/mask; show faces before masking; simple explanation that the room helps him get better.*)
- After leaving the rotavirus room, what is the key step? (*Remove PPE correctly and perform hand hygiene — gloves fail to provide complete protection.*)

13. Chapter MCQs (10)

Q1. The single most important practice to reduce transmission of infection in health care settings is:

- a) Antibiotic prophylaxis
- b) Hand hygiene
- c) Daily linen changes
- d) UV light

Q2. Standard Precautions apply to:

- a) Only strongly suspected infections
- b) All patients
- c) Only patients with fever
- d) Only immunocompromised children

Q3. An infant with measles should be placed on:

- a) Contact Precautions
- b) Droplet Precautions
- c) Airborne Precautions in an airborne infection isolation room (AIIR)
- d) Standard Precautions only

Q4. Droplet Precautions are required for:

- a) Tuberculosis
- b) Varicella
- c) Influenza and pertussis
- d) C. difficile

Q5. MRSA is most importantly transmitted by:

- a) Airborne nuclei
- b) Patient-to-patient via the hands of health care providers
- c) Blood transfusion
- d) Breast milk

Q6. After removing gloves, the nurse should:

- a) Skip hand hygiene because gloves protect fully
- b) Wash hands thoroughly
- c) Only sanitize if visibly soiled
- d) Change gloves once more

Q7. Preschoolers in isolation are most likely to view it as:

- a) A holiday
- b) Punishment
- c) A game
- d) Nothing

Q8. A principle of atraumatic care is:

- a) Minimizing parent–child separation
- b) Restraints before reassurance
- c) Delaying pain relief until the child asks
- d) Masking faces to avoid frightening the child

Q9. Atraumatic care is best defined as care that:

- a) Uses only pharmacological pain control
- b) Minimizes or eliminates psychological and physical distress
- c) Avoids family involvement
- d) Shortens hospital stay at all costs

Q10. Needles should be disposed of:

- a) In the trash bin at the nurse's desk
- b) Uncapped and unbroken, in a rigid puncture-resistant container near the point of use
- c) Recapped and placed in the medication cart
- d) In the laundry

Answers: Q1 b — Q2 b — Q3 c — Q4 c — Q5 b — Q6 b — Q7 b — Q8 a — Q9 b — Q10 b.

14. Key Points

- **Hand hygiene is the single most important infection-control practice.**
- Standard Precautions (PPE for blood, body fluids, nonintact skin, mucous membranes) apply to **all patients**.
- Airborne (AIIR: measles, varicella, TB), Droplet (≤ 3 ft: Hib, meningococcus, pertussis, influenza), Contact (MRSA, C. difficile, rotavirus) — all **added to** Standard Precautions.

- Isolated children — especially preschoolers — need **preparation and play**; they may view isolation as punishment.
- Atraumatic care: minimize physical/psychological stressors, avoid painful procedures, minimize separation, promote family-centered care and sense of control.
- Document objectively, weigh daily, and give structured SBAR handovers.

15. Reference Table: Precautions Summary

Precautions	Key elements	Representative conditions
Standard	Hand hygiene + PPE (gloves, gown, mask, goggles) for all body fluids; cough etiquette; safe sharps	All patients
Airborne	AIIR, special ventilation, respirator	Measles, varicella, TB
Droplet	Surgical mask, close contact, ≤3 feet travel	Hib, meningococcus, pertussis, influenza, mumps, rubella
Contact	Gloves + gown, dedicated equipment	MRSA/MDROs, C. difficile, rotavirus, Shigella, hepatitis A

END OF CHAPTER 10

Core Ward Skills: Hand Hygiene, Atraumatic Care, Isolation, and Documentation

Next: **Chapter 11** — Neonatal Jaundice (Hyperbilirubinemia)

PART II - NICU WARD

CHAPTER 11

Chapter 11 — Neonatal Jaundice (Hyperbilirubinemia)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part II — NICU Ward, Chapter 11

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain the pathophysiology of unconjugated (physiologic) hyperbilirubinemia and distinguish physiologic, breastfeeding-associated, breast-milk, and hemolytic jaundice.
2. Recognize pathologic warning signs: jaundice appearing **within 24 hours**, rapidly rising bilirubin (≥ 5 mg/dL/day), and a positive Coombs test.
3. Provide evidence-based nursing care for the infant under phototherapy (eye protection, position, fluid/growth monitoring, temperature, skin care, charting).
4. Prepare an infant and family for exchange transfusion and monitor for complications/transfusion reactions.
5. Detect early signs of bilirubin encephalopathy (kernicterus) and report promptly.

2. Placement Box

Ward: NICU — one of the two neonatal topics. Jaundice is the most common newborn problem requiring ward admission, phototherapy, and (rarely) exchange transfusion.

Foundation skills reused: CH5 vital signs, CH6 neurologic/reflex assessment, CH7 fluid calculations, CH8 medication safety, CH10 core ward skills.

4. Definition and Pathophysiology

- **Hyperbilirubinemia** is an excessive level of accumulated bilirubin in the blood, characterized by **jaundice (icterus)** — a yellowish discoloration of the skin and other organs.
- Bilirubin is a breakdown product of hemoglobin from RBC destruction. When RBCs are destroyed, hemoglobin splits into **heme** and **globin**; the heme portion is converted to **unconjugated bilirubin** (insoluble, bound to albumin).

- In the liver, bilirubin is detached from albumin and **conjugated with glucuronic acid** (enzyme: glucuronyl transferase) to a water-soluble product that is excreted in bile; bacteria reduce it to **urobilinogen**, the pigment giving stool its color.
- When developmental limitations or disease **interfere with this balance, bilirubin accumulates** in the tissues → jaundice.
- Causes: physiologic (prematurity), breastfeeding/breast-milk association, excess production (hemolysis, bruises), disturbed liver conjugation/excretion, certain diseases (G6PD deficiency, hypothyroidism, galactosemia, infant of diabetic mother), and genetic predisposition.

5. Types of Unconjugated Jaundice

Type	Cause	Onset	Peak	Duration/Therapy
Physiologic	Immature hepatic function + increased bilirubin load from RBC hemolysis; enterohepatic shunting	After 24 h	2nd–5th day (higher in breastfed, preterm, Asian descent)	Declines 5th–7th day; frequent feeding; monitor TcB/TSB; phototherapy if levels rise significantly
Breastfeeding-associated (early onset)	Decreased milk intake before milk is established; enterohepatic shunting; less stooling	3rd–4th day	3rd–5th day	Breastfeed 10–12×/day; avoid water/dextrose supplements; monitor; phototherapy if needed
Breast-milk jaundice (late onset)	Possible breast-milk factors that prevent conjugation; less frequent stooling	4th day	10th–15th day	May remain jaundiced 3–12 wk; reassure; rarely stop breastfeeding (may use temporary home phototherapy)

Type	Cause	Onset	Peak	Duration/Therapy
Hemolytic disease (ABO/Rh)	Blood antigen incompatibility → hemolysis of many RBCs; liver cannot conjugate/excrete excess bilirubin	First 24 h (rises ≥ 5 mg/dL/day)	Variable	Phototherapy; IVIG; tin-mesoporphyrin per protocol; exchange transfusion if severe; prenatal: RhIg to prevent Rh sensitization

Physiologic jaundice details : in formula-fed term infants bilirubin rises to **~5–6 mg/dL by 2–5 days**, then plateaus <3 mg/dL by day 5–7; breastfed infants peak higher (**7–14 mg/dL**); preterm formula-fed infants may peak **10–12 mg/dL** around day 5 and decrease over 2–4 weeks. Only about **half** of newborns with elevated levels show visible jaundice.

6. Assessment and Diagnosis

- **Visual assessment is NOT reliable** to judge severity — jaundice is estimated by **transcutaneous bilirubinometry (TcB)** or **total serum bilirubin (TSB)**; visual/transcutaneous estimates are invalid once phototherapy begins.
- Risk factors for **pathologic** hyperbilirubinemia include jaundice with onset **within 24 hours**, rapidly rising levels, maternal blood group O, **positive Coombs test**, previous sibling with hyperbilirubinemia, and prematurity/illness.
- Use the **hour-specific bilirubin nomogram** to determine the risk zone and the phototherapy/exchange thresholds for infants ≥ 35 weeks.
- **ABO incompatibility:** treat with early detection + phototherapy; diagnosis may be hard because the **direct Coombs test can be negative or weakly reactive**; consider when jaundice appears **within 24 hours**, bilirubin rises, with RBC **spherocytosis**.

7. Complications: Bilirubin Encephalopathy and Kernicterus

- Unconjugated bilirubin is **highly toxic to neurons**; severe hyperbilirubinemia risks **bilirubin encephalopathy** (early, varying, reversible symptoms) and **kernicterus** — yellow staining and necrosis of brain cells with chronic permanent damage.
 - **Early signs:** lethargy, poor feeding, **hypotonia**, **high-pitched cry**, **temperature instability** (some preterm infants may be asymptomatic).
 - **Later signs:** athetoid cerebral palsy, extrapyramidal symptoms, opisthotonos, dental enamel hyperplasia, motor delay, seizures, and **sensorineural hearing loss** — long-term cognitive, motor, and behavioral effects.
 - Safe practice: **serum bilirubin alone does not predict CNS injury**; risk and neurologic signs guide action.
-

8. Medical and Nursing Management

8.1 Phototherapy

- Place the infant under the light source exposing **as much skin as possible**; position and **reposition frequently** to expose all body surfaces.
- Monitor **serum bilirubin every 6–12 hours** after initiation (visual/TcB no longer valid).
- **Eye protection:** opaque mask sized correctly, covering eyes completely **without occluding the nares**; close the eyelids before applying (corneal excoriation); check eyes **every 4–6 hours** for discharge/pressure/irritation; **remove shields during feeding** for visual/sensory stimulation.
- **Temperature:** monitor for temperature instability — phototherapy may raise body temperature; maintain **manufacturer-specified lamp distance** (halogen lights too close cause burns); keep the infant flexed with rolled blankets.
- **Side effects:** loose greenish stools (accelerated bilirubin excretion), transient rashes, mild hyperthermia, increased metabolic rate, priapism, rarely dehydration/hypocalcemia; **meticulous perianal skin care**; add fluids/feedings to compensate for insensible and stool losses; **do not use oily lotions** (increased tanning).
- **Bronze-baby syndrome:** grayish-brown skin/serum/urine, usually with elevated conjugated bilirubin + cholestasis; resolves after discontinuing phototherapy.

- **Rebound effect:** a transient rise in bilirubin after phototherapy stops — usually resolves without resuming therapy.
- **Charting must include:** start/stop times, eye shielding, phototherapy unit type, number of lamps, lamp-to-infant distance, incubator vs open bassinet use, light intensity (μW), and observed side effects.
- **Family support:** reassure parents the infant is warm and comfortable; **remove shields and turn off lights during visits** to facilitate attachment and eye contact.

8.2 Exchange Transfusion

- **Indications:** rapidly rising bilirubin and **hemolysis despite aggressive phototherapy**; also hydrops fetalis or signs of cardiac failure (immediate procedure).
- **Procedure:** sterile surgical technique; catheter inserted into the **umbilical vein** to the inferior vena cava; **5–10 mL** blood withdrawn over 15–20 s and an equal volume of donor blood infused; donor blood typed/cross-matched to the **mother's serum**; usually **double the infant's blood volume (~85 mL/kg)**, replacing ~85% of circulating blood.
- **Nursing during the procedure:** infant stays **NPO** with a peripheral dextrose/electrolyte infusion; document volumes withdrawn/infused, times, and cumulative total; monitor vital signs continuously, correlate with blood removal/infusion; **stop temporarily** for cardiac/respiratory problems and resume when stable; observe for **transfusion reaction** (temperature instability, hypotension, tachycardia, bradycardia, rash); maintain thermoregulation, glucose, and fluid balance.
- **ABO incompatibility rarely needs exchange transfusion** unless phototherapy fails.
- **Family support:** parents often feel guilty about "causing" the incompatibility — reassure and praise their prenatal care.

8.3 Prevention and Adjunctive Therapy

- Prevent Rh sensitization with **Rh immune globulin (RhIg)** given to the Rh-negative mother (prenatal/postpartum).
 - IVIG and **tin-mesoporphyrin** (heme oxygenase inhibitor) are adjuncts used with phototherapy per protocol.
-

9. Nursing Care Plan (Infant Under Phototherapy)

Assessment: 3-day-old term infant, TcB in the high-risk zone on the nomogram; phototherapy initiated; loose green stools; temperature 37.6 °C.

NANDA diagnosis: *Risk for injury (bilirubin encephalopathy)* related to elevated unconjugated bilirubin; *Risk for impaired skin integrity* related to frequent stools and moisture.

Goals: TSB stays below the exchange threshold; normothermia maintained; skin intact; parents informed and involved.

Interventions (with rationale):

1. Monitor TSB **every 6–12 hours** during phototherapy — visual estimation is invalid.
2. Expose maximum skin; reposition frequently; maintain lamp distance per guidelines — maximizes light effect while preventing burns.
3. Apply and check eye shields every 4–6 hours; remove during feedings — prevents corneal injury; provides stimulation.
4. Feed frequently and increase fluids; assess for dehydration and hypocalcemia — compensates for losses.
5. Meticulous perianal care and dry skin; no oily lotions — prevents excoriation/breakdown.
6. Observe for kernicterus signs (hypotonia, high-pitched cry, temperature instability, lethargy) and report immediately — early detection prevents permanent damage.
7. Evaluate: bilirubin decreasing toward the low-risk zone; temperature stable; skin intact; parents verbalize understanding.

10. Procedure — Laboratory Step: Phototherapy Setup and Nursing Care (Checklist)

#	Step	Done
1	Verify the bilirubin level and phototherapy order (type, intensity).	☐
2	Position the light at the manufacturer's distance from the infant.	☐

#	Step	Done
3	Undress the infant down to the diaper; expose maximum skin surface.	☐
4	Close the eyelids; apply a correctly sized opaque eye shield over both eyes (nares patent).	☐
5	Place the infant in a flexed position with rolled blankets for comfort.	☐
6	Check the eyes every 4–6 h for discharge/pressure; remove shields during feeds.	☐
7	Monitor temperature frequently (hyperthermia risk); adjust the warmer/bassinet.	☐
8	Increase feedings/fluids; record intake and urine output.	☐
9	Provide perianal skin care after loose stools; no oily lotions.	☐
10	Chart: start/stop times, unit type, lamp count/distance, intensity, side effects; arrange bilirubin per order.	☐



Infant receiving phototherapy (eye shield, max skin exposure)



Infant receiving phototherapy (eye shield, max skin exposure)

11. Procedure — Hospital Step

- On admission of any jaundiced newborn: obtain TSB or TcB, plot on the **hour-specific nomogram**, weigh, assess feeding and stooling, and check the prenatal history (maternal blood group, Coombs).
- For an infant under phototherapy, follow the checklist; draw ordered bilirubin levels and notify the physician of values crossing nomogram thresholds.
- **Assist with exchange transfusion:** keep NPO, maintain IV access, prepare the (double-volume) blood product cross-matched to the mother, document every mL exchanged, vital signs, and observed reactions; provide family support.
- On discharge: schedule bilirubin follow-up (rebound), teach the family feeding frequency and warning signs (lethargy, high-pitched cry, poor feeding).

12. Case Study (Jaundice at 48 Hours)

Scenario: A 48-hour-old term infant, blood type A (mother O, Rh positive), develops jaundice to the chest. TcB 13.8 mg/dL; TSB 14.2 mg/dL; Coombs weakly positive; RBC spherocytosis noted.

1. Classify the onset. (*Jaundice on day 2 — the critical warning sign is onset within the first 24 hours; at 48 h this is still within the high-risk pathway for hemolytic disease — investigate hemolysis.*)
 2. Which type of jaundice is most likely? (*ABO hemolytic disease — maternal O with infant A, spherocytosis, weakly positive Coombs.*)
 3. Planned treatment? (*Early phototherapy; monitor TSB q6–12 h; IVIG per protocol; exchange transfusion only if phototherapy fails.*)
 4. Which nursing observation suggests early bilirubin encephalopathy? (*Lethargy, hypotonia, high-pitched cry, temperature instability.*)
-

13. Chapter MCQs (10)

Q1. Jaundice that appears **within the first 24 hours** of life is:

- a) Always physiologic
- b) Suggestive of hemolytic disease — needs evaluation
- c) Normal breast-milk jaundice
- d) Due to dehydration only

Q2. The most common type of neonatal jaundice is:

- a) Hemolytic disease
- b) Physiologic jaundice
- c) Breast-milk jaundice
- d) Obstructive jaundice

Q3. Unconjugated bilirubin in the liver is made water-soluble by:

- a) Albumin
- b) Glucuronyl transferase + glucuronic acid conjugation

- c) Urobilinogen
- d) Heme oxygenase

Q4. The serious complication of severe hyperbilirubinemia is:

- a) Colic
- b) Kernicterus (bilirubin encephalopathy)
- c) Dehydration
- d) Osteopenia

Q5. One early sign of bilirubin encephalopathy is:

- a) Rosy cheeks
- b) High-pitched cry and hypotonia
- c) Increased activity
- d) Normal temperature

Q6. During phototherapy, the infant's eyes are protected with an opaque mask that:

- a) Is removed only for 24 hours
- b) Is removed during feedings for stimulation and checked q4–6 h
- c) May occlude the nares
- d) Is unnecessary at night

Q7. A common side effect of phototherapy is:

- a) Loose greenish stools
- b) Constipation
- c) Hypertension
- d) Hyperkalemia

Q8. Serum bilirubin levels during phototherapy should be rechecked:

- a) Weekly
- b) Every 6–12 hours
- c) Only at discharge
- d) Once after 24 hours

Q9. In exchange transfusion, the usual volume exchanged is:

- a) Half the blood volume
- b) Double the blood volume (~85 mL/kg), replacing ~85% of circulating blood

- c) 10 mL/kg
- d) One full volume once

Q10. Birth dose of which vaccine relates to preventing Rh disease?

- a) BCG
- b) RhIg (to the Rh-negative mother) prevents Rh sensitization
- c) OPV
- d) Pentavalent

Answers: Q1 b — Q2 b — Q3 b — Q4 b — Q5 b — Q6 b — Q7 a — Q8 b — Q9 b — Q10 b.

14. Key Points

- Jaundice = accumulated bilirubin; **unconjugated** type dominates in newborns; only ~half of infants with elevated levels are visibly jaundiced.
- Physiologic: onset **after 24 h**; formula-fed term peak ~5–6 mg/dL (day 2–5); breastfed 7–14 mg/dL; preterm 10–12 mg/dL.
- **Pathologic if:** onset <24 h, bilirubin rising ≥ 5 mg/dL/day, positive Coombs, maternal group O + non-O infant, sibling history.
- Phototherapy: max skin exposure, reposition, eye shields (close lids first, check q4–6 h, remove at feeding), temperature + lamp distance, increased fluids, perianal care, chart details; watch **rebound** after stopping.
- Exchange transfusion: double-volume (~85 mL/kg), umbilical vein catheter, NPO, monitor volumes/vitals/reactions, document cumulative.
- Early kernicterus signs: lethargy, hypotonia, high-pitched cry, temperature instability → report immediately.

15. Normal Values / Thresholds Table

Parameter	Value	Source
Physiologic peak (term formula-fed)	~5–6 mg/dL by 2–5 days	

Parameter	Value	Source
Physiologic plateau after day 5	<3 mg/dL	
Breastfed term peak	7–14 mg/dL	
Preterm (formula-fed) peak	10–12 mg/dL about day 5	
"Normal" bilirubin after physiologic phase	≈1 mg/dL by 12–14 days	
Phototherapy/lamp distance	Per manufacturer's guidelines (halogen burn risk)	
Exchange volume	Double blood volume ≈ 85 mL/kg (≈85% replaced)	
Newborn blood volume estimate	≈85 mL/kg	

END OF CHAPTER 11

Neonatal Jaundice (Hyperbilirubinemia)

Next: **Chapter 12** — Respiratory Distress Syndrome (RDS) of the Newborn

PART II - NICU WARD

CHAPTER 12

Chapter 12 — Respiratory Distress Syndrome (RDS) of the Newborn

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part II — NICU Ward, Chapter 12

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain the pathophysiology of RDS (surfactant deficiency → atelectasis → hyaline membrane → hypoxemia/acidosis).
 2. Recognize the clinical signs: tachypnea (>60, then 80–120/min), retractions, expiratory grunting, nasal flaring, fine crackles, and central cyanosis as a late sign.
 3. Describe the therapeutic plan: CPAP/high-flow nasal cannula/mechanical ventilation (PEEP, PIP, MAP, SIMV), surfactant replacement, neutral thermal environment, and fluid/electrolyte support.
 4. State the nursing responsibilities during surfactant administration and ventilator/oxygen therapy.
 5. Complete the laboratory and ward steps of respiratory monitoring and thermoregulation.
-

2. Placement Box

Ward: NICU — second neonatal topic. RDS is the classic preterm emergency; the guide integrates newborn neuro/reflex assessment (CH6), vital signs (CH5), fluid therapy (CH7), and core ward skills (CH10) into the preterm bundle.

4. Definition and Risk Factors

RDS is a **surfactant deficiency** disorder with physiologic immaturity of the thorax; also called **hyaline membrane disease**. It occurs almost exclusively in **preterm infants** but is also associated with multifetal pregnancies, **infants of diabetic mothers**, cesarean delivery, cold stress, asphyxia, and a family history of RDS.

Non-pulmonary causes of similar distress: sepsis, cardiac defects, cold exposure, airway obstruction (atresia), IVH, hypoglycemia, metabolic acidosis, acute blood loss, and drugs; neonatal pneumonia may occur alone or complicate RDS.

5. Pathophysiology

- The final unfolding of alveolar septa occurs in the **last trimester**; preterm infants are born with underdeveloped, partly uninflatable alveoli; the chest wall is highly compliant (cartilage) and the diaphragm fatigues easily.
- **Surfactant** — a surface-active phospholipid secreted by **type II alveolar cells** — is first produced at ~**24 weeks** but the type II cells do not mature until ~**36 weeks**.
- Surfactant reduces the surface tension of alveolar fluid, allowing uniform inflation at low pressure. Deficiency causes **unequal inflation and collapse of alveoli on expiration** (atelectasis); each breath may require 60–75 cm H₂O negative pressure, exhausting the infant.
- Hypoxemia + increased pulmonary vascular pressure cause **transudation of fluid** into alveoli; necrotic cells + fibrin form the **hyaline membrane** that lines the alveoli and reduces compliance.
- **Atelectasis** → **increased pulmonary vascular resistance** → **right-to-left shunting through the ductus arteriosus and foramen ovale**; hypoxemia and hypercapnia → pulmonary vasospasm, worsening PVR.
- Prolonged hypoxemia causes **anaerobic glycolysis** → **lactic acid** → **metabolic acidosis**; the atelectatic lung cannot clear CO₂ → **respiratory acidosis**; low pH worsens vasoconstriction.
- Additional contributors: pulmonary edema (renal immaturity, high fluid intake, PDA, low colloid osmotic pressure, oxygen toxicity), pulmonary interstitial emphysema, and neurologic immaturity (weak gag/cough).

6. Clinical Manifestations

- Distress may be acute or develop over hours. After apparently normal breathing at birth, respirations gradually exceed **60/min** with **retractions** (suprasternal, substernal, supracostal, subcostal, intercostal) from the compliant chest wall.
- Within hours breathing becomes labored with the rate rising to **80–120/min**; infants increase **rate rather than depth**; **fine inspiratory crackles** and an **audible expiratory grunt** (a useful mechanism raising end-expiratory pressure) appear.
- **Nasal flaring** accompanies tachypnea, grunting, and retractions.
- **Central cyanosis** (bluish oral mucous membranes and generalized cyanosis) is a **late and serious sign**; use **pulse oximetry and arterial blood gases** instead of relying

on color.

- Severe RDS → shock-like state (diminished cardiac inflow, low blood pressure). **ELBW/VLBW infants** may have severe RDS at birth.
- Assessment grading: the **Silverman–Anderson scale** grades upper chest/lower chest retractions, xiphoid retractions, nares dilation, and expiratory grunt from 0–2 each.

7. Diagnostic Evaluation

- Laboratory data are nonspecific – **hypoxemia, hypercapnia, acidosis**; test blood glucose (hypoglycemia), blood gases/pH (acidosis), and PaO₂ (hypoxia); **pulse oximetry is an important component**.
- **Chest X-ray**: (1) a diffuse **ground-glass pattern** over both lungs (alveolar atelectasis) and (2) **air bronchograms** (dilated air-filled bronchioles). Distinguish RDS from **pneumonia**.
- **Antenatal lung maturity tests: L/S ratio** (lecithin/sphingomyelin) — an **L/S ratio of 2:1** in nondiabetic mothers predicts no significant RDS risk; **PG** appears in amniotic fluid at ~36 weeks; TDx FLM ≥55 is predictive of maturity; lamellar body counts; "shake"/bubble and tap tests.
- **Antenatal glucocorticoids** enhance fetal lung maturity, especially with postnatal surfactant.

8. Therapeutic Management

Six supportive measures are crucial:

1. **Maintain adequate ventilation and oxygenation** — CPAP, high-flow nasal cannula (HFNC), or mechanical ventilation.
2. **Maintain acid–base balance.**
3. **Maintain a neutral thermal environment.**
4. **Maintain adequate tissue perfusion and oxygenation.**
5. **Prevent hypotension.**

6. Maintain adequate hydration and electrolyte status.

Nipple and gavage feeding are avoided when the respiratory rate is markedly increased (aspiration hazard).

- **CPAP** delivers 3–8 cm H₂O positive pressure via nasal prongs/face mask, preventing alveolar collapse and increasing diffusion time; the infant breathes spontaneously.
- **IMV/PEEP**: mechanical breaths at a preset rate with positive end-expiratory pressure to prevent alveolar collapse; **PIP** is the peak inspiratory pressure; **MAP (mean airway pressure)** is the major determinant of oxygenation in severe RDS.
- **SIMV** synchronizes ventilator breaths with the infant's own breathing.
- Complications to prevent/monitor: those of positive-pressure ventilation, plus PDA/congestive heart failure, retinopathy of prematurity, IVH, bronchopulmonary dysplasia, NEC, and neurologic sequelae.

8.1 Surfactant Replacement

- Exogenous surfactant (natural porcine/bovine or synthetic; e.g., beractant) is given **via the endotracheal tube directly into the trachea**; single vs multiple doses per protocol.
- **Nursing responsibilities**: assist delivery, collect/monitor arterial blood gases, **scrupulously monitor oxygenation and tolerance**; after absorption there is usually a rise in compliance requiring ventilator adjustment (lower MAP) to prevent overinflation/hyperoxemia; **delay suctioning about an hour** (per product/protocol) to allow maximum effect.
- Infant may receive surfactant shortly after birth then be placed on nasal CPAP or HFNC.

9. Nursing Care Plan (Respiratory + Thermoregulation — RDS)

Assessment: 32-week preterm infant, 1 hour old; grunting, tachypnea 70/min, intercostal retractions, nasal flaring; O₂ sat 88% on room air; axillary temperature 35.8 °C.

NANDA diagnoses:

- *Ineffective breathing pattern / Impaired gas exchange* related to surfactant deficiency.

- *Ineffective thermoregulation* related to prematurity.

Goals: Maintain O₂ sat target per protocol; temperature stable; no aspiration; family informed.

Interventions (with rationale):

1. **Position** to facilitate airway expansion (prone may be preferred initially to increase chest expansion and oxygenation).
2. Monitor pulse oximetry, **arterial blood gases**, and clinical signs — grunting, nasal flaring, apnea, tachypnea, retractions, cyanosis — early detection of deterioration.
3. Monitor vital signs and perfusion for decreased cardiac output (poor perfusion, mottling).
4. **Assist with surfactant administration**, monitor tolerance and change in status; prepare to adjust ventilator/MAP.
5. Suction oropharynx/ETT **only as necessary** based on respiratory assessment (not routinely).
6. **Thermoregulation:** place in a thermally controlled incubator or radiant warmer; cap the head; plastic wrap for ELBW; use servo control and humidity; monitor core temperature and correlate with environmental temperature.
7. Evaluate: stable saturation, no worsening retractions; temperature within target; blood gases improving.

10. Procedure — Laboratory Step: Respiratory Monitoring and Oxygen/CPAP Care (Checklist)

#	Step	Done
1	Obtain baseline RR, heart rate, temperature, and SpO ₂ before and during handling.	☐
2	Score the Silverman–Anderson criteria (chest retractions, xiphoid, nares, grunt).	☐
3	Position the infant to maximize chest expansion; avoid neck flexion.	☐
4	Set/verify the O ₂ source (hood/cannula) or CPAP pressure (3–8 cm H ₂ O) per order.	☐
5	Apply SpO ₂ probe correctly; correlate with ABG results as ordered.	☐

#	Step	Done
6	Observe for central cyanosis, apnea, and worsening retractions; report immediately.	☐
7	Suction the airway only as indicated by assessment.	☐
8	Keep the infant NPO/gavage as ordered; monitor for aspiration risk.	☐
9	Provide neutral thermal environment (incubator/warmer, cap, servo control).	☐
10	Chart respiratory pattern, SpO ₂ , ventilator settings, and temperature regularly.	☐



Infant on nasal CPAP



Infant on nasal CPAP

11. Procedure — Hospital Step

- On admission of a preterm infant with distress, obtain blood gases, glucose, chest X-ray, and baseline vitals; plot the severity score.
- **During surfactant therapy:** assist the physician; levels/ABG sampling; adjust MAP after compliance improves; delay suctioning ~1 hour; monitor for rapid desaturation or reflux.
- Maintain IV fluids (CH7) and neutral thermal environment; keep parents informed with simple language and encourage participation when stable.

12. Case Study (Preterm with Distress)

Scenario: At 30 weeks' gestation, a 1,400-g infant is delivered by cesarean (mother diabetic). At 2 hours of age: RR 85/min, expiratory grunt audible, intercostal + subcostal retractions,

nasal flaring, SpO₂ 86% on room air.

1. Which birth variables increase RDS risk here? (*Prematurity + cesarean + infant of a diabetic mother.*)
 2. Why does this infant grunt? (*Grunting raises end-expiratory pressure, preserving alveolar expansion and permitting brief gas exchange in the early stages.*)
 3. What X-ray findings confirm RDS? (*Ground-glass pattern + air bronchograms; rule out pneumonia.*)
 4. Which therapy would you anticipate first? (*Oxygen + nasal CPAP (3–8 cm H₂O), with surfactant via ET tube and ventilator if oxygenation fails; correct acidosis; neutral thermal environment.*)
-

13. Chapter MCQs (10)

Q1. RDS is primarily caused by:

- a) Bacterial pneumonia
- b) Surfactant deficiency and physiologic immaturity of the thorax
- c) Congenital heart disease
- d) Meconium aspiration

Q2. Surfactant is produced by:

- a) Type I alveolar cells
- b) Type II alveolar cells
- c) Bronchial mucosa
- d) Alveolar macrophages

Q3. A **late and serious** sign of RDS is:

- a) Grunting
- b) Nasal flaring
- c) Central cyanosis
- d) Tachypnea >60/min

Q4. Radiographic findings of RDS include:

- a) Hyperinflated lungs

- b) Ground-glass pattern + air bronchograms
- c) Cardiomegaly
- d) Pleural effusion

Q5. An L/S ratio of **2:1** in a nondiabetic mother indicates:

- a) High RDS risk
- b) Virtually no RDS risk for the fetus
- c) Maternal infection
- d) Need for immediate delivery

Q6. CPAP acts by:

- a) Delivering 20 cm H₂O pressure
- b) Applying 3–8 cm H₂O positive pressure to prevent alveolar collapse
- c) Replacing surfactant
- d) Increasing tidal volume only

Q7. After surfactant administration, the nurse should:

- a) Suction immediately
- b) Delay suctioning ~1 hour and monitor oxygenation/ABG; adjust MAP to prevent overinflation
- c) Stop all monitoring
- d) Increase PIP immediately

Q8. Ventilator parameter that is the **major determinant of oxygenation** in severe RDS:

- a) PEEP
- b) Mean airway pressure (MAP)
- c) FiO₂ only
- d) Rate

Q9. Nipple or gavage feeding in a markedly tachypneic infant with RDS is:

- a) Encouraged to save calories
- b) Avoided because of aspiration hazard
- c) Given only cow's milk
- d) Always safe

Q10. Antenatal treatment that enhances fetal lung maturity is:

- a) Antibiotics
- b) Glucocorticoids
- c) Iron
- d) Thyroxine at birth

Answers: Q1 b — Q2 b — Q3 c — Q4 b — Q5 b — Q6 b — Q7 b — Q8 b — Q9 b — Q10 b.

14. Key Points

- RDS = surfactant deficiency → atelectasis → hyaline membrane → PVR/shunt → hypoxemia + metabolic/respiratory acidosis.
- Signs: tachypnea (>60 then 80–120/min), retractions, grunt, nasal flaring, crackles; **central cyanosis is late**; use SpO₂/ABG.
- X-ray: ground-glass pattern + air bronchograms; confirm lung maturity antenatally with **L/S ratio 2:1** or TDx FLM ≥55.
- Therapy: CPAP (3–8 cm H₂O)/HFNC/ventilation (PEEP, PIP, MAP, SIMV), acid-base correction, neutral thermal environment, perfusion/hydration; avoid feeding during marked tachypnea.
- Surfactant via ET tube: monitor ABG/oxygenation, expect compliance rise → lower MAP, delay suctioning ~1 hr.
- Nursing: position for expansion, monitor for poor perfusion/desaturation, suction only as indicated, provide thermoregulation (incubator, cap, plastic wrap, servo control).

15. Normal Values / Reference Table

Parameter	Value	Source
RR (newborn)	30–60/min	
Tachypnea (RDS)	>60, rising to 80–120/min	
L/S ratio (maturity)	≥2:1 (nondiabetic)	

Parameter	Value	Source
TDx FLM (maturity)	≥55	
CPAP pressure	3–8 cm H ₂ O	
Surfactant onset	~24 weeks; mature ~36 weeks	
Newborn temperature target	36.5–37.5 °C	

END OF CHAPTER 12

Respiratory Distress Syndrome (RDS) of the Newborn

Next: **Chapter 13** — Beta Thalassemia (Cooley Anemia)

PART III - BLOOD WARD

CHAPTER 13

Chapter 13 — Beta Thalassemia (Cooley Anemia)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part III — Blood Ward, Chapter 13

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain the genetics (autosomal recessive, 25% risk per pregnancy) and pathophysiology of beta thalassemia major.
2. Recognize the clinical manifestations: anemia, pallor, fever, hepatosplenomegaly, bone changes, and growth failure; and the classic hematologic picture (microcytic/hypochromic RBCs, target cells; Hb electrophoresis for diagnosis).
3. Describe the medical management: transfusion therapy (target Hgb ≈ 9.5 g/dL), iron chelation (deferoxamine, deferasirox, deferiprone), splenectomy, immunization, and cure by HSCT.

4. Implement nursing care: transfusion monitoring, chelation compliance education, prevention of postsplenectomy sepsis, and family/genetic counseling.

2. Placement Box

Ward: Blood ward — one of the three hematology topics. Thalassemia is the classic chronic hemolytic anemia requiring repeated transfusion + chelation.

Foundation skills reused: CH1 communication (chronic illness education), CH5 vital signs, CH7 fluid monitoring, CH8 medication safety (IV therapy), CH10 core ward skills.

4. Definition and Genetics

- **Thalassemia** is the term for inherited blood disorders with a **deficiency in the rate of production of specific globin chains in hemoglobin**; the name derives from the Greek *thalassa* ("sea"), referring to Mediterranean populations.
- **Beta thalassemia** is the most common form, classified as:
- **Thalassemia minor/trait** — heterozygous, usually mild microcytic anemia
- **Thalassemia intermedia** — moderate–severe anemia with splenomegaly
- **Thalassemia major (Cooley anemia)** — homozygous, severe anemia **not compatible with life without transfusion support**.
- **Mode of transmission:** autosomal recessive; when **both parents carry the trait**, each pregnancy carries a **25% chance of a child with thalassemia major**.

5. Pathophysiology

- Normal HbA is composed of two α and two β polypeptide chains. In beta thalassemia there is a **partial or complete deficiency in β -chain synthesis**, with compensatory excess of α -chains and continued γ -chain production → **unstable, defective hemoglobin**.

- The unstable polypeptide disintegrates, damaging RBCs and causing **severe hemolytic anemia**; the bone marrow overproduces erythrocytes unless suppressed by transfusions.
 - **Excess iron** from transfusions and rapid destruction of defective cells is stored in organs (**hemosiderosis**).
-

6. Clinical Manifestations

- **Before diagnosis:** anemia, pallor, unexplained fever, poor feeding, enlarged spleen or liver.
 - **Progressive anemia:** signs of chronic hypoxia, headache, bone pain, decreased exercise tolerance, listlessness, anorexia.
 - **Other features:** small stature, delayed sexual maturation, bronzed/freckled complexion (without chelation).
 - **Bone changes (untreated, older children):** enlarged head, frontal/parietal bossing, prominent malar eminences, flat nasal bridge, enlarged maxilla, protruding upper central incisors/malocclusion, generalized osteoporosis.
 - Onset in major may be insidious, recognized in late infancy or early toddlerhood.
-

7. Diagnostic Evaluation

- CBC/smear: **microcytosis, hypochromia, anisocytosis, poikilocytosis, target cells**, basophilic stippling; low Hgb and Hct in severe anemia.
 - **Hemoglobin electrophoresis confirms the diagnosis** and distinguishes the type by analyzing the quantity/kind of Hb variants.
 - Screen the whole family for the trait; refer for genetic counseling; prenatal diagnosis by amniocentesis or fetal blood sampling is available.
-

8. Therapeutic Management

- **Transfusions are the foundation** of management; goal: maintain Hgb at approximately **9.5 g/dL**, possibly requiring transfusion **every 2–5 weeks**. Benefits: normal activity, decreased cardiomegaly/hepatosplenomegaly, fewer bone changes, near-normal growth, fewer infections.
- **Iron chelation** for hemosiderosis/hemochromatosis (the body cannot eliminate excess iron): **deferoxamine** (parenteral/subcutaneous infusions), **deferasirox** (oral), and **deferiprone** (oral).
- **Splenectomy** may be needed for severe splenomegaly/increased transfusion requirements; **major complication = severe overwhelming infection** — children usually receive prophylactic antibiotics for years plus **pneumococcal and meningococcal vaccines**.
- **NURSING TIP:** the family must notify health personnel of **any fever $\geq 38.5^{\circ}\text{C}$ (101.3°F)** because of **sepsis risk from asplenia**.
- **Cure:** allogeneic **hematopoietic stem cell transplantation (HSCT)** — about 80–97% complication-free survival in children <14 years without excessive iron stores/hepatomegaly; gene-therapy approaches are under study.
- Prognosis: with transfusions + early chelation, survival into adulthood; common causes of death are heart disease, liver cirrhosis, postsplenectomy sepsis, and multiorgan failure from hemochromatosis.

9. Nursing Care Plan (Chronic Care)

Assessment: 6-year-old with thalassemia major, Hgb 7.2 g/dL, attends every 3 weeks for transfusion; missed 2 chelation doses this month; parents concerned about injections.

NANDA diagnoses:

- *Deficient knowledge* (parents/child) related to chelation therapy.
- *Risk for injury* (transfusion reaction / infection) related to repeated transfusions and asplenia.

Goals: Compliance with chelation; complication-free transfusions; parents identify the fever rule; Hgb maintained ≈ 9.5 g/dL.

Interventions (with rationale):

1. Explain the defect, its effect on RBCs, and the consequences of untreated iron overload (delayed growth, **heart disease**) — motivates adherence.
2. Teach and supervise the chelation regimen (deferoxamine infusions / oral agents); acknowledge that needle/infusion anxiety is common and support coping — improves adherence.
3. Monitor every transfusion: baseline vitals, correct blood product, observation for reactions — prevents/early-manages reactions.
4. **Fever ≥ 38.5 °C = report immediately** (asplenia sepsis risk); ensure pneumococcal/meningococcal/regular vaccines.
5. Support chronic-illness adjustment and refer for genetic counseling (25% recurrence risk) — family coping and informed choices.
6. Evaluate: Hgb maintained near 9.5 g/dL; no sepsis/transfusion reactions; chelation doses documented; family states the fever rule.

10. Procedure — Laboratory Step: Transfusion Monitoring (Checklist)

#	Step	Done
1	Verify the order and blood product (cross-match, expiry, two identifiers) with a second nurse.	☐
2	Check the blood product condition (no discoloration, clots, or damage; correct storage).	☐
3	Record baseline vital signs (T, HR, RR, BP, SpO ₂).	☐
4	Start the transfusion at a slow rate as ordered; remain with the child for the first 15–20 min.	☐
5	Monitor vitals per protocol and observe for reactions: fever, chills, rash, dyspnea, back pain.	☐
6	If a reaction is suspected: stop the transfusion, keep the line open, notify the physician, and recheck the product/bag.	☐
7	Chart volume, time, product details (lot), and observations.	☐
8	Educate the family about the transfusion schedule and the fever rule.	☐



Target cells and microcytic/hypochromic RBCs on a blood smear



11. Procedure — Hospital Step

- On admission of a child for transfusion: weigh, obtain CBC, verify the product, and use the transfusion checklist.
- Reinforce chelation teaching and confirm the next chelation dose is given in the ward/pharmacy; document iron studies (ferritin) results and coordinate with the hematologist.
- Provide the postsplenectomy/at-risk immunization record review; schedule pneumococcal/meningococcal vaccines if due.

12. Case Study (Fever in Thalassemia)

Scenario: A 7-year-old boy with thalassemia major (postsplenectomy 1 year ago) is brought to the emergency room with fever 39.1 °C, chills, and listlessness.

1. Why is this an emergency? (*Postsplenectomy + fever = risk of overwhelming sepsis; fever ≥ 38.5 °C must be reported immediately.*)
2. Which preventive measures should have been in place? (*Prophylactic antibiotics, pneumococcal + meningococcal + regular vaccines.*)
3. What is the likely long-term therapy-related risk in this child? (*Iron overload/hemochromatosis → heart and liver disease if chelation is incomplete.*)
4. Which single test confirms thalassemia? (*Hemoglobin electrophoresis.*)

13. Chapter MCQs (10)

Q1. Beta thalassemia is inherited as:

- a) Autosomal dominant

- b) Autosomal recessive (25% risk per pregnancy when both parents carry the trait)
- c) X-linked
- d) Mitochondrial

Q2. The hallmark hematologic finding is:

- a) Macrocytosis
- b) Microcytic/hypochromic RBCs with target cells
- c) Thrombocytosis
- d) Immature white cells only

Q3. The diagnosis is confirmed by:

- a) CBC alone
- b) Hemoglobin electrophoresis
- c) Bone marrow only
- d) Serum iron

Q4. The goal of transfusion therapy is to maintain Hgb at approximately:

- a) 5 g/dL
- b) 9.5 g/dL
- c) 14 g/dL
- d) 12 g/dL

Q5. Complications of repeated transfusions are mainly due to:

- a) Respiratory alkalosis
- b) Iron overload (hemosiderosis/hemochromatosis)
- c) Folate deficiency
- d) Hyponatremia

Q6. Which of the following is an oral iron-chelating agent?

- a) Deferasirox
- b) Ferrous sulfate
- c) Vitamin C
- d) Penicillin

Q7. After splenectomy, the child must be taught to report:

- a) Any fever ≥ 38.5 °C

- b) Mild itching
- c) Repeated hiccups
- d) Cold hands

Q8. A critical adverse outcome of severe postsplenectomy infection is:

- a) Anemia only
- b) Overwhelming sepsis
- c) Jaundice
- d) Osteoporosis

Q9. The curative option for some children with thalassemia major is:

- a) Daily iron tablets
- b) Hematopoietic stem cell transplantation
- c) Splenectomy alone
- d) Transfusion only

Q10. The most common causes of death in untreated/inadequately chelated thalassemia are:

- a) Pneumonia and ear infection
- b) Heart disease and liver cirrhosis from hemochromatosis
- c) Trauma
- d) Dehydration

Answers: Q1 b — Q2 b — Q3 b — Q4 b — Q5 b — Q6 a — Q7 a — Q8 b — Q9 b — Q10 b.

14. Key Points

- Beta thalassemia major (Cooley anemia) = homozygous, autosomal recessive, **severe anemia requiring transfusion support**.
- Key clinical: pallor, fever, hepatosplenomegaly, bone deformities of untreated children; hematology: microcytic/hypochromic + **target cells**; diagnosis by **Hb electrophoresis**.
- Medical care: transfusions to Hgb $\approx$ **9.5 g/dL** (q2–5 wk), **iron chelation** (deferoxamine/deferiprone/deferiprone), splenectomy if needed, immunizations, and HSCT for cure.

- Nursing: transfusion safety, chelation compliance, **fever $\geq 38.5^{\circ}\text{C}$ = emergency (asplenia sepsis)**, genetic counseling (25% risk).

15. Normal Values / Reference Table

Parameter	Normal (reference)	In thalassemia major
Hgb	11–16 g/dL (age-dependent)	Low (maintain ≈ 9.5 g/dL by transfusion)
RBC morphology	Normocytic normochromic	Microcytic, hypochromic, target cells
Incidence	—	~ 4.4 / 10,000 live births worldwide
Recurrence risk (both carriers)	—	25% per pregnancy

END OF CHAPTER 13

Beta Thalassemia (Cooley Anemia)

Next: **Chapter 14** — Acute Lymphoblastic Leukemia (ALL)

PART III - BLOOD WARD

CHAPTER 14

Chapter 14 — Acute Lymphoblastic Leukemia (ALL)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part III — Blood Ward, Chapter 14

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. State the epidemiology of ALL (the most common childhood cancer; peak age 3–5 years; more common in boys) and the main risk factors.
 2. Link the clinical signs (anemia, infection/neutropenia, bleeding, bone pain, lymphadenopathy, hepatosplenomegaly, fever) to bone-marrow infiltration.
 3. Describe the diagnostic work-up (CBC, peripheral smear, bone marrow aspiration with blast percentage, CSF for CNS evaluation) and the three treatment phases (induction, consolidation/intensification, maintenance).
 4. Explain nursing care: prevention of infection, bleeding and tumor lysis syndrome, CNS-directed intrathecal therapy, mouth care, and family education.
-

2. Placement Box

Ward: Blood ward — second hematology topic. ALL is the index childhood cancer; the guide reuses CH1 communication and CH8 medication-safety skills, CH7 fluid therapy (tumor lysis prevention), and CH13 transfusion/chelation principles.

4. Definition and Epidemiology

- **Leukemia** is a broad term for **malignant diseases of the bone marrow, blood, and lymphatic system**. In healthy children the marrow makes blood stem cells that mature into lymphoid or myeloid cells; in acute leukemias, **immature nonfunctional cells predominate**.
 - The two most common types in children are **acute lymphoblastic leukemia (ALL)** and **acute myeloid leukemia (AML)**.
 - **ALL is the most common form of childhood cancer**, ~3,000 new cases annually in the U.S.; **more common in boys (10% more)**, more common in Hispanics; peak age of onset **between 3 and 5 years**.
 - Risk factors: prenatal exposure to x-rays, previous treatment with chemotherapy, and genetic conditions (**Down, Bloom, and Fanconi syndromes**; Li-Fraumeni); some ALL begins prenatally.
-

5. Pathophysiology → Clinical Manifestations

- Signs reflect **infiltration of the bone marrow by nonfunctional leukemic cells (blasts)**. The three main consequences of marrow infiltration:
 1. **Anemia** from decreased RBCs;
 2. **Infection** from neutropenia;
 3. **Bleeding** from decreased platelets.
 - About half of patients have an **elevated WBC count at presentation ($>20,000/\text{mm}^3$)**.
 - Common signs: **hepatosplenomegaly (75%)**, splenomegaly alone (24%), **fever (63%)**, lymphadenopathy (60%), bleeding (petechiae/purpura, 20%), **bone pain (12%)**.
 - Onset ranges from acute to insidious; a "cold that will not go away," pallor, listlessness, irritability, fever, anorexia, weight loss, bruising without cause, and continued bone/joint pain should arouse suspicion.
 - Two important **extramedullary sanctuaries** — sites where leukemic cells hide from systemic therapy: **the CNS and the testes**; they require specific therapy.
-

6. Diagnostic Evaluation

- **Suspected** from history, physical signs, and a **peripheral blood smear containing leukemic blasts**, frequently with low counts.
- Work-up: thorough history and physical exam; laboratory tests (**CBC with differential**, blood chemistries); **bone marrow aspirate or biopsy with cytogenetic analysis and immunophenotyping**.
- A chest x-ray may detect a mediastinal mass.
- **Definitive diagnosis** is based on analysis of the bone marrow sample — typically a **monotonous infiltrate of blast cells**.
- After confirmation, a **lumbar puncture** checks for **CNS involvement**.
- **Prognostic (risk) factors** including age at diagnosis (favorable 1–10 years), **WBC count at diagnosis (favorable $\leq 50,000/\text{mm}^3$)**, CNS/testicular involvement, Down

syndrome, sex, cytogenetics, and response to initial treatment (minimal residual disease, MRD).

- **NURSING ALERT:** children vaccinated 2 weeks before or during chemotherapy are considered unimmunized; revaccinate or give live-virus vaccines only **6 months after chemotherapy stops**, following institutional guidelines.

7. Therapeutic Management

- Long-term disease-free survival now **approaches 90%** at major centers; treatment is **risk-based** (high-risk children get the most intensive therapy).
- Three phases:
 1. **Remission induction** — begins almost immediately after diagnosis, lasts 4–6 weeks. Drugs: **dexamethasone or prednisone, vincristine, L-asparaginase**, with or without an anthracycline. **Complete remission (CR)** = <5% blasts in the marrow and no detectable extramedullary leukemia.
 2. **Consolidation/intensification** — given after CR (all children need post-induction therapy); the very intensive **Berlin–Frankfurt–Münster (BFM) backbone** uses cyclophosphamide, cytarabine, mercaptopurine, then intermittent maintenance with high-dose methotrexate (with or without leucovorin rescue) and **delayed intensification**.
 3. **Maintenance (continuation)** — daily mercaptopurine, weekly low-dose methotrexate, plus vincristine/corticosteroid and continued intrathecal therapy; **adherence <90% increases relapse risk**.
- **CNS-directed therapy** — because leukemic cells in the CNS are protected from systemic drugs by the blood–brain barrier, **all children receive CNS prophylaxis**: intrathecal chemotherapy (**methotrexate** alone or with cytarabine + hydrocortisone) plus CNS-directed systemic therapy (dexamethasone, high-dose methotrexate with leucovorin rescue); cranial radiation for the highest-risk children.
- Four standard treatment modalities: chemotherapy, radiation therapy, chemotherapy with stem cell transplant, and targeted therapy (e.g., TKIs, monoclonal antibodies, proteasome inhibitors).
- **Supportive therapy** (essential in the period immediately after remission when myelosuppression leaves the body defenseless): **transfusion support and**

antibacterial/antifungal agents.

- **Late effects & survivorship:** long-term follow-up per the Children's Oncology Group (COG) guidelines; nurses help families access survivorship and transition care.

8. Nursing Care Plan (supportive care)

Assessment: 7-year-old boy, newly diagnosed ALL, WBC 85,000/mm³, Hgb 6.0 g/dL, platelets 60,000/mm³; hepatomegaly; starting induction tomorrow.

NANDA diagnoses:

- *Risk for infection* related to neutropenia and immunosuppressive therapy.
- *Risk for injury / bleeding* related to thrombocytopenia.
- *Risk for imbalanced fluid volume* related to tumor lysis syndrome.

Goals: No severe infection; no significant bleeding; no tumor lysis; family educated.

Interventions (with rationale):

1. **Neutropenic precautions:** hand washing, frequent bathing, no shared utensils, avoid crowds/ill contacts; monitor for fever, chills, cough, sore throat and report immediately — prevents/early-detects infection.
2. **Skin and mouth care:** keep skin/perianal area clean with mild lotion; frequent mouth care with saline rinses; check for ulcers and gum swelling — prevents mucositis and skin entry of organisms.
3. **Bleeding precautions:** recognize and report bruising/petechiae/nosebleeds; avoid contact sports; use soft toothbrush; monitor counts — prevents serious hemorrhage.
4. **Tumor lysis syndrome prevention: aggressive IV fluids,** monitor serum chemistries frequently, check urine pH with each void, administer a drug to reduce uric acid formation, maintain strict I & O, and monitor WBC count — prevents metabolic catastrophe with high blast counts.
5. **CNS prophylaxis support:** assist with lumbar puncture; obtain consent, monitor vitals, provide comfort/sedation per protocol, watch for bleeding at the puncture site.
6. **Prep for procedures:** explain each diagnostic/therapeutic procedure (bone marrow aspirate, LP), what will happen, and how the child can help; assess the child's understanding — reduces fear and fosters trust.

7. **Diet:** high-protein, high-calorie diet; adequate hydration with high-fiber diet and stool softeners — supports nutrition and prevents constipation.
8. Evaluate: afebrile; no infection/bleeding; chemistry normal; family lists fever-rule and precautions.

9. Procedure — Laboratory Step: Supportive Care for the Child on Chemotherapy (Checklist)

#	Step	Done
1	Verify two identifiers; review the CBC (absolute neutrophil count, platelets, Hgb).	☐
2	Apply infection-prevention protocols (hand hygiene, isolation per policy, no fresh flowers/ill visitors).	☐
3	Inspect the mouth daily for mucositis/ulcers and provide saline rinses/soft brush care.	☐
4	Institute bleeding precautions when platelets are low (soft brush, no IM injections, pressure to puncture sites).	☐
5	Maintain aggressive IV fluids and strict I & O; check urine pH; monitor chemistries per protocol (tumor lysis).	☐
6	Administer the uric-acid–reducing medication and other ordered agents on time.	☐
7	Assist with lumbar puncture/intrathecal chemotherapy: consent, positioning, vitals, post-procedure flat positioning.	☐
8	Document all observations; report fever, chills, bleeding, or altered behavior immediately.	☐



Lymphoblasts on bone marrow aspirate



10. Procedure — Hospital Step

- Admit and complete the diagnostic work-up (CBC with differential, chemistries, bone marrow, LP) with family preparation and consent.
- Begin tumor-lysis precautions the day before induction; monitor counts and chemistries throughout the day.
- Provide CNS-intrathecal chemotherapy support in the treatment room; afterwards keep the child supine as ordered and observe the puncture site.
- Discharge teaching: infection recognition (**fever/chills/cough/sore throat → report immediately**), bleeding precautions, mouth care, hydration, high-protein high-calorie diet, and vaccination advice (no live vaccines until 6 months after stopping chemo).

11. Case Study (New Diagnosis)

Scenario: A 7-year-old boy had bilateral neck swelling for 1 month; now hoarse and sleeping most of the time. CBC: Hgb 6.0 g/dL, WBC 85,000/mm³, platelets 60,000/mm³; bilateral parotid, submental and axillary lymph nodes; hepatomegaly 6 cm. Bone marrow flow cytometry: 70% B-cell blasts — ALL confirmed. Induction chemotherapy starts tomorrow.

1. Which signs represent anemia, infection-risk, bleeding-risk, and infiltration? (*Anemia → pallor/fatigue; neutropenia → infection; thrombocytopenia → bleeding/bruising; infiltration/marrow pressure → bone pain, hepatosplenomegaly, lymphadenopathy.*)
2. Why is he at high risk for tumor lysis syndrome? (*WBC 85,000/mm³ — massive, rapid lysis of blast cells releases intracellular contents causing metabolic abnormalities; prevent with IV fluids, uric-acid drug, urine pH checks, strict I&O.*)
3. Why intrathecal chemotherapy now? (*CNS is a sanctuary where leukemic cells hide behind the blood–brain barrier; all children get CNS prophylaxis with intrathecal methotrexate-containing regimens.*)

4. What discharge teaching is essential? (*Fever/immediate infection reporting, bleeding precautions, mouth and skin care, high-protein high-calorie diet, live-vaccine timing (≥ 6 months after chemo).*)
-

12. Chapter MCQs (10)

Q1. ALL is:

- a) A rare childhood tumor
- b) The most common form of childhood cancer
- c) Inherited in all cases
- d) A skin malignancy

Q2. The peak age of onset of ALL is:

- a) 1–5 years
- b) 3–5 years
- c) 12–15 years
- d) First year of life

Q3. The three main consequences of bone-marrow infiltration are:

- a) Myocarditis, hepatitis, nephritis
- b) Anemia, infection, and bleeding
- c) Hyponatremia, hypoglycemia, acidosis
- d) Arthritis, myositis, dermatitis

Q4. The definitive diagnosis of ALL is made by:

- a) Physical exam alone
- b) Bone marrow aspirate analysis (monotonous blast infiltration)
- c) Urine analysis
- d) Chest x-ray alone

Q5. CNS involvement in ALL is evaluated by:

- a) EEG
- b) Lumbar puncture (CSF cytology)

- c) CT without LP
- d) Fundoscopy

Q6. The three standard phases of ALL treatment are:

- a) Induction, consolidation/intensification, maintenance
- b) Loading, loading-dose, weaning
- c) Surgery, radiation, physiotherapy
- d) Induction, palliation, hospice

Q7. A drug characteristic of remission induction is:

- a) Aminophylline
- b) Vincristine (with dexamethasone/prednisone and L-asparaginase)
- c) Digoxin
- d) Deferoxamine

Q8. All children receive CNS prophylactic therapy because:

- a) The CNS always has blasts
- b) Leukemic cells in the CNS are protected from systemic chemotherapy by the blood–brain barrier
- c) CNS is not a sanctuary
- d) To treat headache

Q9. A child with high WBC (e.g., 85,000/mm³) and beginning induction is at risk for:

- a) Tumor lysis syndrome
- b) Anaphylaxis only
- c) Constipation only
- d) Diarrhea only

Q10. Live-virus vaccines during/soon after immunosuppressive therapy:

- a) Are safe any time
- b) Should be given 6 months after chemotherapy stops (with revaccination rules)
- c) Should be given before starting chemo
- d) Are required immediately

Answers: Q1 b — Q2 b — Q3 b — Q4 b — Q5 b — Q6 a — Q7 b — Q8 b — Q9 a — Q10 b.

13. Key Points

- ALL = most common childhood cancer; peak 3–5 years, more in boys; marrow infiltration → **anemia, neutropenia/infection, thrombocytopenia/bleeding**, bone pain, lymphadenopathy, hepatosplenomegaly, fever.
- Diagnosis: CBC + peripheral smear with blasts, **bone marrow aspirate** (definitive), LP for CNS staging; risk-based on age, WBC, CNS status, cytogenetics, MRD.
- Treatment: **induction** → **consolidation/intensification** → **maintenance**; CNS prophylaxis for all; targeted therapy and HSCT in selected/relapsed cases; supportive transfusions + anti-infectives.
- Nursing: infection/bleeding precautions, mouth and skin care, **tumor-lysis prevention (IV fluids, urine pH, uric-acid drug, I&O, chemistries)**, LP support, family education and survivorship.

14. Normal Values / Reference Table

Parameter	Reference (normal)	In ALL
WBC count	~5–15,000/mm ³ (age-dependent)	Often elevated >20,000, may be low
Favorable WBC at diagnosis	—	≤50,000/mm ³
Platelets	150,000–400,000/mm ³	Low (e.g., 60,000/mm ³)
Marrow blasts	<5%	Complete remission = <5% blasts
MRD after induction	Undetectable	Best prognosis at <10 ⁻⁴
CR definition	—	<5% blasts + no extramedullary disease

Chapter 15 — Hemophilia

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part III — Blood Ward, Chapter 15

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain the inheritance (X-linked recessive) of hemophilia and differentiate hemophilia A (factor VIII), hemophilia B (factor IX), and von Willebrand disease.
2. Recognize the clinical manifestations (Box: prolonged bleeding, hemarthrosis, bruising, hematomas, hematuria) and severe complications (airway obstruction, intracranial hemorrhage).
3. Describe the therapy: factor VIII/IX replacement, DDAVP in mild forms, prophylaxis vs on-demand treatment, RICE for acute bleeds.
4. Provide nursing care: prevent bleeding, implement RICE, avoid aspirin/IM injections, teach home therapy and self-administration, genetic counseling, and school/activity planning.

2. Placement Box

Ward: Blood ward — third hematology topic. Hemophilia demonstrates precise medication-safety and intravenous skill reuse (CH8), transfusion principles (CH13), and

4. Definition and Genetics

- **Hemophilia** is a group of bleeding disorders from **congenital deficiency, dysfunction, or absence of specific coagulation proteins (factors)**.
- About **80% of cases are X-linked recessive**. The two most common forms are:
- **Hemophilia A (classic)** — factor VIII deficiency; prevalence ~**1 in 5,000** live births.
- **Hemophilia B (Christmas disease)** — factor IX deficiency; ~1 in 20,000–30,000 live births.
- **Von Willebrand disease (vWD)** — hereditary bleeding disorder from deficiency/abnormality of the von Willebrand factor (vWF) protein. Factor VIII deficiency accounts for **80–85% of all hemophilia cases**.

5. Pathophysiology

- The basic defect of hemophilia A is **deficiency of factor VIII (anti-hemophilic factor [AHF])**, produced by the liver and needed for thromboplastin formation in phase I of coagulation. **The less factor VIII in the blood, the more severe the disease**.
- Individuals have intact vascular influence and platelets, so they **bleed longer, not faster**.
- Three clinical grades (severe, moderate, mild) by severity of factor deficiency; **60–70% of children have the severe form**. With severe deficiency, hemorrhage may be spontaneous or from minor trauma (e.g., post-circumcision); with milder forms, bleeding may only be noticed when the child begins to walk.
- **Hemarthrosis** (bleeding into joints — knees, elbows, ankles) is the **most frequent form of internal bleeding**; repeated bleeds cause bone changes and crippling deformities. Early signs of a joint bleed: **stiffness, tingling, or achiness**, then warmth, redness, swelling, severe pain, and loss of movement.
- **Serious sites**: neck/mouth/thorax (airway obstruction), **intracranial hemorrhage (a major cause of death)**, GI bleeding (anemia), retroperitoneal hemorrhage, and

spinal hematoma (paralysis).

6. Clinical Manifestations

- Prolonged bleeding anywhere from or in the body.
 - Hemorrhage from any trauma: loss of deciduous teeth, circumcision, cuts, epistaxis, injections.
 - Excessive bruising from even slight injury; subcutaneous and IM hemorrhages.
 - **Hemarthrosis** (knees, ankles, elbows); **hematomas** (pain, swelling, limited motion).
 - **Spontaneous hematuria**.
 - Maintain a high index of suspicion for **internal bleeding** with: **headache, slurred speech, loss of consciousness** (cerebral) and **black tarry stools** (GI).
-

7. Diagnostic Evaluation

- Diagnosis from the **history of bleeding episodes**, evidence of **X-linked inheritance** (only ~1/3 are new mutations), and laboratory findings.
 - Specific tests: **factor VIII and factor IX assays** (performed in specialized laboratories).
 - General screening test depending on specific factors: the **partial thromboplastin time (PTT)**.
 - **DNA (carrier) testing** allows detection as early as classic hemophilia and is important for families where female offspring may have inherited the trait.
-

8. Therapeutic Management

- **Primary therapy = replacement of the missing clotting factor.** Products: **factor VIII concentrates** (recombinant by genetic engineering, or plasma-derived), **reconstituted with sterile water immediately before use.**

- **DDAVP (1-deamino-8-d-arginine vasopressin)** — synthetic vasopressin that **increases plasma factor VIII activity; treatment of choice in mild hemophilia and vWD type I** if the child shows an appropriate response (expect a 3–4-fold rise). **Not effective in severe hemophilia A, severe vWD, or hemophilia B.**
- **Corticosteroids** for hematuria, acute hemarthrosis, and chronic synovitis.
- **Avoid aspirin and traditional NSAIDs (e.g., ibuprofen)** because they inhibit platelet function → GI bleeding; selective **COX-2 inhibitors** may be suitable for hemophilic arthropathy.
- **ε-aminocaproic acid (Amicar)** prevents clot destruction; limited to mouth trauma or surgery, with factor concentrate given first.
- **Exercise and physical therapy** within reasonable limits strengthen muscles around joints and may reduce spontaneous bleeds.
- **Home treatment:** most children are treated at home because early treatment gives faster recovery and fewer complications; family is taught **venipuncture and AHF administration** for children older than 2–3 years; the child learns **self-administration at 8–12 years.**
- **Prophylaxis:** periodic factor replacement for severe hemophilia to prevent **arthropathy and spontaneous life-threatening bleeds; primary prophylaxis** (regular factor VIII infusion before joint damage) vs **secondary prophylaxis** (after first joint bleed); infusions every other day to 3 times weekly.
- **Prognosis:** no cure, but symptoms controlled and deformities minimized; today's children using recombinant products are at virtually no risk of HIV from treatment (90% of severe hemophiliacs treated 1979–1985 seroconverted to HIV-positive before modern purification). Gene therapy is under development.

9. Nursing Care Plan

Assessment: 5-year-old boy with severe hemophilia A presents with a swollen, warm right knee after a fall; earlier he had a "stiff feeling" in the joint.

NANDA diagnoses:

- *Risk for injury* (bleeding episodes) related to factor VIII deficiency.
- *Impaired physical mobility* related to hemarthrosis and joint pain.

- *Deficient knowledge* (family) related to home factor-replacement and RICE.

Goals: Bleed treated early; joint function preserved; family demonstrates home care; no aspirin/IM injuries.

Interventions (with rationale):

1. **RICE: rest, ice, compression, elevation** together with factor replacement (cold packs kept in the freezer for emergencies); these measures do **not** replace factor therapy.
2. **Recognize internal bleeding early:** headache, slurred speech, unconsciousness, black tarry stools, unusual swelling/pain — minimizes serious complications.
3. **Prevent bleeding:** soft-bristle or water-irrigation dental care, **electric shaver (no safety razors)**, medical identification, supervised play with protective equipment; avoid contact sports (non-contact: swimming, walking, jogging, tennis, bowling, golf, yoga).
4. **Medication precautions: no aspirin or aspirin-containing compounds;** acetaminophen is a suitable substitute; avoid **IM injections** — use subcutaneous or IV routes; prefer venipuncture (less bleeding than finger/heel).
5. **Immobilize and elevate the affected joint during a bleed; institute active range-of-motion after the acute phase;** monitor weight to reduce joint strain.
6. **Teach home therapy:** reconstitute factor with sterile water immediately before use; family performs venipuncture/AHF infusion; child self-administers at 8–12 years.
7. **Genetic counseling ASAP** (support the mother; DNA carrier testing available); teach about safe sexual practices for those exposed before 1990 purification.
8. Evaluate: bleeding stopped; joint ROM preserved; family performs RICE + factor infusion; school and play plan in place.

10. Procedure — Laboratory Step: Care of Acute Bleeding / RICE (Checklist)

#	Step	Done
1	Recognize the bleed early (stiffness/tingling/achiness in joint; visible swelling, warmth, pain limitation).	☐

#	Step	Done
2	Notify the physician and prepare factor replacement per protocol.	☐
3	Institute RICE: Rest — immobilize the joint in a neutral comfortable position.	☐
4	Ice — cold pack for 15–20 min every 2–3 h (cloth barrier, no direct skin contact).	☐
5	Compression — light elastic wrap; Elevation — raise above heart level when feasible.	☐
6	After the acute phase, begin active range-of-motion exercises supervised.	☐
7	Reinforce aspirin avoidance and safe activities; document the episode.	☐
8	Teach the family/home plan (factor reconstitution, venipuncture, emergency numbers).	☐



Hemarthrosis of the knee



Hemarthrosis of the knee

11. Procedure — Hospital Step

- On admission for a joint/cerebral/GI bleed: establish IV access by venipuncture (avoid IM injections), obtain factor level + PTT, transfuse/reconstitute factor concentrate, and apply RICE.
- Watch neurologic status (headache, slurred speech, LOC) and stools (tarry). After stabilization, review home therapy and update the school activity plan and medical ID.

12. Case Study (First Severe Joint Bleed)

Scenario: A 3-year-old with severe hemophilia A falls at home; his mother finds the right knee warm, swollen, and he refuses to bear weight. PTT prolonged; factor VIII level 2%.

1. What is the priority action? (*Factor VIII replacement without delay — early treatment = faster recovery, fewer complications; the lab finding confirms severe disease.*)
 2. What first-aid measures accompany the infusion? (*RICE — rest, ice, compression, elevation; never instead of factor replacement.*)
 3. Which medications are contraindicated? (*Aspirin and traditional NSAIDs (ibuprofen); acetaminophen is an alternative.*)
 4. Why is IM injection avoided? (*IM hemorrhage risk; use subcutaneous or IV routes.*)
 5. What long-term plan prevents arthropathy? (*Prophylactic factor replacement — primary prophylaxis before joint damage — plus exercise.*)
-

13. Chapter MCQs (10)

Q1. Hemophilia A is a deficiency of:

- a) Factor IX
- b) Factor VIII (anti-hemophilic factor)
- c) Factor VII
- d) vWF

Q2. The inheritance pattern of hemophilia is:

- a) Autosomal dominant
- b) X-linked recessive (about 80% of cases)
- c) Autosomal recessive
- d) Mitochondrial

Q3. The most common form of internal bleeding in hemophilia is:

- a) Intracranial
- b) Hemarthrosis (especially knees, elbows, ankles)
- c) GI bleeding
- d) Retroperitoneal

Q4. Hemophilia is severe when the factor level is markedly low, present in approximately:

- a) 10% of children
- b) 60–70% of children

- c) 100% of children
- d) 30% of children

Q5. A screening test for hemophilia is:

- a) Partial thromboplastin time (PTT)
- b) Platelet count only
- c) Bleeding time only
- d) INR only

Q6. DDAVP is the treatment of choice for:

- a) Severe hemophilia A
- b) Mild hemophilia and vWD type I with an appropriate response
- c) Hemophilia B
- d) Any GI bleed

Q7. Which medication is contraindicated in hemophilia?

- a) Acetaminophen
- b) Aspirin and traditional NSAIDs (ibuprofen)
- c) Paracetamol
- d) Oral rehydration salts

Q8. RICE stands for:

- a) Rest, Ice, Compression, Elevation
- b) Rotation, Inhalation, Compression, Exhalation
- c) Reposition, Infuse, Cover, Evaluate
- d) Rest, IV, Cold, Exercise

Q9. To prevent bleeding from injections, the nurse should:

- a) Use the IM route routinely
- b) Prefer venipuncture, avoid IM injections
- c) Use heel pricks for all labs
- d) Give aspirin first

Q10. Factor VIII concentrates are reconstituted:

- a) Days before use
- b) Immediately before use, with sterile water

- c) With saline and stored
- d) With tap water

Answers: Q1 b — Q2 b — Q3 b — Q4 b — Q5 a — Q6 b — Q7 b — Q8 a — Q9 b — Q10 b.

14. Key Points

- Hemophilia = congenital factor deficiency; **A = factor VIII (classic, ~1:5000), B = factor IX (Christmas disease)**; X-linked recessive; severe form in 60–70%.
- Bleed longer, not faster; **hemarthrosis most common**; internal-bleeding red flags: headache, slurred speech, LOC, tarry stools.
- Diagnosis: history + inheritance + **factor assays / PTT**; DNA carrier testing available.
- Therapy: **factor VIII/IX replacement**, DDAVP for mild, corticosteroids for hematuria/arthropathy, COX-2 for analgesia; **avoid aspirin/NSAIDs and IM injections**.
- Nursing: RICE with factor therapy, prophylaxis (primary/secondary), home therapy teaching + self-administration at 8–12 years, school/activity planning, genetic counseling.

15. Normal Values / Reference Table

Parameter	Normal	In hemophilia
Factor VIII level	~50–150% activity	Low (severe <1%)
PTT	Normal reference range	Prolonged
Prevalence A / B	—	1:5,000 / 1:20,000–30,000 live births
Severe form	—	60–70% of children
DDAVP factor-VIII rise	—	3–4-fold if responsive

Chapter 16 — Pneumonia in Children

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part IV — Medical Ward, Chapter 16

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Define pneumonia and classify it by etiologic agent (viral, bacterial, mycoplasmal/atypical, aspiration) and by age-group organisms.
2. Recognize the clinical manifestations (Box: fever, cough, tachypnea, crackles/rhonchi, dullness, retractions, nasal flaring, cyanosis, behavior and GI changes).
3. Describe the therapeutic plan (antibiotics, rest, fluids, antipyretics, oxygen) and the indications for hospitalization.
4. Provide supportive nursing care: respiratory assessment, oxygen, fluids, fever control, airway clearance with suctioning, positioning, and family education.

2. Placement Box

Ward: Medical ward — first of two medical topics. The guide reuses CH6 respiratory assessment and CH10 core ward skills (hand hygiene, standard/droplet precautions), CH7 fluid therapy, and CH8 medication safety.

4. Definition and Classification

- **Pneumonia** = **inflammation of the pulmonary parenchyma**, common in childhood, more frequent in infancy and early childhood; may occur as a primary disease or complication of another illness.
- The most useful classification is by **etiologic agent**: viral, bacterial, mycoplasmal (atypical), or aspiration. Agents are usually introduced through inhalation or from the bloodstream.
- **Pneumonitis** is a localized acute inflammation without the toxemia of lobar pneumonia.
- Common organisms by age:
- **Neonates**: group B streptococci, gram-negative enteric bacteria, CMV, *Ureaplasma urealyticum*, *Listeria*, *C. trachomatis*.
- **Infants and preschool**: RSV, parainfluenza, influenza, adenovirus, *S. pneumoniae*, *H. influenzae*, *M. pneumoniae*, group A streptococci.
- **School-age**: *M. pneumoniae*, *C. pneumoniae*, *M. tuberculosis*, group A streptococci, *S. aureus*, respiratory viruses.

5. Types

5.1 Viral Pneumonia

- Occur more frequently than bacterial pneumonias; often associated with viral URIs; **interstitial pneumonitis** with mucosal inflammation of bronchi/bronchioles.
- Onset acute or insidious: mild fever, slight cough, malaise → high fever, severe cough, fatigue; cough early unproductive or with small amounts of whitish sputum; radiography: **diffuse or patchy infiltration with peribronchial distribution**.
- Prognosis good, but viral infection predisposes to **secondary bacterial invasion** (denuded bronchial mucosa). Treatment is **symptomatic** — oxygen, chest percussion/postural drainage, antipyretics, fluid intake; **antibiotics only when bacterial infection is demonstrated**.

5.2 Primary Atypical Pneumonia

- Caused by pathogens other than the common bacteria; **M. pneumoniae** is the most common cause of community-acquired pneumonia in children 5 years or older; more prevalent in fall/winter and crowded conditions.
- Onset sudden/insidious with systemic symptoms: fever, chills (older children), headache, malaise, anorexia, myalgia; then rhinitis, sore throat, and **dry hacking cough** producing seromucoid sputum that becomes mucopurulent or blood-streaked; fever may last several days to 2 weeks; dyspnea infrequent.
- Radiography may show pneumonia before physical signs; recovery in 7–10 days with symptomatic treatment; incubation 2–3 weeks; cough may last weeks; hospitalization rarely needed; **erythromycin, azithromycin, clarithromycin** are primary agents.

5.3 Bacterial Pneumonia

- **S. pneumoniae** is the most common bacterial pathogen for community-acquired pneumonia in children and adults; others: group A streptococcus, *S. aureus*, *M. catarrhalis*, *M. pneumoniae*, *C. pneumoniae*.
- **Neonatal:** infection within 3–5 days of birth usually from maternal organisms (transplacental or aspirated amniotic fluid); onset abrupt and nonspecific — may include respiratory distress.
- Beyond the neonatal period, onset often **follows a viral infection**; in 3-month–5-year group: *S. pneumoniae*, *M. catarrhalis*, group A streptococci common; *H. influenzae* type B less common since Hib vaccine; *S. aureus* is rare but aggressively progressive.
- **The child appears ill:** fever, malaise, rapid shallow respirations, cough, chest pain (may be referred to the abdomen in young children); chills and meningism common; cough may persist weeks to months.

6. Clinical Manifestations

- **Fever:** usually high.
- **Respiratory:** cough (unproductive → productive whitish sputum); tachypnea; breath sounds crackles/rhonchi; dullness with percussion; chest pain; **retractions; nasal**

flaring; pallor to cyanosis (severity-dependent).

- **Chest radiography:** diffuse/patchy infiltration with peribronchial distribution.
 - **Behavior:** irritability, restlessness, lethargy.
 - **GI:** anorexia, vomiting, diarrhea, abdominal pain.
 - **Infants:** more severe symptoms; **cyanosis and apnea common**; abrupt fever, vomiting, diarrhea, abdominal distention; suspect bacterial pneumonia in **all neonates with respiratory symptoms** (high morbidity/mortality).
-

7. Diagnostic Evaluation

- Clinical history, age, health history, physical examination, **radiography**, and laboratory examination help identify the etiologic agent.
 - Radiograph: diffuse/patchy infiltration, peribronchial distribution; pleural effusion/empyema detection by radiograph; needle aspiration/thoracentesis if fluid is present.
-

8. Therapeutic Management

- Most older children are treated **at home** with: **antibiotic therapy, rest, liberal oral fluids, and antipyretics for fever**; chest percussion/postural drainage may be indicated (limited evidence in CAP).
- **Hospitalization indications:** pleural effusion or **empyema**; moderate/severe respiratory distress or deoxygenation; poor compliance; **infants younger than 6 months**; chronic illnesses (CHD, BPD); IV fluids for hydration; **oxygen if in respiratory distress**; some need parenteral antibiotics initially.
- **Complications:** empyema, **pyopneumothorax or tension pneumothorax** (especially staphylococcal pneumonia in infants), AOM and pleural effusion with pneumococcal pneumonia; pneumococcal vaccination is important prevention.
- **Pleural effusion/empyema management:** thoracentesis; continuous closed chest drainage with chest tube for 5–7 days; intrapleural fibrinolytics or video-assisted thoracoscopy; rare open debridement.

9. Nursing Care Plan

Assessment: 4-month-old, RSV-positive infant with pneumonia: RR 68/min, subcostal retractions, copious nasal secretions, refusing feeds, temperature 39 °C.

NANDA diagnoses:

- *Ineffective airway clearance* related to secretions.
- *Impaired gas exchange* related to inflammation and airway obstruction.
- *Deficient fluid volume / risk for aspiration* related to poor feeding and secretions.

Goals: Patent airway; O₂ sat within target; adequate hydration; temperature controlled; family empowered.

Interventions (with rationale):

1. **Frequency of assessment:** monitor respiratory rate and status, oxygenation, general disposition, and activity level frequently — early detection of deterioration.
2. **Oxygenation:** administer O₂ by nasal cannula, face mask, blow-by, or face tent; pulse oximetry as indicated.
3. **Positioning:** semierect position (child's comfort); lying on the affected side for unilateral pneumonia ("**good lung up**") splints the chest and reduces pleural rubbing.
4. **Airway clearance:** suction ineffectual secretions with bulb syringe/mechanical suction (especially in infants); postural drainage/percussion/nebulized bronchodilators if ordered.
5. **Fluids:** IV fluids during the acute phase; NG tube for hydration/antibiotics if oral intake poor; small frequent fluids (5–10 mL per medication syringe every 10 min) for infants; offer fluids even if solids are refused.
6. **Fever control:** cool environment + antipyretics as prescribed.
7. **Infection control:** hand hygiene, PPE; limit crowding; droplet precautions per policy.
8. **Family education:** observe for worsening symptoms; antibiotic/antipyretic administration; encourage fluids; involve the family to reduce child anxiety.
9. **Chest-tube care (if effusion):** monitor drainage, vacuum, kinks, intact dressing, water seal, tube below entry site; encourage movement; analgesia for discomfort.

10. Evaluate: RR improves, SpO₂ stable, secretions cleared, urine output adequate, fever down, parent reports concerns.

10. Procedure — Laboratory Step: Respiratory Care of the Child With Pneumonia (Checklist)

#	Step	Done
1	Baseline: RR, HR, BP, temperature, SpO ₂ ; note retractions, nasal flaring, grunting.	☐
2	Auscultate chest: note crackles, rhonchi, diminished breath sounds, dullness.	☐
3	Position semierect for comfort and ventilation; allow the child's position of comfort.	☐
4	Clear secretions: bulb syringe/mechanical suction for infants; encourage cough for older children.	☐
5	Give oxygen per order (nasal cannula, face mask, blow-by, face tent); monitor SpO ₂ .	☐
6	Administer antibiotics, antipyretics, and nebulized bronchodilator per order; verify dose by weight.	☐
7	Encourage oral fluids or maintain IV/NG hydration; monitor I & O.	☐
8	Apply hand hygiene + droplet precautions as policy; keep the environment calm.	☐
9	Teach the family: signs to report immediately, medication schedule, fluid intake.	☐
10	Document assessment, oxygen, fluids, and family education.	☐



Chest radiograph — pneumonic consolidation



11. Procedure — Hospital Step

- Admit using hospitalization criteria; obtain baseline vitals/SpO₂ and chest X-ray; start O₂ and IV/NG fluids per protocol.
- In empyema: prepare for thoracentesis/chest tube; provide closed-drainage monitoring and analgesia.
- Before discharge: complete family teaching (antibiotics, fluids, return criteria) and confirm the pneumococcal vaccination status.

12. Case Study (Infant With Pneumonia)

Scenario: A 5-month-old boy has fever 39.5 °C, RR 70/min with retractions and copious nasal discharge, poor feeding for two days, and a chest X-ray showing patchy peribronchial infiltration. WBC 4,000/mm³.

1. Why classify this as likely viral initially? (*Patchy peribronchial infiltration + normal/low WBC favors viral; RSV being the common infant pathogen; antibiotics reserved for demonstrated bacterial infection.*)
2. Why plan for fluids aggressively? (*Dehydration risk: poor intake, vomiting, increased losses with tachypnea/fever; IV/NG routes if oral intake poor.*)
3. Why suction before feeds? (*Nasal obstruction impairs breathing and feeding; suction mucus with bulb syringe before feedings and bedtime.*)
4. Which signs indicate deterioration? (*Increased work of breathing, apnea (infants), desaturation, cyanosis — requires O₂ and possible admission.*)
5. Why is this infant hospitalized? (*Younger than 6 months + respiratory distress → criteria met.*)

13. Chapter MCQs (10)

Q1. Pneumonia is defined as:

- a) Upper-airway inflammation
- b) Inflammation of the pulmonary parenchyma
- c) Bronchospasm only
- d) Pleural inflammation only

Q2. The most common bacterial pathogen in community-acquired pneumonia is:

- a) *S. pneumoniae*
- b) *M. catarrhalis* only
- c) *C. trachomatis*
- d) *Listeria monocytogenes*

Q3. The most common cause of atypical (community-acquired) pneumonia in children ≥ 5 years is:

- a) RFV
- b) *M. pneumoniae*
- c) Adenovirus
- d) CMV

Q4. In viral pneumonia, antibiotic therapy is:

- a) Always given
- b) Reserved for demonstrated bacterial infection
- c) Never used
- d) Replaced by steroids

Q5. A common early manifestation of pneumonia in infants is:

- a) Apnea and cyanosis
- b) Seizures
- c) Fractures
- d) Polyuria

Q6. A sign included in the general signs of pneumonia is:

- a) Bradycardia

- b) Nasal flaring
- c) Hypertension
- d) Hyperglycemia

Q7. Which infant should be hospitalized when treated for pneumonia?

- a) Age 5 months
- b) Age 8 years
- c) Any child with mild cough only
- d) Fully vaccinated child only

Q8. The primary drugs for atypical pneumonia are:

- a) Macrolides (erythromycin, azithromycin, clarithromycin)
- b) Aminoglycosides only
- c) Antifungals
- d) Sulfonamides only

Q9. "Good lung up" positioning is used to:

- a) Increase gastric emptying
- b) Splint the chest on the affected side and reduce pleural rubbing in unilateral pneumonia
- c) Reduce fevers
- d) Prevent dehydration

Q10. Antibiotic hiatus in infants: suspect bacterial pneumonia in:

- a) All neonates with respiratory symptoms
- b) Toddlers with runny nose only
- c) School-age children with mild cough
- d) Infants with normal X-ray only

Answers: Q1 b — Q2 a — Q3 b — Q4 b — Q5 a — Q6 b — Q7 a — Q8 a — Q9 b — Q10 a.

14. Key Points

- Pneumonia = inflammation of the pulmonary parenchyma; classify by organism (viral > bacterial; atypical/mycoplasmal; aspiration); organisms vary by age (neonates: GBS/enteric; infants/preschool: RSV/S. pneumoniae; school-age: M. pneumoniae).
- Signs: high fever, cough, tachypnea, crackles/rhonchi, dullness, retractions, nasal flaring, cyanosis (severity), irritability, GI symptoms; infants: apnea/cyanosis.
- Therapy: antibiotics (macrolides for atypical), rest, fluids, antipyretics; O₂ and support; hospitalize if effusion/empyema, distress/desaturation, age <6 months, poor compliance, chronic disease.
- Nursing: frequent respiratory assessment, O₂/fluids/fever control, suctioning, positioning (semierect; "good lung up"), infection control, chest-tube care for empyema, and thorough family education.

15. Normal Values / Reference Table

Parameter	Normal	In pneumonia
RR (infant)	30–60/min	Tachypnea >60/min
SpO ₂	95–100%	May be low → O ₂
WBC	5–15,000/mm ³	Viral: often normal/low; bacterial: may be elevated
Fever	<38 °C	Usually high
X-ray	Clear lung fields	Diffuse/patchy peribronchial infiltration

END OF CHAPTER 16

Pneumonia in Children

Next: **Chapter 17** — Acute Diarrhea, Vomiting, and Dehydration

PART IV - MEDICAL WARD

CHAPTER 17

Chapter 17 — Acute Diarrhea, Vomiting, and Dehydration

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. State the common causes of acute gastroenteritis (rotavirus is the leading cause of serious gastroenteritis) and its pathophysiology (secretory vs invasive diarrhea → dehydration, acidosis, shock).
2. Classify dehydration as mild (5–6%), moderate (7–9%), or severe (>9%) and identify the signs of each.
3. Apply the rehydration protocol: ORS 50 mL/kg (mild) and 100 mL/kg (moderate) within 4 hours; IV lactated Ringer 40 mL/kg for severe dehydration until pulse/consciousness improve.
4. Provide nursing care: small frequent ORS feedings even while vomiting, ongoing stool-loss replacement, I&O and daily weight monitoring, early reintroduction of diet, breastfeeding continuation, and prevention education (hand hygiene, rotavirus vaccine).

2. Placement Box

Ward: Medical ward — second medical topic. This chapter expands the fluid/electrolyte calculations from CH7 and applies the oral/IV rehydration protocol, monitoring, and communication skills from the foundation chapters.

4. Definition, Causes, and Transmission

- Most pathogens causing diarrhea spread by the **fecal-oral route** (contaminated food/water) or person-to-person close contact; risk factors: **lack of clean water, crowding, poor hygiene, nutritional deficiency, poor sanitation**. Infants are especially susceptible because their immune systems have not acquired protective antibodies.
- Worldwide, acute gastroenteritis is caused by viruses, bacteria, and parasites; **rotavirus is the most important cause of serious gastroenteritis in children**, spreads fecal-orally, and almost all children are infected at least once by age 5; it is the most common cause of diarrhea-associated hospitalization.
- Common bacterial pathogens: *Salmonella*, *Campylobacter*, *Yersinia*, *Shigella* (raw/undercooked food or contaminated water).
- **Antibiotic-associated diarrhea**: antibiotics alter intestinal flora; *C. difficile* is the most common bacterial overgrowth, usually acquired in health care facilities (20% community/daycare).

5. Pathophysiology

- Pathogens attach to mucosal cells; diarrhea is **secretory (noninvasive, toxin-producing, noninflammatory — the most common)** when an enterotoxin acts after mucosal attachment, or **systemic (invasive)** when the organism penetrates the mucosa.
- The most serious immediate disturbances of severe diarrheal disease: **dehydration, acid-base imbalance with acidosis, and shock** when dehydration seriously impairs circulation.

6. Clinical Manifestations — Assessment of Dehydration

Degree	Signs and Symptoms
Mild (5–6%)	Increased thirst; slightly dry buccal mucous membranes

Degree	Signs and Symptoms
Moderate (7–9%)	Loss of skin turgor; dry buccal mucous membranes; sunken eyes; sunken fontanel
Severe (>9%)	Signs of moderate dehydration plus one of: rapid thready pulse, cyanosis, rapid breathing, lethargy, coma

If no signs of dehydration are present, rehydration therapy is not necessary — proceed with maintenance therapy and replacement of stool losses.

Additional moderate–severe findings: **decreased tear production, sunken orbits, dry mucous membranes, listlessness/lethargy, skin tenting**; heart rate increases in moderate dehydration but blood pressure stays normal until it decreases in severe dehydration.

7. Diagnostic Evaluation

- Usually clinical (history of diarrhea/vomiting + examination); inspect **stool color/consistency**, auscultate **hyperactive bowel sounds** in gastroenteritis vs hypoactive in obstruction, palpate for tenderness.
- Identify pathogens when indicated (stool culture/PCR) especially for persistent, bloody, or daycare-associated diarrhea.

8. Therapeutic Management

8.1 Oral Rehydration Therapy (ORT) — the mainstay

- **ORS should contain 75 mmol/L sodium chloride and 13.5 g/L glucose** (standard products: Pedialyte, Infalyte); **tap water, milk, undiluted fruit juice, soup, and broth are NOT appropriate** for rehydration.
- **Mild dehydration: ORS 50 mL/kg within 4 hours.**
- **Moderate dehydration: ORS 100 mL/kg within 4 hours**; after reevaluation continue ORS if still dehydrated.

- **Vomiting is NOT a contraindication to ORS:** give ORS at frequent intervals in small amounts — 5–10 mL increments (spoon or small syringe) every 2–5 minutes for young children; ORS may also be given by NG or gastrostomy tube.
- **Replacement of ongoing stool losses:** ORS 10 mL/kg (infants) or 150–250 mL (older children) **for each diarrheal stool**; if stool volume is unknown, about 10 mL/kg is the guide.
- **Maintenance nutrition:** breastfeeding continues; regular formula if tolerated (lactose-free formula for suspected lactose intolerance); children on solids continue their usual diet; **early reintroduction of the normal diet lessens duration of illness and improves weight gain.**
- **Dietary cautions:** avoid high-carbohydrate fluids (fruit juice, Kool-Aid — low in electrolytes, decrease stool transit time); discourage milk/high-sugar fluids during the acute phase.

8.2 IV (Parenteral) Therapy

- IV fluids are for **severe dehydration, uncontrollable vomiting, inability to drink, or failure of ORT.**
- **Severe dehydration:** IV **lactated Ringer (or normal saline) 40 mL/kg until pulse and state of consciousness return to normal**, then ORS 50–100 mL/kg can be given. Initial bolus 20 mL/kg of NS or LR, then reassess hydration.
- Parenteral rehydration proceeds in phases: initial deficit replacement, maintenance, and replacement of ongoing losses.

9. Vomiting Management

- Vomiting is a **symptom of some underlying condition**; therapeutic management usually involves **slow oral rehydration** and possibly antiemetics.
- **Withhold oral feeding for 1–2 hours after emesis**, then start ORT; give the infant/child **0.5–2 oz (about 15–60 mL) of ORS every 15 minutes** depending on age and size; most retain small amounts; larger amounts as improvement occurs.
- If oral rehydration is impossible because of continued nausea/vomiting → **IV fluids**; antiemetics (e.g., ondansetron — preferred over promethazine for fewer side effects) may control vomiting.

- **Bilious vomiting** is the main symptom of bowel obstruction (e.g., malrotation with volvulus) — emergency.

10. Worked Scenario Examples (Calculations)

Example 1 — Moderate dehydration, ORS volume : A 10-kg toddler has moderate dehydration (7–9%).

- ORS needed in 4 h = $100 \text{ mL/kg} \times 10 \text{ kg} = \mathbf{1000 \text{ mL over 4 hours}}$.
- Ongoing stool-replacement = $10 \text{ mL/kg per stool} = \mathbf{100 \text{ mL ORS per diarrheal stool}}$.

Example 2 — Mild dehydration, ORS volume: A 6-kg infant with mild dehydration.

- ORS in 4 h = $50 \text{ mL/kg} \times 6 \text{ kg} = \mathbf{300 \text{ mL over 4 hours}}$.
- Give 5–10 mL every 2–5 min by syringe to minimize vomiting.

Example 3 — Severe dehydration, IV bolus: A 12-kg child with severe dehydration (rapid thready pulse, lethargy).

- IV **lactated Ringer 40 mL/kg** = $12 \times 40 = \mathbf{480 \text{ mL}}$ until pulse and consciousness normalize, then ORS 50–100 mL/kg.

Example 4 — Maintenance (review from CH7): Holliday–Segar: for 10 kg = 1000 mL/day + ongoing stool losses each replaced 1:1 with ORS (10 mL/kg per stool).

11. Nursing Care Plan

Assessment: 8-month-old, 3 days of watery stools (10/day), vomiting after feeds, decreased tears, dry mucous membranes, sunken fontanel, skin turgor tents; temperature 38.5 °C; input very low.

NANDA diagnoses:

- *Deficient fluid volume* related to diarrhea and vomiting.
- *Imbalanced nutrition: less than body requirements* related to poor intake and losses.
- *Risk for impaired skin integrity* (diaper area).

Goals: Rehydrated within 4 h; urine output adequate; no worsening; skin intact; parent competent in ORT.

Interventions (with rationale):

- 1. **Give ORS 100 mL/kg over 4 h** in 5–10 mL increments by spoon/syringe every 2–5 min; continue despite vomiting — effective rehydration even with emesis.
- 2. **Replace each diarrheal stool** with ORS 10 mL/kg — prevents recurrent deficit.
- 3. **Assess hydration and daily weight;** strict I & O; assess skin turgor, oral mucosa, tears, fontanel **every 4–8 h**; daily weight is one of the best indicators of fluid status — early detection of failure.
- 4. **Monitor urine output** (0.5–1 mL/kg/h adequate) — end-organ perfusion gauge.
- 5. **Continue breastfeeding;** resume formula/usual diet early after rehydration — nutrition, lessens illness duration.
- 6. **Skin care:** barrier ointment in the diaper area; frequent diaper changes — prevents excoriation.
- 7. **IV fluids** if severe dehydration, uncontrollable vomiting, or failed ORT; initial bolus 20 mL/kg NS/LR then reassess — restores circulation.
- 8. **Prevention education:** hand hygiene; rotavirus vaccination (first dose by 15 weeks of age); avoid sharing cups/utensils — breaks the fecal-oral cycle.
- 9. Evaluate: 4-h weight gain, urine output adequate, tolerates feeding, parent demonstrates ORS technique.

12. Procedure — Laboratory Step: Oral Rehydration Demonstration (Checklist)

#	Step	Done
1	Check the ORS packet/oral mix product against the recommended composition (Na 75 mmol/L, glucose 13.5 g/L).	☐
2	Prepare/measure the ORS fresh; never dilute with water/juice; discard unused after the stated time.	☐
3	Assess hydration (skin turgor, mucous membranes, eyes/fontanel, tears, pulse) and grade dehydration.	☐

#	Step	Done
4	Calculate the 4-hour volume (50 or 100 mL/kg) and the per-stool replacement (10 mL/kg).	☐
5	Offer ORS 5–10 mL by spoon/small syringe every 2–5 min (0.5–2 oz every 15 min for vomiting child).	☐
6	Continue giving ORS despite vomiting (vomiting is not a contraindication).	☐
7	Continue breastfeeding/formula; reintroduce the normal diet early; avoid juice and high-sugar drinks.	☐
8	Record intake/output and daily weight; observe for worsening (lethargy, rapid pulse, cyanosis).	☐
9	Apply diaper-area skin care; teach hand hygiene and rotavirus-vaccine prevention.	☐



Signs of dehydration (sunken fontanel, dry mucosa, tenting) — diagnostic illustration



Signs of dehydration (sunken fontanel, dry mucosa, tenting) — diagnostic illustration

13. Procedure — Hospital Step

- On admission, grade dehydration, weigh the child, initiate ORT (or IV LR 40 mL/kg if severe), and start strict I&O with stool-loss replacement.
- Monitor weight trends and reassess hydration every 4–8 h; escalate to IV when ORT fails.
- Teach the mother ORS preparation/dosing, early feeding, signs to return, and prevention (hand hygiene + rotavirus vaccine).

14. Case Study (Severe Dehydration)

Scenario: A 2-year-old (12 kg) has 2 days of profuse watery diarrhea and frequent vomiting. He is listless, with a rapid thready pulse, deep rapid breathing, dry mouth, and tenting skin;

no urine output in 8 hours.

1. How do you grade the dehydration? (*Severe (>9%): moderate signs + rapid thready pulse and rapid breathing/lethargy.*)
2. What is the first treatment? (*IV lactated Ringer 40 mL/kg = 480 mL until pulse and consciousness improve; then ORS; initial bolus 20 mL/kg then reassess.*)
3. Why is acidosis present? (*Severe diarrheal disease → dehydration plus acid-base imbalance with acidosis from bicarbonate loss; shock when circulation is impaired.*)
4. After stabilization, what fluids? (*ORS 50–100 mL/kg, 5–10 mL increments; replace each stool loss 10 mL/kg; keep breastfeeding.*)
5. Which vaccine prevents the leading viral cause? (*Rotavirus vaccine — rotavirus is the most important cause of serious gastroenteritis.*)

15. Chapter MCQs (10)

Q1. The most important cause of serious gastroenteritis in children is:

- a) *Salmonella*
- b) Rotavirus
- c) Giardia
- d) *Yersinia*

Q2. Most diarrhea pathogens are spread by:

- a) The airborne route
- b) The fecal-oral route
- c) Skin contact only
- d) Blood transfusion

Q3. Moderate dehydration is best defined as:

- a) 5–6% with slight thirst
- b) 7–9% with skin turgor loss, sunken eyes/fontanel
- c) >9% with shock
- d) No clinical signs

Q4. ORS for moderate dehydration is given over 4 hours as:

- a) 100 mL/kg
- b) 50 mL/kg
- c) 10 mL/kg
- d) 200 mL/kg

Q5. Severe dehydration (>9%) first requires:

- a) IV lactated Ringer 40 mL/kg until pulse and consciousness normalize
- b) Immediate solid food
- c) 20 mL ORS only
- d) Antibiotics only

Q6. A child who is vomiting should:

- a) Never receive ORS
- b) Receive ORS in small frequent amounts (vomiting is not a contraindication)
- c) Only drink juice
- d) Be given undiluted milk

Q7. Ongoing losses for each diarrheal stool are replaced with ORS:

- a) 10 mL/kg
- b) 50 mL/kg
- c) 100 mL/kg
- d) 1 mL/kg

Q8. Standard ORS composition is:

- a) 75 mmol/L sodium + 13.5 g/L glucose
- b) No sodium
- c) 150 mmol/L sodium only
- d) Fruit juice base

Q9. Which fluid is NOT appropriate for rehydration?

- a) Undiluted fruit juice
- b) ORS
- c) Pedialyte
- d) Breast milk (as maintenance)

Q10. The most common bacterial overgrowth following antibiotics is:

- a) *Clostridium difficile*
- b) Rotavirus
- c) *Giardia lamblia*
- d) *Shigella*

Answers: Q1 b — Q2 b — Q3 b — Q4 a — Q5 a — Q6 b — Q7 a — Q8 a — Q9 a — Q10 a.

16. Key Points

- Acute gastroenteritis: viruses/bacteria/parasites, fecal-oral spread; **rotavirus = leading viral cause**; secretory diarrhea most common → **dehydration, acidosis, shock**.
- Grading: mild 5–6% (thirst), moderate 7–9% (turgor loss, sunken eyes/fontanel), severe >9% (thready pulse, cyanosis, rapid breathing, lethargy/coma).
- Rehydration: ORS 50 mL/kg (mild) or 100 mL/kg (moderate) over 4 h in 5–10 mL increments; replace stools 10 mL/kg; severe → IV LR 40 mL/kg then ORS; ORS composition Na 75 mmol/L, glucose 13.5 g/L.
- Vomiting management: 1–2 h nothing by mouth then slow ORS; antiemetics (ondansetron preferred); IV if ORT fails.
- Nursing: I&O, daily weight, urine output, skin care, early feeds/breastfeeding, prevention (hand hygiene, rotavirus vaccine).
- **Never give tap water, milk, juice, soup, or broth for rehydration.**

17. Normal Values / Reference Table

Parameter	Normal	Alteration in dehydration
Urine output	0.5–1 mL/kg/h	Decreased → anuria in severe
Pulse	Age-based	Increases (moderate), becomes thready (severe)
Blood pressure	Age-based	Normal until severe, then decreases

Parameter	Normal	Alteration in dehydration
Skin turgor	Instant recoil	Tenting (moderate–severe)
Weight	Baseline	Daily weight tracks deficit
Tears	Present	Decreased (moderate–severe)

END OF CHAPTER 17

Acute Diarrhea, Vomiting, and Dehydration

Next: **Chapter 18** — Hirschsprung Disease (Congenital Aganglionic Megacolon)

PART V - SURGICAL WARD

CHAPTER 18

Chapter 18 — Hirschsprung Disease (Congenital Aganglionic Megacolon)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part V — Surgical Ward, Chapter 18

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain the pathophysiology of Hirschsprung disease (HD): absence of ganglion cells in the myenteric/submucosal plexuses → lack of intestinal relaxation and peristalsis.
2. Recognize the clinical manifestations in the newborn (delayed meconium, bilious vomiting, distention), infancy (FTT, constipation, enterocolitis), and childhood (ribbonlike stools, palpable fecal mass).
3. Describe the diagnostic work-up (contrast enema with transition zone, **rectal biopsy for absence of ganglion cells**, anorectal manometry) and the surgical pull-through procedures (Swenson, Duhamel, Soave).

4. Provide nursing care: preoperative stabilization (enemas, fluids/electrolytes, antibiotics, colonic irrigation), **measurement of abdominal circumference**, monitoring for enterocolitis and bowel perforation, postoperative care, and home teaching (anal dilations, colostomy care).

2. Placement Box

Ward: Surgical ward — first surgical topic. This chapter applies the surgical-preparation and procedure skills (CH20) plus core ward skills (CH10), fluid monitoring (CH7), and post-op pain/nutrition care.

4. Definition and Epidemiology

- **Hirschsprung disease (HD)** is a congenital anomaly resulting in **mechanical obstruction from inadequate motility of part of the intestine**. It accounts for about **one-fourth of all neonatal intestinal obstructions**; incidence **1 in 5,000 live births**, four times more common in males, small familial pattern (**RET proto-oncogene mutations**).

5. Pathophysiology

- The pathology relates to the **absence of ganglion cells** in the affected intestine (in the **myenteric plexus of Auerbach and submucosal plexus of Meissner**), with loss of the rectosphincteric reflex.
- Absence of enteric nervous system stimulation **decreases the internal sphincter's ability to relax**; unopposed sympathetic stimulation increases intestinal tone; peristalsis is lost and the internal sphincter does not relax when a stool bolus enters the rectum.
- In about 80% of cases the aganglionic segment is confined to the **rectum and part of the distal colon (short-segment disease)**; the whole colon or part of the small

intestine can be involved (**long-segment**, ~20%); **total intestinal aganglionosis** in ~5%.

6. Clinical Manifestations

Newborn period:

- **Failure to pass meconium within 24–48 hours after birth**; refusal to feed; **bilious vomiting**; abdominal distention.

Infancy:

- Failure to thrive; constipation; abdominal distention; episodes of diarrhea and vomiting.
- **Signs of enterocolitis: explosive, watery diarrhea; fever; appears significantly ill.**

Childhood:

- Constipation; **ribbonlike, foul-smelling stools**; abdominal distention; **visible peristalsis**; easily palpable fecal mass; undernourished, anemic appearance.

Note: 99% of term infants pass meconium within the first 48 hours, whereas few infants with HD do so.

7. Diagnostic Evaluation

- Suspected in the neonate from signs of obstruction or failure to pass meconium; in infants/children, the history of a chronic pattern of constipation; on examination the **rectum is empty of feces, the internal sphincter is tight**, and leakage of liquid stool/gas may occur in short-segment disease.
- **Contrast enema** often demonstrates the **transition zone between the dilated proximal colon (megacolon) and the aganglionic distal segment** (may not develop until age 2 months).
- **Rectal biopsy** (surgical full-thickness or **suction biopsy**) confirms the diagnosis by demonstrating the **absence of ganglion cells**.

- **Anorectal manometry** records the sphincter reflex response to balloon distention: in HD the internal sphincter **fails to relax** while the external sphincter contracts normally.
-

8. Therapeutic Management

- The majority require **surgery**; one of three procedures: the **Swenson** (pull-through), the **Duhamel** procedure, and the **Soave** (endorectal pull-through).
 - Surgery is performed after the child is stabilized with **fluid and electrolyte replacement** and **colonic cleansing with enemas**; the transanal **Soave endorectal pull-through** pulls normal bowel through the rectal muscular sleeve from which aganglionic mucosa has been removed; earlier diagnosis allows a **primary one-stage procedure** without a temporary colostomy.
 - After pull-through, most children achieve fecal continence; possible complications: **anal stricture, recurrent enterocolitis, prolapse, perianal abscess, incontinence** — needing dilations or bowel-retraining therapy.
-

9. Nursing Care Plan

Assessment: 3-day-old male, failed to pass meconium, bilious vomiting, distended abdomen, refusing feeds, temperature unstable.

NANDA diagnoses:

- *Impaired elimination / risk for constipation* related to aganglionosis.
- *Risk for deficient fluid volume* related to vomiting and withheld feedings.
- *Risk for infection (enterocolitis, perforation)* after surgery.
- *Deficient knowledge* (parents) related to home care.

Goals: Meconium/stool passage achieved or colostomy functioning; no enterocolitis; abdomen decompressed; parents competent in home care.

Interventions (with rationale):

1. **Preoperative stabilization:** monitor vitals/BP for signs of shock; monitor fluid/electrolyte replacement and plasma/blood derivatives; give enemas, oral/systemic antibiotics, and colonic irrigations per protocol — prevents toxicity and prepares the bowel.
2. **Observe for perforation:** fever, increasing abdominal distention, vomiting, increased tenderness, irritability, dyspnea, cyanosis — early detection.
3. **Measure abdominal circumference** serially at a marked point (e.g., umbilicus/widest part) with paper tape; leave the tape in place for the acutely ill child — progressive distention is serious.
4. **Feeding/fluids:** NPO with IV fluids before surgery; resume feeds postoperatively as ordered and involve parents in feedings.
5. **Postoperative care** as for any abdominal surgery: wound observation, analgesia, early ambulation per order, inspect for wound infection and irregular passage of stool.
6. **Teach parents:** signs of enterocolitis, fecal incontinence, and obstruction; **daily anal dilations** to avoid anastomotic strictures (parents often perform at home with encouragement and detailed instructions); **colostomy care** with meticulous skin care when a diverting colostomy is used.
7. **Support the parents** adjusting to a congenital defect; foster infant–parent bonding.
8. Evaluate: stable vitals, decreasing abdominal girth, stool/colostomy output present, no enterocolitis; parents demonstrate dilation/colostomy care.

10. Procedure — Laboratory Step: Abdominal Circumference + Preoperative Bowel Prep (Checklist)

#	Step	Done
1	Weigh the child and record baseline vital signs.	☐
2	Mark the measurement level (umbilicus or widest part) with a pen.	☐
3	Measure abdominal circumference with paper tape at the marked level with each vital-sign check; record in series.	☐
4	Keep the tape in place beneath the child if frequent measurements are needed.	☐

#	Step	Done
5	Monitor I & O, IV fluids, and electrolytes; report signs of shock (thready pulse, hypotension).	☐
6	Give enemas and colonic irrigations with antibiotic solution as ordered; record effects.	☐
7	Observe for enterocolitis (explosive watery diarrhea, fever) and perforation signs; report immediately.	☐
8	Teach parents the disease, the surgical plan, and home care (dilations/colostomy, wound signs).	☐



Hirschsprung disease — transition zone and aganglionic segment



Hirschsprung disease — transition zone and aganglionic segment

11. Procedure — Hospital Step

- On admission: NPO, IV fluids, nasogastric decompression, serial abdominal girth, and baseline labs (CBC, electrolytes, cross-match).
- Perform the contrast enema and prepare for suction rectal biopsy; obtain consent for the pull-through or one-stage procedure.
- Postoperation: monitor vital signs, wound, and stool/colostomy output; begin parents' teaching before discharge (dilations, enterocolitis warning signs, perianal skin care).

12. Case Study (Neonatal HD)

Scenario: A 2-day-old term boy has not passed meconium, has a distended abdomen, and vomits bilious-stained fluid. Rectal exam shows an empty, tight rectum. Contrast enema

shows a transition zone.

1. Why is the history of meconium delay significant? *(99% of term infants pass meconium within 48 h; delay is a hallmark of HD.)*
 2. What confirms the diagnosis? *(Rectal biopsy showing absence of ganglion cells; manometry shows failure of internal sphincter relaxation.)*
 3. Which complication is the most serious? *(Enterocolitis — explosive watery diarrhea, fever, ill appearance — the most serious complication of HD.)*
 4. Why is he NPO with IV fluids and enemas preoperatively? *(Stabilize fluids/electrolytes and cleanse the colon before pull-through surgery.)*
 5. What must the parents be taught for home? *(Anal dilations to prevent stricture, enterocolitis warning signs, and wound/stool observation; colostomy care if performed.)*
-

13. Chapter MCQs (10)

Q1. The primary defect in Hirschsprung disease is:

- a) Absence of ganglion cells (aganglionosis)
- b) Excessive villi
- c) Pancreatic insufficiency
- d) Biliary atresia

Q2. HD accounts for approximately what fraction of neonatal intestinal obstruction?

- a) 1/4
- b) 3/4
- c) 1/10
- d) 1/100

Q3. A classic newborn sign of HD is:

- a) Passage of meconium at birth
- b) Failure to pass meconium within 24–48 hours
- c) Excessive jaundice only
- d) Urinary retention

Q4. The most serious complication of HD is:

- a) Enterocolitis
- b) Otitis media
- c) Tracheitis
- d) Pericarditis

Q5. The definitive diagnostic test is:

- a) Rectal biopsy (absence of ganglion cells)
- b) CBC
- c) Urine catecholamines
- d) Chest X-ray

Q6. In anorectal manometry, HD demonstrates:

- a) Normal internal sphincter relaxation
- b) Internal sphincter fails to relax
- c) Absent external sphincter
- d) Hyperactive relaxation

Q7. Standard surgical treatment is:

- a) Pull-through procedures (Swenson, Duhamel, Soave)
- b) Appendectomy
- c) Splenectomy
- d) Heart repair

Q8. Preoperative bowel preparation includes:

- a) Enemas + antibiotics/colonic irrigation
- b) Salt-free diet only
- c) Immediate solid feeds
- d) Chest physiotherapy

Q9. Progressive abdominal distention after diagnosis of HD is managed by:

- a) Serial abdominal girth measurements at a marked site
- b) Not measuring
- c) Only drinking juice
- d) Waiting for weight gain

Q10. Common childhood presentation of undiagnosed HD includes:

- a) Ribbonlike foul-smelling stools and palpable fecal mass
- b) Watery stool at birth always
- c) Polyuria
- d) Short stature only

Answers: Q1 a — Q2 a — Q3 b — Q4 a — Q5 a — Q6 b — Q7 a — Q8 a — Q9 a — Q10 a.

14. Key Points

- HD = congenital aganglionosis (Auerbach/Meissner plexuses) → loss of rectosphincteric relaxation and peristalsis → functional obstruction; 1:5000, M > F, RET gene.
- Newborn: delayed meconium (>24–48 h), bilious vomiting, distention; infancy: FTT, constipation, enterocolitis; childhood: ribbonlike stools, palpable mass.
- Diagnosis: contrast enema (transition zone), **rectal biopsy** (definitive), manometry.
- Treatment: stabilization (fluids/electrolytes, enemas, antibiotics) then **pull-through surgery** (Swenson/Duhamel/Soave).
- Nursing: monitor for shock/perforation/enterocolitis, serial abdominal girth (marked level), post-op care, parent teaching (anal dilations, colostomy/skin care).

15. Normal Values / Reference Table

Parameter	Normal	In HD
Meconium passage	Within first 24–48 h	Delayed/failure
Rectal biopsy	Ganglion cells present	Ganglion cells absent definitive
Anorectal manometry	Internal sphincter relaxes with distention	Internal sphincter fails to relax

Parameter	Normal	In HD
Contrast enema	No transition zone	Transition zone (megacolon → narrow segment)
Incidence	—	1 in 5,000 live births

END OF CHAPTER 18

Hirschsprung Disease (Congenital Aganglionic Megacolon)

Next: **Chapter 19** — Hypertrophic Pyloric Stenosis (HPS)

PART V - SURGICAL WARD

CHAPTER 19

Chapter 19 — Hypertrophic Pyloric Stenosis (HPS)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part V — Surgical Ward, Chapter 19

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Explain the pathology of HPS: circular pyloric muscle hypertrophy → elongation and narrowing of the pyloric canal → gastric outlet obstruction.
2. Identify the classic presentation: **nonbilious projectile vomiting beginning at 3–6 weeks**, an avaricious hungry infant, and the **olive-shaped mass** in the epigastrium.
3. Interpret the metabolic picture: hypochloremia, hypokalemia, hyponatremia, **metabolic alkalosis**, elevated BUN.
4. Describe the management: preoperative rehydration/electrolyte correction (NPO, IV fluids, NG decompression), **pyloromyotomy (Fredet–Ramstedt)**, and staged postoperative feeding.

5. Provide nursing care: observation, fluid therapy and I&O monitoring, NG tube patency, parental reassurance, and post-op feeding/wound care.

2. Placement Box

Ward: Surgical ward — second surgical topic and the final ministerial chapter. HPS brilliantly ties together vomiting management (CH17) and fluid/electrolyte monitoring (CH7) with surgical care (CH20).

4. Definition and Epidemiology

- **Hypertrophic pyloric stenosis (HPS)** occurs when the **circumferential muscle of the pyloric sphincter thickens**, elongating and narrowing the pyloric canal → **gastric outlet obstruction** with compensatory dilation, hypertrophy, and hyperperistalsis of the stomach.
- Develops in the first few weeks of life, causing **nonbilious vomiting after feeding; projectile vomiting** may develop, and the infant is fussy and hungry after vomiting; untreated → **dehydration, metabolic alkalosis, and failure to thrive**.
- **Boys are affected 4–6 times more often than girls**; more common in white infants; incidence **2–3.5 per 1,000 live births**, with 30–40% first-born infants; etiology multifactorial.
- **HPS is not a congenital disorder**; local innervation may be involved; usually an isolated lesion, but may associate with malrotation, esophageal/duodenal atresia, and anorectal anomalies.

5. Pathophysiology

- The circular pyloric muscle thickens → severe narrowing of the pyloric canal between stomach and duodenum; the lumen is partially obstructed, and over time **inflammation and edema further reduce the opening → complete obstruction**.

- The hypertrophied pylorus is **palpable as an olive-like mass** in the upper abdomen.
-

6. Clinical Manifestations

- **Projectile vomiting:** may be ejected 3–4 feet when side-lying, 1 foot or more when back-lying; usually shortly after a feeding; may follow each feeding or appear intermittently; **nonbilious vomitus that may be blood tinged.**
 - **Infant hungry, avid nurser** — eagerly accepts a second feeding after vomiting; no evidence of pain except chronic hunger; **poor weight gain**; signs of dehydration.
 - Distended upper abdomen; **readily palpable olive-shaped tumor in the epigastrium just to the right of the umbilicus; visible gastric peristaltic waves moving left to right across the epigastrium.**
 - Vomiting usually begins at **4 weeks of age** (range 1 week to 5 months), occurring 10–30 minutes after feeding and becoming projectile as obstruction progresses.
 - Mild **indirect hyperbilirubinemia** (decreased glucuronyl transferase) may occur and resolves with surgical correction.
-

7. Diagnostic Evaluation

- Diagnosis is often made after history and physical examination; the **olive is most easily palpated when the stomach is empty, the infant is quiet, and abdominal muscles relaxed.**
 - If inconclusive: **ultrasonography** demonstrates an elongated mass surrounding a long pyloric canal; if US is negative, **upper GI radiography** rules out other causes.
 - A pacifier or nipple dipped in glucose water may soothe the irritable infant enough to complete the abdominal examination.
 - **Laboratory findings (dehydration):** decreased serum **sodium and potassium** (may be masked by hemoconcentration); more diagnostically useful — **decrease in serum chloride and increases in pH and bicarbonate (metabolic alkalosis); elevated BUN** from dehydration.
-

8. Therapeutic Management

- **Surgical relief by pyloromyotomy is the standard therapy.** Preoperatively the infant must be **rehydrated and the metabolic alkalosis corrected with parenteral fluids and electrolytes**; replacement therapy usually delays surgery 24–48 hours; **stomach decompressed with an NG tube** if vomiting continues; infants without imbalance go to surgery without delay.
- Surgery (often laparoscopic): **longitudinal incision through the circular muscle fibers down to, but not including, the submucosa — pyloromyotomy (Fredet–Ramstedt operation)**; high success rate.
- **Feedings:** usually begun **4–6 hours postoperatively** with small, frequent feedings of water or electrolyte solution; if retained, formula in small increments about 24 hours after surgery; gradually increase until full schedule (~48 hours).
- **Prognosis:** excellent with early diagnosis; mortality 0–0.5%; small percentage develop GERD.

9. Nursing Care Plan

Assessment: 5-week-old male, 1-week history of projectile nonbilious vomiting after feeds, hungry between episodes, weight falling (birth 3.6 kg → now 3.4 kg), signs of dehydration.

NANDA diagnoses:

- *Deficient fluid volume* related to vomiting.
- *Imbalanced nutrition: less than body requirements* related to gastric outlet obstruction.
- *Risk for aspiration* / ineffective airway related to projectile vomiting.
- *Deficient knowledge* (parents) related to the disorder and post-op care.

Goals: Rehydrated and alkalosis corrected preoperatively; successful pyloromyotomy; staged feeds tolerated; parents confident and reassured.

Interventions (with rationale):

1. **Preoperative care — restore hydration/electrolytes:** NPO, IV fluids with glucose and electrolytes based on serum values; careful monitoring of IV fluids and **strict I &**

- O**; accurate description of vomiting and stools — corrects alkalosis before surgery.
2. **Monitor electrolytes:** vitals especially for fluid/electrolyte imbalance; metabolic alkalosis from loss of H^+ and depletion of K, Na, Cl in gastric secretions; assess skin/mucous membranes.
 3. **NG decompression:** keep the tube patent and functioning; measure/record type and amount of drainage if used preoperatively.
 4. **Assist in diagnosis:** observe feeding behavior, peristaltic waves, and palpation of the olive (empty stomach, quiet, glucose-water pacifier).
 5. **Reassure parents:** the condition is a structural problem, not poor parenting; encourage visits and involvement in care — reduces anxiety.
 6. **Postoperative feeding:** start 12–24 h (water/electrolytes 4–6 h per protocol), small frequent amounts; give expressed breast milk or limit nursing time with gradual increase; record feedings and responses.
 7. **Postoperative observation:** expect some vomiting in the first 24–48 h; monitor vitals, IV fluids, I&O; give around-the-clock analgesics (pain is continuous); inspect incisions for drainage/erythema and report infection signs.
 8. **Discharge teaching:** incision/dressing care, feeding schedule and expectations, when to call the surgeon.
 9. Evaluate: urine output adequate, alkalosis corrected, feedings retained without projectile vomiting, parents verbalize confidence.

10. Procedure — Laboratory Step: Preoperative Monitoring + Post-op Feeding (Checklist)

#	Step	Done
1	Weigh daily and record baseline vital signs.	☐
2	Maintain NPO with IV glucose/electrolyte fluids; document IV rate and site.	☐
3	Monitor electrolytes (especially chloride, potassium, pH/bicarbonate) and BUN per protocol.	☐
4	Maintain NG tube patency; measure and chart gastric drainage.	☐

#	Step	Done
5	Record all vomiting episodes (time, amount, projectile/nonbilious); count stools.	☐
6	Palpate the olive when the stomach is empty; use the glucose-water pacifier if needed.	☐
7	Post-op: begin small frequent water/electrolyte feeds, then formula/breast milk per schedule; record tolerance.	☐
8	Administer around-the-clock analgesia; inspect the incision for drainage/erythema.	☐
9	Teach parents home care (feedings, incision, when to call).	☐



Barium swallow/gastric outlet show and pyloric canal narrowing



Barium swallow/gastric outlet show and pyloric canal narrowing

11. Procedure — Hospital Step

- On admission: NPO, IV access, baseline electrolytes/BUN, weight, and abdominal exam; NG decompression if vomiting.
- Confirm with ultrasound (or upper GI series); obtain surgical consent; correct alkalosis before pyloromyotomy.
- Postoperation: staged feedings, analgesics, wound observation; discharge with feeding plan and incision care instructions.

12. Case Study (Projectile Vomiting)

Scenario: A 4-week-old boy began projectile, nonbilious vomiting 10–30 minutes after feeds, is always hungry, and now has a palpable olive felt just right of the umbilicus. Serum chloride 88 mEq/L, bicarbonate 34 mEq/L, K 3.2 mEq/L, BUN 24.

1. Which diagnosis fits? (*HPS — classic age, projectile nonbilious vomiting, hungry infant, olive mass.*)
 2. Interpret the labs. (*Hypochloremic, hypokalemic metabolic alkalosis with elevated BUN from vomiting.*)
 3. Why is surgery delayed? (*To rehydrate and correct the alkalosis with parenteral fluids/electrolytes — ~24–48 h.*)
 4. What operation is planned? (*Pyloromyotomy — Fredet–Ramstedt; longitudinal division of the circular muscle to the submucosa; laparoscopic approach possible.*)
 5. How is feeding resumed? (*Small frequent water/electrolyte feeds 4–6 h post-op (12–24 h per protocol), then formula/breast milk in small increments, increased over ~48 h.*)
-

13. Chapter MCQs (10)

Q1. HPS is caused by:

- a) Hypertrophy of the circular pyloric muscle → narrowed pyloric canal
- b) Absence of ganglion cells
- c) Malrotation of the bowel
- d) Esophageal atresia

Q2. The classic vomiting pattern is:

- a) Nonbilious, projectile, soon after feeding
- b) Bilious and gradual
- c) Only at night
- d) Bloody from birth

Q3. The typical age of onset is:

- a) 3–6 weeks (usually 4 weeks)
- b) Birth
- c) 6 months
- d) 2 years

Q4. The classic abdominal finding is:

- a) An olive-shaped mass in the epigastrium just right of the umbilicus
- b) A mass in the left lower quadrant
- c) No abdominal finding
- d) Distended groin

Q5. The characteristic metabolic disturbance is:

- a) Metabolic alkalosis with hypochloremia and hypokalemia
- b) Metabolic acidosis with hyperkalemia
- c) Respiratory acidosis
- d) Hypernatremia alone

Q6. Definitive treatment for HPS is:

- a) Pyloromyotomy (Fredet–Ramstedt operation)
- b) Antibiotics
- c) Iron therapy
- d) Phototherapy

Q7. Before surgery the infant should be:

- a) Rehydrated and the alkalosis corrected, with NG decompression if vomiting
- b) Fed immediately
- c) Given oral ORS only
- d) Taken directly to surgery regardless

Q8. Postoperative vomiting in the first 24–48 hours is:

- a) Common and expected
- b) Always a failed surgery
- c) A contraindication to feeds
- d) Never seen

Q9. Which maneuver helps examine an irritable infant?

- a) Pacifier or nipple dipped in glucose water
- b) Forceful palpation
- c) Feeding a full bottle
- d) Cold compress on the abdomen

Q10. HPS is:

- a) Not a congenital disorder (acquired postnatal)
- b) Always present at birth
- c) Caused by vaccination
- d) X-linked lethal

Answers: Q1 a — Q2 a — Q3 a — Q4 a — Q5 a — Q6 a — Q7 a — Q8 a — Q9 a — Q10 a.

14. Key Points

- HPS = pyloric circular muscle hypertrophy → pyloric canal narrowing → gastric outlet obstruction; boys 4–6× girls; 2–3.5/1,000.
- Classic: nonbilious **projectile vomiting** (3–6 weeks), hungry avid infant, olive mass in epigastrium, visible peristaltic waves; untreated → dehydration, **hypochloremic hypokalemic metabolic alkalosis**, FTT.
- Diagnosis: history + physical (empty stomach, glucose-water pacifier), **ultrasound**, upper GI series; labs: ↓Cl, ↓K, ↑pH/CO₂, ↑BUN.
- Therapy: rehydrate/correct alkalosis (NPO, IV, NG) → **pyloromyotomy**; feedings start small and frequent 4–6 h post-op; prognosis excellent.
- Nursing: strict I&O, electrolyte monitoring, NG patency, post-op analgesics/feeding/wound observation, parental reassurance (not a parenting failure).

15. Normal Values / Reference Table

Parameter	Normal	In HPS
Serum chloride	~98–106 mEq/L	Decreased → hypochloremia
pH / bicarbonate	7.35–7.45 / 22–28 mEq/L	Increased → metabolic alkalosis
Potassium	~3.5–5/ mEq/L	Decreased → hypokalemia
BUN	~5–18 mg/dL	Elevated (dehydration)

Parameter	Normal	In HPS
Birth incidence	—	2–3.5/1,000 live births
Feeding onset	—	4–6 h post-op (water/electrolytes)

END OF CHAPTER 19

Hypertrophic Pyloric Stenosis (HPS)

Next: **Chapter 20** — Core Clinical Procedures (Diagnosis, Prevention, and Treatment Skills)

PART VI - CORE CLINICAL PROCEDURES

CHAPTER 20

Chapter 20 — Core Clinical Procedures (Diagnosis, Prevention, and Treatment Skills)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part VI — Core Clinical Procedures, Chapter 20

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Perform the **15 core ward procedures** listed below safely, in the correct order, and with the correct equipment for the child's age.
2. Apply the principles of **standard and transmission-based precautions** to every procedure.
3. **Calculate** pediatric drug doses and IV infusion/ORS volumes correctly and recheck the arithmetic.
4. **Verify** tube, cannula, and blood-product placement/identity before feeding, infusing, or transfusing.
5. **Document** every procedure (route, site, volumes, responses) and recognize the **danger signs** that require immediate reporting.

2. Placement Box

Part VI contains the practical "how-to" core every ward nurse performs daily. It consolidates the technical skills taught across Parts I–V (hand hygiene, vitals, injections, ORS, phototherapy, transfusions, surgery) into a single competency checklist. Each procedure has its own sub-file with a full **purpose** → **principles** → **equipment** → **step checklist** → **figure** → **documentation** → **verification** structure.

4. Procedure Index (15 sub-files)

#	Procedure	File
20.1	Hand hygiene and PPE	20.01-Hand-Hygiene-PPE.md
20.2	Newborn bathing and umbilical cord care	20.02-Newborn-Bath-Cord-Care.md
20.3	Temperature (axillary/tympanic)	20.03-Temperature.md
20.4	Apical pulse and respiratory count	20.04-Apical-Pulse-Respiration.md
20.5	Blood pressure (correct cuff)	20.05-Blood-Pressure.md
20.6	Anthropometrics and growth-chart plotting	20.06-Anthropometrics-Charts.md
20.7	NG tube insertion and gavage feeding	20.07-NG-Gavage-Feeding.md
20.8	Oxygen therapy (hood, cannula, mask)	20.08-Oxygen-Therapy.md
20.9	Phototherapy	20.09-Phototherapy.md
20.10	ORS administration and reassessment	20.10-ORS-Administration.md
20.11	IV cannulation and infusion-rate control	20.11-IV-Cannulation-Rate.md
20.12	Blood transfusion	20.12-Blood-Transfusion.md
20.13	SC/IM/ID injections and dose calculation	20.13-Injections-Dose-Calculation.md
20.14	Colostomy care	20.14-Colostomy-Care.md
20.15	Preoperative and postoperative care	20.15-PrePostoperative-Care.md

5. General Principles That Apply to All Procedures

- **Attraumatic care** (CH1): prepare the child/age-appropriate, use distraction and local/anesthetic comfort before painful steps, and involve the parents.
- **Infection control first** (20.1): hand hygiene and correct PPE before every task.
- **Two-nurse check for dose-critical steps** — drug doses, IV rates, ORS volumes, and blood products (20.12, 20.13).
- **Verify before acting:** NG placement before feeding (20.7), oxygen flow at the patient end (20.8), transfusion identity by two staff (20.12).
- **Document continuously:** route, site, gauges, volumes, temperature, and the child's response; use the I&O and vital-sign flow sheets.

6. Normal Values Summary (with file source)

Vital sign	Newborn	Infant	Child	Source
Temperature (°C)	36.5–37.5	36.5–37.5	36.5–37.5	20.3, CH5
Heart rate (bpm)	110–160	100–160	70–120 (older)	20.4, CH5
Respiratory rate (/min)	30–60	20–40	16–25	20.4, CH5
BP (cuff width)	40% arm circ.	40% arm circ.	40% arm circ.	20.5, CH5

Note: full age-by-age bands (infant → adolescent) are in **Appendix A — Normal Vital Signs and Laboratory Reference Tables**, the authoritative table for this guide.

7. Multiple-Choice Questions (Key at the end)

1. The single most important practice to prevent cross-infection in the ward is:

- A. wearing a gown for every procedure · B. **hand hygiene** · C. isolation of every admitted child
· D. fumigation of the ward.

1. The first bath of the newborn should be delayed until:

A. the cord falls off · B. the baby is 24 h old · C. **temperature is stable for at least 2–4 h** · D. meconium is passed.

1. For a gavage feed, the feeding tube is measured from:

A. tip of the nose to the umbilicus · B. **nose → ear → xiphoid (NEX)** · C. mouth to the sternal notch · D. ear lobe to the ankle.

1. Gavage feeding is temporarily **contraindicated** in the infant who:

A. is jaundiced · B. is <1500 g · C. **has marked tachypnea/apnea (aspiration risk)** · D. is receiving phototherapy.

1. The correct cuff width for BP in a child is about:

A. **40% of the upper-arm circumference** · B. half the arm length · C. the full diameter of the antecubital fossa · D. one-third of the thigh.

1. In moderate dehydration, the ORS volume (replacement + maintenance) given over 4 h is approximately:

A. 10 mL/kg · B. 25 mL/kg · C. **100 mL/kg** · D. 300 mL/kg.

1. A newborn's axillary temperature of **38.5 °C** should be:

A. rechecked in 6 h · B. treated with antipyretics by the nurse · C. **reported immediately as a medical concern** · D. ignored because newborns run warm.

1. Which IV set is preferred in pediatrics for precise rate control?

A. macrodrip 15 gtt/mL · B. glass burette · C. **microdrip 60 gtt/mL or infusion pump** · D. drip chamber without calibration.

1. If a transfusion reaction is suspected, the FIRST action is:

A. continue at a slower rate · B. give antihistamine · C. **stop the transfusion, keep the line open with normal saline, and notify the physician** · D. warm the blood.

1. The preferred IM site in infants <3 years is:

A. dorsogluteal · B. deltoid · C. **vastus lateralis** · D. rectus femoris.

Key: 1-B, 2-C, 3-B, 4-C, 5-A, 6-C, 7-C, 8-C, 9-C, 10-C.

8. Key Points

- Fifteen procedures make up the Chapter 20 competency checklist; each is trained in the skills lab then observed on the ward.
- Calculation errors are the most common medication-error cause in children — always **recalculate and double-check** doses, rates, and volumes.
- **Verification** (placement, identity, patency) and **documentation** are non-negotiable parts of every procedure.

END OF CHAPTER 20

Core Clinical Procedures (Diagnosis, Prevention, and Treatment Skills)

Next: **Procedure 20.01** — Hand Hygiene and Personal Protective Equipment (PPE)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.01

Procedure 20.01 — Hand Hygiene and Personal Protective Equipment (PPE)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Core ward procedure · **Time:** ≈5 min · **Applies to:** all ages · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Prevent health-care-associated infection (HAI) and cross-transmission of organisms between children, families, and staff.
- Apply the single most effective measure in infection control (the 5 moments of hand hygiene).

Assessment — What to check before you start

- Hands: free of jewelry; nails short, natural, and intact skin.
- The task to be performed and the precaution category in place (airborne / droplet / contact).
- Which of the 5 moments apply (before contact, before aseptic task, after body-fluid exposure, after contact, after surroundings).

Equipment — What you need

- Soap (plain or antiseptic) or alcohol-based hand rub; single-use towels.
- Gloves (clean exam, sterile as indicated), gown, mask/respirator, face shield/goggles.
- Rigid, puncture-resistant, labeled sharps container.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Remove jewelry; keep nails short and natural.	Rings and long nails harbor organisms and tear gloves.	☐
2	Remove watches/clothing barriers before patient contact.	Reduces the surfaces that carry organisms between patients.	☐
3	Perform hand hygiene at the 5 moments (before/after touching the child, before clean or aseptic tasks, after body-fluid exposure, after surroundings).	These moments mark when hands are most likely to transfer pathogens.	☐
4	Apply the alcohol rub, or wash with soap and water, covering all hand surfaces for the recommended duration.	Alcohol rubs are fast and effective on clean hands; soap and water are needed for visibly soiled hands or spore-forming organisms.	☐
5	Don gloves when contact with blood/body fluids, mucous membranes, or broken skin is anticipated; change between patients and between tasks.	Gloves form a barrier; changing them between tasks prevents cross-contamination.	☐
6	Don gown, mask/respirator, and eye protection when splashing or contact with secretions is anticipated.	PPE matched to the route of transmission protects staff and patients.	☐
7	Dispose of sharps into a rigid, puncture-resistant, labeled container immediately after use.	Immediate disposal prevents needle-stick injuries.	☐

#	Action	Rationale / precision point	Done
8	Remove PPE in the correct order, then perform hand hygiene last.	The final hand hygiene removes contamination picked up while doffing.	☐
9	Remove gloves by peeling inside-out without touching bare skin.	Inside-out removal keeps contaminated surfaces away from the hands.	☐



The 5 moments of hand hygiene poster

Normal values — Benchmarks for this skill

- Alcohol hand rub: **≈20–30 seconds** to cover and dry all hand surfaces (WHO).
- Soap-and-water hand wash: **≈40–60 seconds** (WHO technique).
- Standard Precautions apply to **all patients**, regardless of diagnosis.

Safety alerts — Protect the child and yourself

- Use **soap and water, not hand rub**, when hands are visibly soiled or after exposure to spore-forming organisms (*C. difficile*).
- **Never recap needles two-handed**; dispose of sharps immediately.
- Always perform hand hygiene **after removing gloves** — gloves are not a substitute.

Documentation — What to record

- Skill log: date, observed 5-moments compliance, PPE demonstration.

Verification — How you will be checked

- Student demonstrates the hand-washing technique (observed duration and coverage) and PPE donning/doffing order **without contamination**.

END OF PROCEDURE 20.01

Hand Hygiene and Personal Protective Equipment (PPE)

Next: **Procedure 20.02** — Newborn Bathing and Umbilical Cord Care

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.02

Procedure 20.02 — Newborn Bathing and Umbilical Cord Care

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Newborn care · **Time:** ≈10 min · **Applies to:** healthy newborn (first bath after cord check)
· **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Maintain skin integrity and hygiene of the newborn.
- Prevent hypothermia and promote safe, physiologic cord separation without introducing infection.

Assessment — What to check before you start

- Temperature stability: delay the first bath until the newborn has been **stable for at least 2–4 hours**.
- Skin condition (rashes, birthmarks, desquamation) and cord status (dry, clean, no redness, odor, or discharge).
- Risk factors: prematurity, cold stress, or an unstable newborn — postpone the bath.

Equipment — What you need

- Warm water (tested on the inner wrist), hooded towel, washcloth.
- Basin, clean diaper and clothes, cotton balls/gauze for the cord area.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Check the crib/radiant-heater temperature; gather supplies; close windows.	A warm environment prevents drafts and cold stress.	☐
2	Verify the water temperature (comfortably warm — test on the inner wrist); fill the basin.	Water too hot burns newborn skin; too cold causes hypothermia.	☐
3	Undress the baby keeping the trunk covered; bathe one part at a time with plain warm water (sponge bath).	Uncovering one area at a time preserves body heat.	☐
4	Wash the face (no soap), then the scalp; dry immediately with the hooded towel.	Prompt drying minimizes evaporative heat loss.	☐
5	Wash the body flexures (neck, axillae, groin); dry thoroughly; avoid immersion until the cord falls off.	Flexures collect moisture and organisms; immersion delays cord separation.	☐
6	Cord care: keep the stump clean and dry; fold the diaper below the stump; do not apply antiseptics or powder routinely.	Dry care (WHO) prevents infection and speeds separation.	☐

#	Action	Rationale / precision point	Done
7	Observe the stump for odor, redness, or discharge; report abnormalities.	These are early signs of cord infection (omphalitis).	☐
8	Dress the baby and recheck temperature after the bath.	Detects hypothermia induced by the bath.	☐
9	Chart the bath, temperature, skin condition, and cord status.	Provides a baseline and documents care.	☐



Sponge bath with cord stump exposed to air

Normal values — Benchmarks for this skill

- First bath delayed until temperature is **stable for $\geq 2\text{--}4$ hours** after birth.
- Axillary temperature maintained **$36.5\text{--}37.5$ °C** before and after the bath.
- Cord separation usually occurs within **$\approx 7\text{--}14$ days** (within the first 2 weeks).

Safety alerts — Protect the child

- Never leave the baby unattended in or near water.
- Do not **immerse** the newborn before cord separation.
- Never place the baby in hot water — always test on the inner wrist.
- Keep the cord area dry; fever, redness, or foul odor around the cord requires medical review.

Documentation — What to record

- Temperature before and after the bath, skin condition, and cord appearance.

Verification — How you will be checked

- Student demonstrates correct water-temperature testing, the sponge-bath order, and dry cord care without immersion before cord separation.

END OF PROCEDURE 20.02

Newborn Bathing and Umbilical Cord Care

Next: **Procedure 20.03** — Temperature Measurement (Axillary / Tympanic)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.03

Procedure 20.03 — Temperature Measurement (Axillary / Tympanic)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Vital-sign skill · **Time:** ≈2 min · **Applies to:** all ages (route by age) · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Obtain an accurate estimate of core temperature safely in children of different ages.
- Avoid the **rectal route** (discomfort and risk of injury) in newborns and in immunocompromised or bleeding-risk children.

Assessment — What to check before you start

- Age (axillary preferred for newborns; tympanic only >3 months).
- Ear findings: wax, drainage, or recent ear surgery — do not use the tympanic route.
- History of fever, previous temperature readings, and the child's activity/behavior.

Equipment — What you need

- Digital thermometer with clean probe covers; tympanic probe tips; lubricant not required for the axillary route.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Explain to the child/family; place a clean probe cover.	Consent and infection control; a fresh cover prevents cross-infection.	☐
2	Axillary route: place the probe tip high in the dry axilla; hold the arm adducted across the chest until the signal.	High contact with the skin folds over the probe gives the most accurate axillary reading.	☐
3	Tympanic route (>3 months): gently pull the pinna back toward the occiput; insert the probe snugly without occluding; press the button.	Straightening the canal aims the beam at the tympanic membrane.	☐
4	Read and record the temperature with the route and time.	The route is essential for correct interpretation (axillary reads lower than core).	☐
5	Compare with the age-based normal range; report values below 36.5 °C or above 38 °C (newborn) / per unit protocol.	Out-of-range values trigger medical evaluation in the newborn.	☐
6	Clean the thermometer; perform hand hygiene.	Prevents transmission between patients.	☐



Axillary temperature in an infant

Normal values — Benchmarks for this skill

- Axillary normal range: **36.5–37.5 °C**.
- Axillary readings typically run **0.5–1 °C lower** than core.
- Fever (rectal equivalent): **>38 °C**; newborn <36.5 °C or >38 °C needs medical evaluation.

Safety alerts — Protect the child

- Do not use the **rectal** route in newborns, immunocompromised children, diarrhea, or bleeding disorders.
- Do not use the **tympanic** route in an ear with wax, drainage, infection, or recent surgery.
- Document the route used — values cannot be compared across routes.

Documentation — What to record

- Temperature, route, and time on the vital-sign flow sheet.

Verification — How you will be checked

- Student obtains an axillary temperature correctly (probe placement and duration) and identifies the fever thresholds.

END OF PROCEDURE 20.03

Temperature Measurement (Axillary / Tympanic)

Next: **Procedure 20.04** — Apical Pulse and Respiratory Count

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.04

Procedure 20.04 — Apical Pulse and Respiratory Count

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Vital-sign skill · **Time:** ≈2 min (full-minute counts) · **Applies to:** infants and young children <2 years · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Accurately measure heart and respiratory rates in infants and young children.
- Use the apex as the preferred pulse site after checking respirations before disturbing the child.

Assessment — What to check before you start

- Activity state (sleeping / awake-quiet / crying) — affects the readings.
- Previous vital signs, medications, and signs of distress (retractions, grunting, cyanosis).

Equipment — What you need

- Stethoscope with the correct bell/diaphragm; watch with a second hand.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Allow the child to settle; approach quietly; position comfortably.	A quiet state avoids fear-induced tachycardia and tachypnea.	☐
2	Count respirations for a full minute while appearing to take other vital signs; note rhythm, depth, retractions, or grunting.	Infant breathing is irregular with periodic pauses; counting before handling and for a full minute is accurate.	☐
3	Place the stethoscope at the apex (left chest, 5th intercostal space, midclavicular line).	The apex is the point of maximal impulse and the preferred site in young children.	☐
4	Count the apical pulse for a full minute; note the rhythm and any extra sounds.	A full-minute count detects irregularities missed in short counts.	☐
5	Compare with the age-based normal range; report bradycardia, tachycardia, or irregular rhythm.	Early recognition of arrhythmia is essential in children.	☐
6	Record HR, RR, and the activity state with the time.	The activity state is needed to interpret the values.	☐



Apical pulse auscultation site in an infant

Normal values — Benchmarks for this skill

- Heart rate: **newborn 110–160 bpm** (up to 180 while crying); **infant 100–160 bpm**; **older child 70–120 bpm**.
- Respiratory rate: **newborn 30–60/min**; **infant 20–40/min**; **child 16–25/min**.
- Full age-by-age tables: **Appendix A** (authoritative table for this guide).

Safety alerts — Protect the child

- Report bradycardia, sustained tachycardia, or an irregular rhythm.
- Report increased work of breathing (retractions, grunting, flaring) with the respiratory count.
- Graph every count with the child's state — a "fast" pulse may be normal if the child is crying.

Documentation — What to record

- Both full-minute counts with the child's activity state and time.

Verification — How you will be checked

- Student obtains a correct apex count and 60-second respiratory count and correctly interprets age-based norms.

END OF PROCEDURE 20.04

Apical Pulse and Respiratory Count

Next: **Procedure 20.05** — Blood Pressure Measurement (Correct Cuff, Child Technique)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.05

Procedure 20.05 — Blood Pressure Measurement (Correct Cuff, Child Technique)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Vital-sign skill · **Time:** ≈3–5 min (after 5 min of quiet) · **Applies to:** all ages · **Chapter 20**
— Core Clinical Procedures

Purpose — Why we do this procedure

- Measure blood pressure accurately in children using the correct cuff size.
- Cuff size is **the single most important determinant** of a reliable reading.

Assessment — What to check before you start

- Upper-arm circumference to select the cuff.
- Child state: seated or supine, **calm and quiet for 5 minutes** before measurement.
- Previous readings, age percentiles, and medications.

Equipment — What you need

- Age-appropriate cuff (neonatal, infant, child, small-adult sizes), manual sphygmomanometer or validated oscillometric device.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Select a cuff whose bladder width $\approx$ 40% of the arm circumference; confirm the bladder encircles the limb.	A cuff too small overestimates and one too large underestimates BP.	☐
2	Seat the child with the arm supported at heart level (newborn: supine).	Arm position at heart level gives the correct hydrostatic reference.	☐
3	Place the cuff 2–3 cm above the antecubital fossa over the brachial artery; ensure a snug fit.	Correct placement over the artery gives a reliable reading.	☐
4	Palpate the brachial pulse; inflate 20–30 mmHg above the point where the pulse disappears (manual) or use the device setting.	Correct inflation level prevents errors (too slow/too low).	☐
5	Deflate slowly (2–3 mmHg/s); record systolic (first Korotkoff sound) and diastolic (muffling/disappearance) in the child.	Slow deflation captures the Korotkoff phases correctly.	☐
6	Compare with the age-based percentile; report elevated readings with the correct cuff size documented.	Interpretation requires the percentile and the method used.	☐
7	Record BP, site, cuff size, and position used.	Reproducibility depends on documenting the method.	☐



Correct vs incorrect cuff sizes (40% of arm circumference)

Normal values — Benchmarks for this skill

- Correct cuff width **≈40% of the upper-arm circumference** (bladder encircles 80–100% of the arm).
- Hypertension in children = BP **≥90th/95th percentile** for age and height.
- An isolated elevated reading may reflect stress, not hypertension — recheck with correct technique.

Safety alerts — Protect the child

- A cuff that is **too small reads falsely high** — always measure the arm circumference first.
- Do not take BP in an arm with an IV, infusion, or recent trauma.
- For oscillometric devices, still use the correct cuff size; an arm cuff used on the thigh needs a larger cuff.

Documentation — What to record

- BP with cuff size, site, and child's position on the vital-sign flow sheet.

Verification — How you will be checked

- Student selects the correct cuff by measurement and obtains a reading at heart level.

END OF PROCEDURE 20.05

Blood Pressure Measurement (Correct Cuff, Child Technique)

Next: **Procedure 20.06** — Anthropometrics and Growth-Chart Plotting

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.06

Procedure 20.06 — Anthropometrics and Growth-Chart Plotting

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Assessment skill · **Time:** ≈5 min · **Applies to:** all ages (technique varies by age) · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Measure weight, length/height, head circumference (HC), and mid-upper-arm circumference (MUAC) accurately.
- Plot and interpret results on WHO growth charts to detect normal growth, growth failure, or malnutrition.

Assessment — What to check before you start

- Age (determines recumbent length vs standing height; MUAC age range 6–59 months).
- Equipment calibration and the appropriate WHO chart set (weight-for-age, length-for-age, HC-for-age, weight-for-length).
- Previous measurements for serial comparison (growth velocity).

Equipment — What you need

- Calibrated infant/standing scales, measuring board/stadiometer, non-stretchable tape, WHO chart kit.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Zero the scale; ask the parent to remove shoes/outer clothes (infant: diaper only).	Light clothing and a zeroed scale give an accurate weight.	☐
2	Weigh (infant supine on the pan; older child standing still); read to the nearest 10 g/0.1 kg.	Precise reading detects small changes (daily weight is a fluid-status indicator).	☐
3	Measure recumbent length (<24 months) with two people on a firm board — crown to heels, legs straight.	Recumbent length is used for infants; proper positioning avoids underestimating.	☐
4	Measure standing height (≥24 months) without shoes, heels together against the stadiometer.	Standing height is used from 24 months; technique affects the value.	☐
5	Measure HC: wrap the tape snugly over the occiput and just above the eyebrows (largest circumference).	HC is the index of brain growth in the first years.	☐
6	Measure MUAC at the midpoint of the left upper arm (between acromion and olecranon); record to the nearest 0.1 cm.	MUAC is a rapid malnutrition screen (6–59 months).	☐
7	Plot measurements on the WHO charts; record percentile/z-score.	Placement against percentiles assigns the growth status.	☐
8	Compare serial values (velocity) — plot the trend over time; report deviations across percentiles.	A child crossing percentiles signals growth failure or excess.	☐



Measuring recumbent length and HC on the WHO chart

Normal values — Benchmarks for this skill

- MUAC (6–59 months): ≥ 12.5 cm = **normal**; 11.5–12.4 cm = **moderate** acute malnutrition; < 11.5 cm = **severe** acute malnutrition.
- Weight-for-age, length/height-for-age, HC-for-age, and weight-for-length plotted on the **WHO growth charts**.
- Serial plotting: any **drop across percentile lines** should be flagged and reported.

Safety alerts — Protect the child

- Use a **firm, stable** board for length — never stretch the child.
- Keep the infant warm and supported while weighing.
- Use a fresh disposable liner/towel on the scale and board between children.

Documentation — What to record

- Measurements, the plot date, and the chart type; flag any deviation from the previous percentile.

Verification — How you will be checked

- Student performs all four measurements and plots two serial points correctly.

END OF PROCEDURE 20.06

Anthropometrics and Growth-Chart Plotting

Next: **Procedure 20.07** — NG Tube Insertion and Gavage Feeding

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.07

Procedure 20.07 — NG Tube Insertion and Gavage Feeding

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Feeding skill · **Time:** ≈15–20 min (insertion ≈5 min, feed ≈20–30 min) · **Applies to:** infants/children unable to feed orally · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Provide nutrition to the infant/child who cannot coordinate suck–swallow–breathe, has poor endurance, or is NPO with an intact gut (e.g., RDS with tachypnea, growth failure, pyloric stenosis pre-op decompression).

Assessment — What to check before you start

- Feeding order (volume, method, frequency) and the current NPO/fasting plan.
- Respiratory status — gavage is avoided during **marked tachypnea/apnea** (aspiration risk).
- Weight and the planned feed volume; gastric residual if the tube is in place.

Equipment — What you need

- Tube (5–8 Fr), syringe (for bolus), water/lubricant, stethoscope (confirmatory), pH strips, tape, warm formula/breast milk, pacifier.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Verify the feeding order (volume, method, frequency).	Correct volume and method prevent overfeeding and aspiration.	☐
2	Measure nose-to-ear-to-xiphoid (NEX); mark the tube.	NEX approximates the distance to the stomach for the infant.	☐
3	Position the infant supine with the head slightly elevated (or side-lying); insert gently via nose or mouth.	Head elevation reduces reflux and aspiration risk.	☐
4	Verify placement: aspirate gastric contents; check pH (gastric usually <5.5); confirm if unsure.	pH testing is the reliable bedside check — do not rely on auscultation alone.	☐
5	Secure the tube to the cheek with tape; re-verify position before every feeding.	The tube can migrate; re-checking prevents feeding into the airway.	☐
6	Hold the infant (if stable) during feeding; give warm milk/formula by slow gravity drip over ≈20–30 min; allow pacifier sucking.	Slow gravity feeding and nonnutritive sucking support feeding readiness and tolerance.	☐
7	Position upright/right-side or burp after feeding; monitor for choking, regurgitation, or apnea.	Upright/right-side position aids gastric emptying and airway protection.	☐
8	Flush with a small amount of water if the tube remains in place; close/cap between feeds.	Flushing maintains patency and prevents clogging.	☐
9	Document volume, tolerance, residuals if checked, and the route.	Accurate intake records guide the feeding plan.	☐



Gavage feeding — tube measurement (NEX) and gravity feeding

Normal values — Benchmarks for this skill

- Feeding tube size: **5–8 Fr** for infants.
- Gastric aspirate pH: **usually <5.5** (confirms gastric placement).
- Bolus gavage delivery: **slow gravity drip over ≈20–30 min.**

Safety alerts — Protect the child

- **Avoid gavage/nipple feeding** during marked tachypnea or apnea (aspiration risk in RDS).
- **Verify placement before every feed** — never rely on auscultation alone.
- Stop the feed and notify the physician on choking, cyanosis, or persistent regurgitation.

Documentation — What to record

- Type/volume/viscosity of the feed, placement-verification method, residuals, and tolerance.

Verification — How you will be checked

- Student inserts a tube on the manikin, verifies placement, and administers a full gavage feed slowly without aspiration.

END OF PROCEDURE 20.07

NG Tube Insertion and Gavage Feeding

Next: **Procedure 20.08** — Oxygen Therapy (Hood, Nasal Cannula, Face Mask)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.08

Procedure 20.08 — Oxygen Therapy (Hood, Nasal Cannula, Face Mask)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Respiratory support · **Time:** ≈10 min (set-up) · **Applies to:** hypoxemic children (device by age) · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Provide supplemental oxygen to the hypoxemic child in a safe, humidified, age-appropriate delivery system while monitoring the response.

Assessment — What to check before you start

- The ordered device, flow/FiO₂, and the target SpO₂ range.
- Respiratory effort (grunting, retractions, nasal flaring) and skin color — central cyanosis is late.
- Oxygen source and humidifier water level; the correct device size for the child.

Equipment — What you need

- Wall/source O₂ with flowmeter and regulator; humidifier (bubbler); hood/cannula/mask of the correct size; extension tubing; pulse oximeter.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Verify the order: device, flow/FiO ₂ , and target SpO ₂ range.	Oxygen is a medication — give exactly what is ordered.	☐
2	Set the flowmeter (L/min) and humidifier; confirm the water level.	Humidification prevents mucosal drying, especially at higher flows.	☐
3	Select the correct-size device; check the fit (hood on the shoulders with a neck seal, cannula prongs in the nares, mask seal).	A correct fit delivers the intended FiO ₂ without leaks.	☐
4	Attach the device firmly (secure the line away from the neck); verify free-flow at the patient end.	Prevents tubing kinks and accidental disconnection.	☐
5	Place the pulse-oximeter probe; target the ordered SpO ₂ range; check HR and RR.	Oximetry confirms the effect of oxygen therapy.	☐
6	Observe for comfort and signs of hypoxia (retractions, grunting, flaring, color change); treat cyanosis as an emergency.	Cyanosis is a late sign — monitor clinical effort, not just the reading.	☐
7	Prevent skin breakdown (hood edges, cannula cheeks); humidify as ordered.	Pressure and dryness cause skin and mucosal injury.	☐
8	Chart device, flow/FiO ₂ , SpO ₂ , and changes; wean only per order.	Accurate records guide weaning and avoid hyperoxia.	☐



Oxygen hood in an infant

Normal values — Benchmarks for this skill

- Nasal cannula: **0.25–2 L/min** (humidified in infants).
- Humidification for flows **>2 L/min** or prolonged use.
- SpO₂ target per unit protocol; **avoid hyperoxia/hyperoxemia in neonates** (retinopathy/BPD risk).

Safety alerts — Protect the child

- Oxygen is a **medication** — verify the ordered concentration/flow each time.
- **No smoking/open flame** near oxygen; check the source and tubing.
- Central cyanosis is a **late** sign — treat it as an emergency.
- In RDS, oxygenation is governed by mean airway pressure, not just FiO₂.

Documentation — What to record

- Device, flow, FiO₂, SpO₂, and clinical response each shift.

Verification — How you will be checked

- Student sets the flow, chooses the correct device size, applies pulse oximetry, and states the hazards of hyperoxia in neonates.

END OF PROCEDURE 20.08

Oxygen Therapy (Hood, Nasal Cannula, Face Mask)

Next: **Procedure 20.09** — Phototherapy for Neonatal Hyperbilirubinemia

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.09

Procedure 20.09 — Phototherapy for Neonatal Hyperbilirubinemia

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

NICU treatment skill · **Time:** continuous (checks every shift) · **Applies to:** jaundiced neonate with phototherapy order · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Lower unconjugated (indirect) bilirubin by photo-isomerization in exposed skin to forms excreted in bile/urine.
- Prevent kernicterus (see Chapter 11 — Jaundice).

Assessment — What to check before you start

- The phototherapy order and the bilirubin threshold being treated.
- Unit irradiance ($\mu\text{W}/\text{cm}^2/\text{nm}$) and the correct distance from the light to the infant.
- Weight, hydration state, and baseline temperature.

Equipment — What you need

- Phototherapy unit (bili lights / bili blanket), eye shields (patches), diaper, incubator/warmer if needed, bilirubinometer/lab slips.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Verify the phototherapy order and the bilirubin threshold being treated.	Treatment is threshold-based; confirm the unit is at the correct irradiance.	☐
2	Confirm the unit irradiance ($\mu\text{W}/\text{cm}^2/\text{nm}$) and the correct distance from infant to light.	The more skin exposed and the higher the irradiance, the more effective the therapy.	☐
3	Undress the infant to the diaper; apply eye shields firmly but not tightly over both eyes.	Eye shields protect the retina; a snug but not tight fit avoids pressure injury.	☐
4	Position the infant under the light for maximal skin exposure; remove shields only during feeds/care.	Maximal exposure maximizes the rate of bilirubin breakdown.	☐
5	Maintain a neutral thermal environment (incubator/warmer); check temperature frequently.	Phototherapy units generate heat; temperature instability is a common side effect.	☐
6	Maintain fluid intake (breast milk/formula/IV) to compensate for losses; chart input-output and weight.	Insensible water loss increases under the lamp; fluids protect against dehydration.	☐
7	Monitor bilirubin levels per order (q6–12 h) and observe stools (loose green = expected), skin color, and hydration.	Loose green stools reflect increased bilirubin excretion — a good sign.	☐
8	Protect staff/parents from the light (no direct eye exposure); keep the parents informed.	Light can damage adult eyes as well; parental teaching reduces anxiety.	☐

#	Action	Rationale / precision point	Done
9	On stopping: recheck bilirubin for rebound as ordered; document the last irradiance and level.	Rebound hyperbilirubinemia occurs after therapy stops.	☐



Phototherapy unit setup with eye shields

Normal values — Benchmarks for this skill

- Irradiance per the unit protocol (commonly measured in $\mu\text{W}/\text{cm}^2/\text{nm}$).
- Bilirubin rechecked **every 6–12 hours** during intensive therapy.
- Expected: **loose, green stools** and increased fluid needs.

Safety alerts — Protect the child

- **Never** shine the light directly at the infant's eyes; use eye shields correctly.
- Report **temperature instability**, dehydration, or a greenish-brown skin discoloration (**bronze baby syndrome**, rare).

- Advise parents to keep **both eyes shielded** while the light is on.

Documentation — What to record

- Bilirubin values, irradiance, time under light, hydration, stool frequency, and temperature.

Verification — How you will be checked

- Student applies eye shields correctly, positions the light, and states the monitoring schedule and rebound check.

END OF PROCEDURE 20.09

Phototherapy for Neonatal Hyperbilirubinemia

Next: **Procedure 20.10** — ORS Administration and Reassessment

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.10

Procedure 20.10 — ORS Administration and Reassessment

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Rehydration skill · **Time:** ≈15 min (4-hour plan) · **Applies to:** child with mild/moderate dehydration · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Rehydrate the child with dehydration using oral rehydration solution (ORT) and reassess to prevent progression to hypovolemic shock (see Chapter 17).

Assessment — What to check before you start

- Weight and the current dehydration grade (mild / moderate / severe).
- Hydration indicators: skin turgor, oral mucosa, tears, fontanel, pulse, urine output.
- Ability to drink and presence of vomiting (vomiting is not a contraindication).

Equipment — What you need

- ORS packet/mix + clean water, cup, spoon or oral syringe, basin, graduated measuring cup, weight scale, chart.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Weigh the child; assess and grade dehydration (mild/moderate/severe).	The grade determines the ORS rate and the need for IV therapy.	☐
2	Prepare fresh ORS per the packet instructions (correct water amount; discard after the stated period).	Correct dilution and freshness are essential for safe rehydration.	☐
3	Calculate the 4-hour volume (50 or 100 mL/kg) and per-stool replacement (10 mL/kg).	Volumes are weight-based and re-checked against the grade.	☐
4	Offer ORS 5–10 mL every 2–5 min by spoon/syringe; continue despite vomiting.	Small frequent amounts are tolerated even when vomiting.	☐
5	After the first hour, offer larger amounts as tolerated; keep a strict intake record.	Increasing amounts replace losses once tolerance improves.	☐
6	Reassess hydration every 4–8 h (turgor, mucosa, tears, fontanel, pulse, urine output) and weigh daily.	Reassessment detects failure of oral rehydration early.	☐
7	If ORT fails (vomits all, worsening lethargy, no urine): stop ORT, notify the physician, prepare for IV fluids.	Failure of oral therapy requires the IV route.	☐
8	Continue breastfeeding/formula; reintroduce the normal diet early; avoid juice/high-sugar drinks.	Early feeding shortens illness; high-sugar drinks worsen diarrhea.	☐

#	Action	Rationale / precision point	Done
9	Teach the caregiver ORS preparation, dosing, and the danger signs to return.	Caregiver competence sustains rehydration at home.	☐

Administering ORS with a syringe to a dehydrated infant

Normal values — Benchmarks for this skill

- ORS composition: **75 mmol/L sodium** and **13.5 g/L glucose**.
- Volume plan: **mild 50 mL/kg, moderate 100 mL/kg** within 4 h.
- Replace each diarrheal stool with **ORS 10 mL/kg**; expected urine output **0.5–1 mL/kg/h**.

Safety alerts — Protect the child

- Tap water, milk, juice, soup, or broth are **not** appropriate for rehydration.
- **Severe dehydration:** IV lactated Ringer **40 mL/kg** (bolus 20 mL/kg then reassess) until pulse/consciousness normalize, then ORS.

- Danger signs requiring return: vomiting everything, worsening lethargy, no urine, blood in stool.

Documentation — What to record

- Fluid intake by mouth/IV, stool/emesis output, urine output, daily weight, and reassessment findings.

Verification — How you will be checked

- Student prepares ORS correctly, performs the 4-hour reassessment, and recognizes ORT failure requiring escalation.

END OF PROCEDURE 20.10

ORS Administration and Reassessment

Next: **Procedure 20.11** — IV Cannulation and Infusion-Rate Control (Pediatric)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.11

Procedure 20.11 — IV Cannulation and Infusion-Rate Control (Pediatric)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Parenteral therapy · **Time:** ≈10–15 min · **Applies to:** all ages (site/gauge by age) ·

Chapter 20 — Core Clinical Procedures

Purpose — Why we do this procedure

- Establish venous access for fluids, medications, or blood.
- Deliver infusions at exactly the calculated rate using pediatric microdrip sets or infusion pumps.

Assessment — What to check before you start

- The order: solution, volume, and rate (mL/h) or the infusion duration.

- Site selection by age (hand/forearm any age; scalp veins in infants; avoid the dominant hand when possible).
- Child preparation: topical anesthetic if ordered, distraction, and a helper for holding.

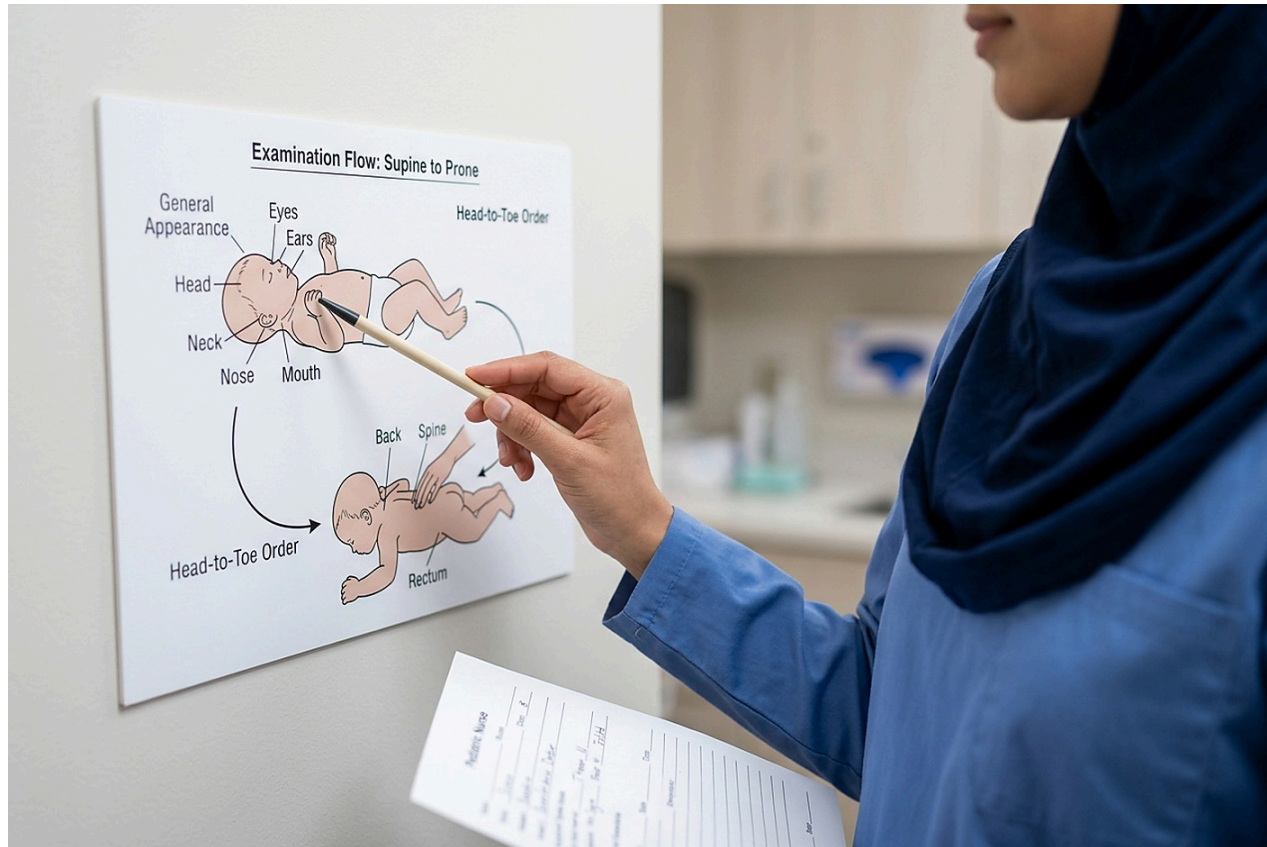
Equipment — What you need

- Small-gauge over-the-needle catheter (24/26 G infants), tourniquet, antiseptic, sterile dressing, T-connector/extension, microdrip set or pump, tape.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Calculate the rate: convert the ordered mL/h or compute gtt/min using the drop factor.	gtt/min = (volume (mL) × drop factor) ÷ time (min); or mL/h = total volume ÷ hours [CH7].	☐
2	Explain to the child/parent with age-appropriate distraction; apply topical anesthetic if ordered.	Atraumatic care reduces pain and fear (CH1, CH3).	☐
3	Select the site; apply a tourniquet or use a helper's hand/gentle hold; don PPE/gloves.	Age-appropriate site and restraint prevent movement injury.	☐
4	Clean with antiseptic; allow it to dry; insert the catheter at 10–30°; observe flashback.	Drying antiseptic reduces stinging; flashback confirms vein entry.	☐
5	Advance, remove the needle, and confirm flow; secure with a sterile dressing.	Confirming free flow prevents infiltration.	☐
6	Set the infusion pump (or count drops) to exactly the calculated rate; prime the tubing to remove air.	Air in tubing risks air embolism if disconnected.	☐

#	Action	Rationale / precision point	Done
7	Monitor hourly: rate, site (no swelling/pain/redness), patency, and total volume against the plan.	Early detection of infiltration/extravasation and phlebitis.	☐
8	Document the site, gauge, rate, and any pump alarms resolved.	Accurate records guide rate verification and troubleshooting.	☐



Peripheral IV on an infant's hand with pump

Normal values — Benchmarks for this skill

- Pediatric infusions use **microdrip (60 gtt/mL) sets or infusion pumps** for precise control.
- **gtt/min = (volume (mL) × drop factor) ÷ time (min); mL/h = total volume ÷ hours.**
- Catheter gauge: **24/26 G for infants**; small-gauge for children.

Safety alerts — Protect the child

- **Fluid is a medication in pediatrics** — verify the ordered maintenance (Holliday–Segar) volume [CH7].
- Check the site **hourly** for swelling, pain, redness, or coolness (infiltration/extravasation).
- Secure the line to prevent **disconnection** (risk of air embolism).

Documentation — What to record

- Cannula gauge/site, IV rate, and hourly totals.

Verification — How you will be checked

- Student performs the gtt/min and mL/h calculation for a given order and sets/checks the pump accordingly.

END OF PROCEDURE 20.11

IV Cannulation and Infusion-Rate Control (Pediatric)

Next: **Procedure 20.12** — Blood Transfusion (Fresh Whole Blood / Packed Red Cells)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.12

Procedure 20.12 — Blood Transfusion (Fresh Whole Blood / Packed Red Cells)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Product administration · **Time:** ≤4 h per unit (starts slowly) · **Applies to:** children needing blood products · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Safely administer blood products (packed red cells / fresh whole blood) to restore oxygen-carrying capacity in anemia (thalassemia, sickle cell disease, blood loss) while preventing and promptly recognizing transfusion reactions.

Assessment — What to check before you start

- Physician's order, cross-match compatibility, and product appropriateness.
- Two patient identifiers and the blood-bank wristband/cross-match label [CH13].
- Baseline vital signs (T, HR, RR, BP) and current IV access.

Equipment — What you need

- Blood product (chilled, undamaged, correct expiry), IV access (microdrip/pump), blood transfusion tubing with filter, normal saline, vital-sign monitor, observations chart.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Verify the physician's order, cross-match compatibility, and product appropriateness.	Transfusion begins with an order and a compatible product.	☐
2	Two-identifier bedside check by two staff: product, unit number, recipient, expiry; inspect the bag (no clots/discoloration/damage).	Checking by two licensed staff prevents giving the wrong product.	☐
3	Take baseline vital signs (T, HR, RR, BP) and record.	Baseline values are the reference for detecting reactions.	☐
4	Prime the transfusion set with the product via filter; connect only to a normal-saline side port.	The filter removes clots/wBCs; normal saline is the only compatible flush.	☐
5	Run slowly initially; stay with the child for the first 15–20 min.	Most transfusion reactions occur early; close observation is critical.	☐
6	Monitor vitals at the ordered intervals and watch for fever, chills, rash, dyspnea, back pain, or hypotension.	Early detection of hemolytic/allergic/febrile reactions.	☐
7	If a reaction is suspected: stop immediately, keep the line open with normal saline, notify	Stopping first, then keeping the vein with saline, limits the	☐

#	Action	Rationale / precision point	Done
	the physician, return the bag/unit for investigation, document.	reaction and preserves access.	
8	Complete the unit within the ordered duration (usually ≤ 4 h); record the volume transfused and any reactions.	Long transfusion times increase the risk of bacterial growth.	☐
9	Document: unit number, start/stop time, volume, vital signs, and observations.	Full documentation is a legal and clinical requirement.	☐



Blood-transfusion bedside two-identifier check

Normal values — Benchmarks for this skill

- Start slowly: often ≈ 5 mL/kg/h (or 1–2 mL/min) per protocol.
- Close observation during the **first 15–20 minutes**.
- Complete each unit **within ≤ 4 hours**.

Safety alerts — Protect the child

- Check the product by **two licensed staff**; never store blood at the bedside.
- Do not mix **medications or IV fluids (except normal saline)** with the unit.
- Suspected reaction: **stop** → **keep IV patent with normal saline** → **notify physician** → **recheck/return the unit** [CH13].

Documentation — What to record

- Product details, start/stop times, volumes, vital signs, and any reaction.

Verification — How you will be checked

- Student performs the two-identifier check, starts the transfusion slowly, and correctly reacts (stop/IV saline/notify) to a simulated reaction.

END OF PROCEDURE 20.12

Blood Transfusion (Fresh Whole Blood / Packed Red Cells)

Next: **Procedure 20.13** — SC, IM, and ID Injections plus Drug-Dose Calculation

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.13

Procedure 20.13 — SC, IM, and ID Injections plus Drug-Dose Calculation

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Medication administration · **Time:** ≈10 min · **Applies to:** all ages (site by age) · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Administer medications safely by subcutaneous (SC), intramuscular (IM), and intradermal (ID) routes in children.
- Verify that every dose is calculated and checked before administration.

Assessment — What to check before you start

- The 8 rights: right patient, drug, dose, route, time, documentation, assessment, right to refuse/education.
- Weight-based dose (mg/kg) vs the ordered dose — recheck the arithmetic with a second staff member.
- Site by age: **vastus lateralis preferred <3 years**; deltoid after $\approx$ 15 months with enough muscle; dorsogluteal avoided in children.

Equipment — What you need

- Correct syringe (1 mL tuberculin for ID/subunit vaccines; 2–3 mL IM), correct-gauge needle (IM 22–25 G; SC 25–26 G; ID 26–27 G), alcohol swab, gloves, sharps container, drug/vial and the original order.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Calculate the dose: $\text{ordered mg} \div \text{available mg/mL} = \text{volume (mL)}$; recheck against the mg/kg recommendation and have it double-checked.	Volume (mL) = ordered dose (mg) $\div$ drug concentration (mg/mL) — arithmetic errors are the main cause of drug errors.	☐
2	Verify the 8 rights using the original order + physician/doc read-back; check expiry and appearance.	Independent verification prevents wrong drug/dose/patient errors.	☐
3	Prepare and draw the drug under aseptic technique; expel air; cap the needle.	Asepsis and air expulsion maintain sterility and correct dose.	☐
4	Select and palpate the correct site for the age (vastus lateralis for infants; deltoid >3 yr; inner forearm for ID).	Age-appropriate sites avoid nerves and vessels.	☐
5	Clean the site with antiseptic; allow it to dry.	Drying antiseptic reduces stinging and ensures antisepsis.	☐

#	Action	Rationale / precision point	Done
6	IM: stretch/Z-track, insert at 90°; inject slowly (do not aspirate — current vaccine guidance [CH09]); remove and apply pressure (no rubbing for IM vaccines).	Deep 90° insertion reaches the muscle; not rubbing prevents injection-site reactions.	☐
7	SC: pinch the skin; insert at 45–90°; inject slowly; remove without rubbing.	A skin fold avoids the muscle; angle depends on tissue depth.	☐
8	ID: stretch the skin; insert at 5–15° bevel-up; inject a small bleb/wheel; do not massage or wipe (Mantoux: mark/read on schedule).	ID injection stays in the dermis; a wheal confirms technique.	☐
9	Dispose of the sharps immediately in the puncture-resistant container; never recap two-handed.	Immediate disposal prevents needle-stick injury.	☐
10	Document the drug, dose, route, site, time, and the child's response; observe for local/systemic reaction.	Documentation completes the medication cycle and tracks effects.	☐



Normal values — Benchmarks for this skill

- Volume limits: IM ≤ 2 mL in a single site (older children); ≤ 1 mL in infants/smaller children; SC **up to 1 mL**; ID **small (≤ 0.1 mL)**.
- Angles: **IM 90°**; **SC 45–90°**; **ID 5–15°**.
- Dose formula: **Volume (mL) = ordered dose (mg) $\div$ concentration (mg/mL)** [CH7, CH8].

Safety alerts — Protect the child

- **Two-nurse dose double-check** for high-alert drugs.
- **Do not aspirate for vaccines** (current vaccine guidance [CH09]).
- **Dorsogluteal site is avoided in children** (sciatic-nerve risk, poor muscle development).
- **Never recap needlessly/two-handed**; use the sharps container immediately.

Documentation — What to record

- Drug, dose (mg and mL), route, site, time, and signature.

Verification — How you will be checked

- Student calculates a weight-based dose correctly, selects the correct IM/SC/ID site and angle for the age, and performs the injection on a manikin with correct technique.

END OF PROCEDURE 20.13

SC, IM, and ID Injections plus Drug-Dose Calculation

Next: **Procedure 20.14** — Colostomy Care (Pouch Change, Skin, Stoma Assessment)

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.14

Procedure 20.14 — Colostomy Care (Pouch Change, Skin, Stoma Assessment)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Stoma care · **Time:** ≈10 min · **Applies to:** child with a colostomy (e.g., Hirschsprung) · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Maintain a healthy peristomal area, support bowel-function recovery, and prevent complications after colostomy for Hirschsprung disease (and other causes) — see Chapter 18.

Assessment — What to check before you start

- Stoma color and condition (healthy stoma: **red, moist, mucosal**).
- Peristomal skin: redness, excoriation, moisture, or bleeding.
- Output amount, color, and consistency; the pouch fill level.

Equipment — What you need

- Clean pouch/pediatric pouch (correct size), skin barrier wipes/paste, warm water, gauze, gloves, measuring guide, disposal bag.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Explain to the child/family; prepare gloves and supplies; note the stoma color and condition.	Preparation and observation guide a safe, quick change.	☐
2	Empty the existing pouch through the bottom and rinse; gently remove the pouch from the skin outwards.	Emptying first avoids spillage; gentle removal protects the skin.	☐

#	Action	Rationale / precision point	Done
3	Clean the peristomal skin with warm water and gauze; shape/trim the skin barrier to the stoma opening.	Warm water (not harsh soap) and a correctly trimmed opening protect the skin.	☐
4	Dry the skin completely.	Moisture causes maceration and skin breakdown.	☐
5	Apply the barrier/paste around the stoma and press the pouch onto it, ensuring a leak-proof seal.	A secure seal prevents leakage and skin irritation.	☐
6	Observe for redness, excoriation, bleeding, or change in stoma color/character; report abnormalities.	Early complication detection prevents serious stoma/skin problems.	☐
7	Document stool output (amount, color, frequency), skin condition, and the pouch-change interval.	Output records guide hydration and nutrition.	☐
8	Teach the caregiver the change procedure, skin-care rules, and when to return (bleeding, dark stoma, no stool/obstruction).	Caregiver competence sustains safe home care.	☐



Applying a pediatric colostomy pouch over a skin barrier with trimmed opening

Normal values — Benchmarks for this skill

- Healthy stoma: **red, moist, mucosal**; pale, dusky, black, or bleeding needs immediate attention.
- Change the pouch when **1/3 to 1/2 full** or leaking; empty through the bottom for each stool.
- Stool may be semi-formed or formed depending on the level of the stoma.

Safety alerts — Protect the child

- **Pale, dusky, black, or bleeding stoma** → **immediate attention** (ischemia/necrosis).
- Clean with **warm water only** — no harsh soaps, detergents, or alcohol wipes on the peristomal skin.
- Involve the stoma care nurse; avoid tight clothing or belts over the pouch.

Documentation — What to record

- Stoma appearance, output, skin condition, and pouch changes.

Verification — How you will be checked

- Student performs a pouch change on a manikin, trims the barrier correctly, and reports stoma complications.

END OF PROCEDURE 20.14

Colostomy Care (Pouch Change, Skin, Stoma Assessment)

Next: **Procedure 20.15** — Preoperative and Postoperative Care of the Child

PART VI - CORE CLINICAL PROCEDURES

PROCEDURE 20.15

Procedure 20.15 — Preoperative and Postoperative Care of the Child

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Perioperative care · **Time:** pre-op ≈ 15 min; post-op continuous monitoring · **Applies to:** child scheduled for surgery (see Chapters 18, 19) · **Chapter 20 — Core Clinical Procedures**

Purpose — Why we do this procedure

- Prepare the child and family for surgery safely (fasting, consent, psychological preparation).
- Provide systematic postoperative monitoring, pain control, wound care, and early feeding per the surgical condition (see Chapters 18 and 19).

Assessment — What to check before you start

- **Pre-op:** NPO start time, consent/explanations, allergies, surgical site, identification band, baseline vitals and weight.
- **Post-op:** vital signs, SpO₂, level of consciousness, pain (FLACC/Wong-Baker/0–10 scale by age), wound/dressing, drains/tubes, and input-output.

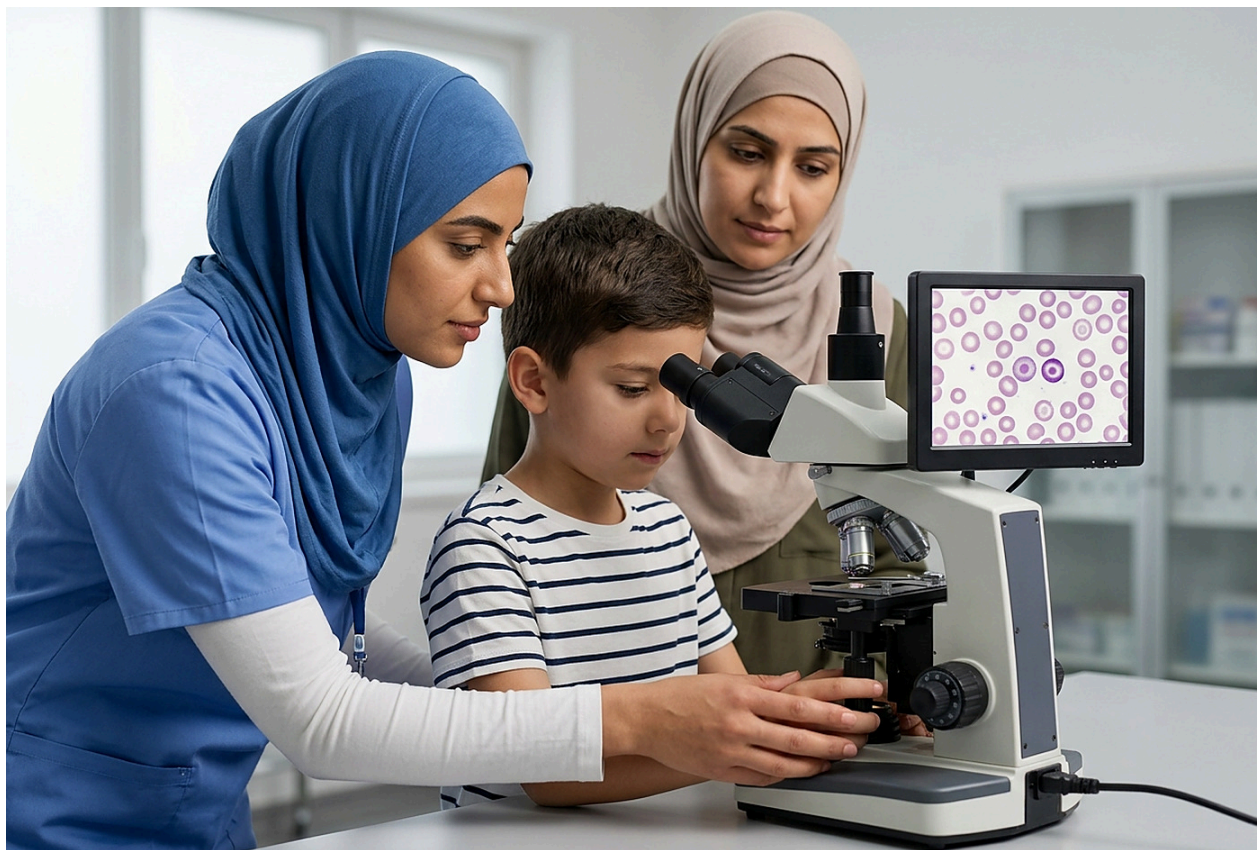
Equipment — What you need

- Vital-sign monitor/pulse oximetry, pain scale, IV pump, sterile dressing tray, NG tube + suction (if ordered), emesis basin, appropriate documentation sheet.

Procedure Steps

#	Action	Rationale / precision point	Done
1	Pre-operative: verify NPO start, consent/explanations, allergies; confirm the surgical site and identification band.	NPO and site verification prevent aspiration and wrong-site surgery.	☐
2	Use age-appropriate communication/play to reduce anxiety; document baseline vitals and weight.	Psychological preparation and a baseline support safe care.	☐
3	Confirm the pre-op medication/antibiotic orders and admission data.	Correct pre-op orders prevent omission and drug errors.	☐
4	Post-operative: receive report and monitor vitals, SpO ₂ , and level of consciousness at the ordered frequency.	Systematic monitoring detects early complications (e.g., airway, bleeding).	☐
5	Assess pain with the age-appropriate scale; give the ordered analgesic and re-assess 30–60 min after.	Re-assessment confirms the analgesic worked and guides the next dose.	☐
6	Monitor the wound/dressing (exudate, bleeding, swelling) and drains/tubes (volume, patency).	Early detection of bleeding, infection, or blocked drains.	☐
7	Begin fluids/feeds per the operation-specific plan (NG decompression → gradual feeds; staged feeding for pyloromyotomy).	Feeding plans are procedure-specific (CH19) and prevent aspiration/ileus.	☐
8	Prevent complications: reposition 2-hourly, observe for fever/ileus/vomiting; document input-output.	Repositioning prevents pressure injury; close observation detects ileus and infection.	☐

#	Action	Rationale / precision point	Done
9	Document every assessment and the plan; involve the family in discharge teaching (wound care, analgesia, return-if signs).	Teaching ensures safe recovery at home.	☐



Postoperative infant with pulse oximetry and staged feeds

Normal values — Benchmarks for this skill

- **Preoperative fasting (NPO):** clear liquids ≈2 h; breast milk ≈4 h; formula/light meal ≈6 h; solids ≈8 h (per guidelines/unit protocol).
- Pain re-assessment **30–60 min after** analgesia.
- Repositioning **every 2 hours** postoperatively.

Safety alerts — Protect the child

- Confirm the **NPO start time, surgical site, and identification band** before surgery.

- Monitor for opioid **respiratory depression** when opioids are used for severe pain (CH8).
- Wound checks: redness, swelling, discharge, or **dehiscence** must be reported immediately.

Documentation — What to record

- NPO start, vitals, pain scores, wound status, intake-output, and feeding progression.

Verification — How you will be checked

- Student completes a pre-op checklist and performs systematic post-op monitoring with correct pain-scale selection and documentation of staged feeding.

END OF PROCEDURE 20.15

Preoperative and Postoperative Care of the Child

Next: **Chapter 21** — The Nursing Care-Plan Model (Problem-Based Care Planning)

PART VII - NURSING CARE-PLAN MODEL

CHAPTER 21

Chapter 21 — The Nursing Care-Plan Model (Problem-Based Care Planning)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Part VII — Care-Plan Model, Chapter 21

1. Learning Objectives

By the end of this chapter, the student will be able to:

1. Structure a nursing care plan using the **NANDA-I nursing process**: assessment → nursing diagnosis → goals → interventions (with rationale) → evaluation.

2. Build a care plan for any of the ward topics in Parts II–V (jaundice, RDS, thalassemia, ALL, hemophilia, pneumonia, diarrhea, Hirschsprung, HPS).
3. Write **measurable, time-bound goals** and select **age-appropriate interventions**.
4. Use the standard care-plan template attached to every clinical chapter.

2. The Nursing Process in Pediatrics

Phase	What you do	Example (diarrhea)
1. Assessment	Gather data (history, exam, labs, growth)	2-year-old, 8 stools 24 h, skin turgor slow, no tears, weight falls 10%
2. Nursing diagnosis	State the problem (NANDA-I)	<i>Deficient fluid volume</i> related to excessive GI loss
3. Goals/outcomes	Expected, measurable, timed	Rehydrated (normal turgor, tears, urine ≥ 1 mL/kg/h) within 8 h
4. Interventions	Actions with rationale	Oral rehydration per ORS plan — replaces fluids/electrolytes
5. Evaluation	Did the goal happen? Record	Urine output normal, weight stable → continuing care plan updated

Every clinical chapter (CH11–CH19) already includes a "Nursing Care Plan" section written in exactly this template.

3. Common Pediatric Nursing Diagnoses by Ward

Ward/topic	Frequently used NANDA diagnoses
Jaundice (NICU)	<i>Risk for injury / Hyperbilirubinemia</i> related to excess unconjugated bilirubin
RDS (NICU)	<i>Ineffective breathing pattern; Impaired gas exchange; Risk for aspiration</i>

Ward/topic	Frequently used NANDA diagnoses
Thalassemia / ALL	<i>Risk for infection; Fatigue; Imbalanced nutrition: less than body requirements; Deficient knowledge</i>
Hemophilia	<i>Risk for bleeding; Risk for injury; Deficient knowledge</i>
Pneumonia	<i>Ineffective airway clearance; Hyperthermia; Ineffective breathing pattern</i>
Diarrhea/dehydration	<i>Deficient fluid volume; Risk for impaired skin integrity</i>
Hirschsprung	<i>Constipation; Risk for infection (enterocolitis); Deficient knowledge (colostomy)</i>
Pyloric stenosis	<i>Deficient fluid volume; Imbalanced nutrition; Risk for aspiration</i>

4. Writing SMART Goals

- **Specific** — "Pass urine at least 1 mL/kg/h" (not "better hydration").
- **Measurable** — use numbers (urine output, weight, temperature, SaO₂, pain scale).
- **Achievable, Relevant** to the diagnosis, **Time-bound** ("within 8 h", "by discharge").

Example (fever in pneumonia): goal — "Body temperature within the age normal range (<38 °C axillary) within 12 h of starting antipyretics and fluids." **Interventions:** antipyretic per order; sponge bath only if very high and per protocol; observe for shivering; increase oral fluids; recheck temperature q2–4 h.

5. Care-Plan Template (use in ward assignments)

1. **Patient context:** age, weight, admitting diagnosis, relevant history.
2. **Problem list (from assessment):** data + nursing diagnosis.
3. **Goals (SMART):** short-term + long-term.
4. **Interventions with rationale:** for each, state what you do and why (cite the chapter source).
5. **Evaluation:** outcome measures + revision.

6. Case-Study Practice

Give the care plan for: a 4-month-old with bronchiolitis/pneumonia (fever 39 °C, RR 58/min, SpO₂ 90% on room air, feeding poorly).

Sample answer frame: diagnosis *Ineffective airway clearance* (secretions/inflammation) → goal: RR 30–40/min, SpO₂ ≥95% on O₂ in 24 h → interventions: O₂ per 20.8, position/pulmonary toilet, chest physiotherapy after feeding, antipyretic, fluids, suction only if needed, monitor retractions, escalate if apnea.

7. Multiple-Choice Questions (Key at the end)

1. The correct sequence of the nursing process is:

A. diagnosis → assessment → planning → implementation → evaluation · B. **assessment** → **diagnosis** → **goals** → **interventions** → **evaluation** · C. planning → diagnosis → assessment → evaluation → intervention · D. evaluation → planning → diagnosis → assessment.

1. "Pass urine 1 mL/kg/h by 8 h" is an example of a:

A. nursing diagnosis · B. **SMART goal** · C. intervention · D. rationale.

1. Which phrase belongs in a NANDA-I nursing diagnosis?

A. "The nurse will give ORS" · B. "The risk is 20%" · C. **"Deficient fluid volume related to excessive GI losses"** · D. "Administer oxygen 2 L/min".

1. The rationale for giving ORS in moderate dehydration is:

A. to stop diarrhea · B. to sedate the child · C. **to replace water and electrolytes lost** · D. to prevent vomiting.

Key: 1-B, 2-B, 3-C, 4-C.

8. Key Points

- The care plan is the record of **your clinical thinking**, not a paperwork exercise — it guides and evidences nursing care.
- Always **re-assess**: evaluation is the step that closes the loop and revises the plan.

APPENDICES

APPENDIX A

Appendix A — Normal Vital Signs and Laboratory Reference Tables (Pediatric)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Appendix A — age-based reference values for vital signs, growth, and common labs used in the ward chapters.

A.1 Vital Signs by Age

Age	Temperature (°C axillary)	Heart rate (bpm, awake)	Respiratory rate (/min)	Systolic BP (mmHg) — approx
Newborn	36.5–37.5	110–160 (crying to 180)	30–60	60–80
Infant (1–12 mo)	36.5–37.5	100–160 (90–140 sleeping)	20–40	70–100
Toddler (1–3 y)	36.5–37.5	90–150	20–30	80–110
Preschool (3–5 y)	36.5–37.5	80–140	20–25	90–110
School (6–12 y)	36.5–37.5	70–120	16–25	90–120
Adolescent	36.5–37.5	60–100	12–20	100–130

Temperature: axillary preferred in children; rectal avoided in newborns. Fever = $>38^{\circ}\text{C}$ (age-appropriate route).

A.2 Growth Reference (WHO Child Growth Standards)

Measure	Newborn	6 mo	12 mo	2 y	5 y
Weight (kg, median boy/girl)	~3.3/3.2	~7.9/7.3	~9.6/8.9	~12.2/11.5	~18/17
Length/height (cm)	~50	~66	~75	~87	~110
Head circumference (cm)	~34.5	~43.5	~46	~48.5	~50

MUAC (6–59 mo, WHO screening): ≥ 12.5 cm = normal; 11.5–12.4 cm = moderate acute malnutrition; < 11.5 cm = severe acute malnutrition.

A.3 Selected Laboratory Reference (used in CH11–CH19)

Test	Normal value (child)	Used in
Hemoglobin (infant ~6 mo)	10.5–14 g/dL	CH13, CH14
WBC (child)	5,000–15,000 /mm ³	CH14
Platelets	150,000–450,000 /mm ³	CH14, CH15
Total bilirubin (term newborn day 3–5)	< 15 mg/dL (physiologic curve on chart)	CH11
pH (arterial)	7.35–7.45	CH12, CH19
Serum Na ⁺	135–145 mmol/L	CH17, CH19
Serum K ⁺	3.5–5.5 mmol/L	CH17, CH19
Serum Cl ⁻	98–106 mmol/L	CH19
BUN	5–18 mg/dL	CH17, CH19
Glucose (capillary)	60–100 mg/dL (fasting)	CH12, CH14

A.4 Fluid Maintenance (Holliday–Segar) and ORS

Weight	Daily maintenance
First 10 kg	100 mL/kg/day
10–20 kg	1000 mL + 50 mL/kg for every kg >10
>20 kg	1500 mL + 20 mL/kg for every kg >20

ORS: **75 mmol/L Na, 13.5 g/L glucose** — mild 50 mL/kg, moderate 100 mL/kg over 4 h + 10 mL/kg per stool (CH17, 20.10).

A.5 WHO Growth-Chart Use (CH4, CH20.6)

- Sex-specific WHO charts: **weight-for-age, length/height-for-age, head-circumference-for-age, weight-for-length/height, and BMI-for-age.**
- Plot each measurement; use the same chart type every visit; interpret by **percentile and z-score.**
- Flags: crossing >2 major percentile lines, drop >1 z-score, weight-for-length <−2SD (wasting), height-for-age <−2SD (stunting), MUAC <11.5 cm (6–59 mo).
- WHO MUAC screening (6–59 mo): ≥12.5 cm normal; 11.5–12.4 cm moderate; <11.5 cm severe acute malnutrition.

A.6 Representative EPI-Type Schedule (CH9 companion)

Birth	2 mo	4 mo	6 mo	9 mo	12 mo	18 mo	School age
BCG, HepB-1, OPV-0	DTP-1/Hib, OPV-1, PCV-1, RV	DTP-2/Hib, OPV-2, PCV-2, RV	DTP-3/Hib, OPV-3, HepB-3	Measles-1 (per protocol)	MMR, PCV booster	DTP booster	OPV booster / Td (per MoH)

Representative chart for teaching only — always apply the current official MoH of Iraq schedule; verify locally (CH9).



Growth charts, bilirubin nomogram and the normal vital-sign table

END OF APPENDIX A

Normal Vital Signs and Laboratory Reference Tables (Pediatric)

Next: **Appendix B** — Dehydration and Oral Rehydration (ORS) Quick-Reference

APPENDICES

APPENDIX B

Appendix B — Dehydration and Oral Rehydration (ORS) Quick-Reference

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Appendix B — rapid dehydration grading, ORS volume planning, and IV fluids (complements CH17 and Procedures 20.10–20.11)

B.1 Dehydration Grading

Sign	Mild	Moderate	Severe
Weight loss	3–5%	6–9%	≥10%
General state	Alert	Restless, irritable	Lethargic/limp
Eyes / fontanel	Normal; moist	Somewhat sunken	Very sunken
Tears / skin turgor	Present; normal	Decreased; slow turgor	Absent; very slow (tenting)
Urine output	Normal	Decreased	Oliguria/anuria

B.2 ORS Composition and Volumes

- ORS: **75 mmol/L sodium + 13.5 g/L glucose**; tap water, juice, soup, or broth are **not** rehydration fluids.
- **Plan A (no dehyd.)**: and continue feeding.
- **Plan B — mild**: 50 mL/kg over 4 h. **Moderate**: 100 mL/kg over 4 h — give **5–10 mL by spoon/syringe every 2–5 min**; vomiting is not a contraindication.
- **Stool-loss replacement**: ~10 mL/kg ORS per diarrheal stool.
- **Plan C — severe**: IV lactated Ringer **40 mL/kg** (bolus 20 mL/kg then reassess), then ORS once stable.
- Continue **breastfeeding** throughout; reassess every 4–8 h, weigh daily.

B.3 IV Fluids and Monitoring

Need	Choice of fluid	Note
Maintenance	D5 0.45% Na (per institution)	mL/h by Holliday–Segar (App. A.4)
Severe dehydration	Lactated Ringer 40 mL/kg (20 mL/kg bolus)	High-alert; monitor urine output

Need	Choice of fluid	Note
Alkalosis correction (HPS)	Parenteral fluids/electrolytes (Cl, K)	Correct before surgery (CH19)

I&O checklist: hourly IV rate; urine output (target 0.5–1 mL/kg/h); daily weight on the same scale; stool and emesis counts; shift summary.



WHO dehydration-assessment poster and ORS preparation steps

END OF APPENDIX B

Dehydration and Oral Rehydration (ORS) Quick-Reference

Next: **Appendix C** — Drug-Dose Quick Reference and Pediatric Calculations

APPENDICES

APPENDIX C

Appendix C — Drug-Dose Quick Reference and Pediatric Calculations

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

C.1 Formulas

Calculation	Formula
Dose volume	Volume (mL) = ordered dose (mg) ÷ concentration (mg/mL)
IV rate	mL/h = volume (mL) ÷ time (h)
Drip rate	gtt/min = (mL × drop factor) ÷ time (min)
Weight-based dose	Ordered mg/kg/day ÷ frequency = per-administration dose
Maintenance fluids	Holliday–Segar (Appendix A.4)

C.2 Worked Examples

1. Paracetamol 120 mg PO q6h; suspension 120 mg/5 mL → $120 \div (120/5) = \mathbf{5 \text{ mL}}$.
2. 20 mg/kg/day amoxicillin ÷ q8h, child 15 kg → $15 \times 20 \div 3 = \mathbf{100 \text{ mg q8h}}$; 125 mg/5 mL → $100 \div 25 = \mathbf{4 \text{ mL}}$.
3. 300 mL over 12 h, microdrip 60 gtt/mL → **25 mL/h** and **25 gtt/min**.
4. IM ampicillin 200 mg/kg/day ÷ q6h, child 12 kg, 500 mg/2 mL → 600 mg q6h = **2.4 mL** (recheck vs site-volume limit $\leq 2 \text{ mL}$).

C.3 Safe-Dose Verification (checklist)

- Dose within the mg/kg range in the reference; arithmetic rechecked by a second nurse.
- Volume $\leq$ site-safe volume (IM infant $\leq 1 \text{ mL}$; child $\leq 2 \text{ mL}$).
- Correct reconstitution/dilution; expiry checked; right route/time; documented.

C.4 Common Pediatric Drugs in the Ward Chapters

Drug	Use (chapter)	Route	Nursing points
Paracetamol	Fever/pain (CH16, CH15)	PO/PR	Max dose by weight; hepatic caution
Ibuprofen	Fever/pain	PO	Avoid in bleeding disorders (CH15)
ORS	Rehydration (CH17, 20.10)	PO	75 mmol/L Na + glucose 13.5 g/L
Ondansetron	Vomiting (CH17)	PO/IV	Give before ORS in persistent vomiting
Deferoxamine / deferasirox	Iron chelation (CH13)	SC/PO	Monitor hearing/vision
Folic acid / vitamin K	Thalassemia support (CH13)	PO/IM	Per panel schedule
Amoxicillin	Pneumonia / prophylaxis (CH16)	PO	Complete the course
Azithromycin (macrolide)	Mycoplasma pneumonia (CH16)	PO	GI upset; QT caution
Surfactant	RDS (CH12)	Endotracheal	Physician-given; no suction ~1 h
Factor VIII/IX concentrate	Hemophilia (CH15)	IV	Reconstitute immediately before use

All doses are for study — administer per the drug insert and the current order in the clinical setting.



Pediatric dose-calculation reminder poster

END OF APPENDIX C

Drug-Dose Quick Reference and Pediatric Calculations

Next: **Appendix D** — Glossary of Key Terms (English–English)

APPENDICES

APPENDIX D

Appendix D — Glossary of Key Terms (English–English)

CLINICAL PEDIATRIC NURSING · FOR WARD AND LABORATORY TRAINING

Book: Clinical Pediatric Nursing: A Comprehensive Practical Guide

Appendix D — alphabetized glossary of the technical terms used in the ward chapters and procedures.

Term	Definition
Aganglioneosis	Absence of the enteric ganglion cells (Auerbach/Meissner plexuses) — the cause of Hirschsprung disease [CH18].

Term	Definition
Atelectasis	Collapse of small areas of lung; in RDS caused by surfactant deficiency [CH12].
Attraumatic care	Care that minimizes physical and psychological stress (pain, separation, loss of control) [CH1, CH3].
Chelation	Removal of excess iron from the body (deferoxamine/deferasirox) in thalassemia [CH13].
CNS prophylaxis	Preventive treatment of the central nervous system (intrathecal chemotherapy) in ALL [CH14].
Cold chain	The refrigerated system (2–8 °C) that preserves vaccine potency from manufacturer to use [CH9].
Colostomy	An opening of the colon onto the abdominal wall for stool drainage [CH18, 20.14].
Consolidation	(a) Chemotherapy phase in ALL; (b) solidification of lung tissue in pneumonia [CH14, CH16].
DDAVP (desmopressin)	Drug that raises factor VIII/vWF; used in mild hemophilia A and von Willebrand disease [CH15].
Deferasirox	Oral iron chelator for thalassemia; alternatives: deferoxamine (SC), deferiprone (PO) [CH13].
Enterocolitis	Severe inflammatory complication of Hirschsprung disease (fever, distension, loose bloody stools) [CH18].
Exchange transfusion	Replacement of the infant's blood to lower bilirubin/antibody load [CH11].
Factor concentrate	Purified clotting factor VIII or IX given IV for bleeding in hemophilia [CH15].
Gavage feeding	Feeding by a tube (nasoro/gastric) when the infant cannot feed safely by mouth [20.7].
Ground-glass appearance	Hazy lung opacity on the RDS chest radiograph [CH12].

Term	Definition
Grunting	Forced expiration against a partially closed glottis — a respiratory distress sign [CH12, CH16].
Hemarthrosis	Bleeding into a joint — the most common bleeding manifestation of hemophilia [CH15].
Hyperbilirubinemia	Elevation of serum bilirubin; physiologic vs pathologic [CH11].
Hypochloremia	Low serum chloride — characteristic of HPS alkalosis [CH19].
Induction	The first intensive chemotherapy phase of ALL [CH14].
Infantile reflexes	Automatic responses of the newborn (Moro, rooting, grasp) that disappear with maturation [CH6].
Intrathecal	Into the cerebrospinal fluid (e.g., methotrexate for CNS prophylaxis) [CH14].
Kernicterus	Bilirubin encephalopathy — permanent brain damage from very high unconjugated bilirubin [CH11].
Maintenance	Prolonged lower-intensity phase of ALL chemotherapy [CH14].
Metabolic alkalosis	Elevated blood pH/bicarbonate from loss of gastric acid — classic in HPS [CH19].
Neutropenia	Low neutrophil count → high infection risk (ALL chemotherapy) [CH14].
Olive-shaped mass	The palpable hypertrophied pylorus in HPS [CH19].
ORS	Oral rehydration solution — 75 mmol/L sodium + 13.5 g/L glucose [CH17, 20.10].
Phototherapy	Blue-green light therapy that converts bilirubin to exercisable isomers [CH11, 20.9].
Projectile vomiting	Forceful, nonbilious vomiting that travels a distance — classic of HPS [CH19].
Pull-through	Definitive surgical repair of Hirschsprung disease (Swenson, Duhamel, Soave) [CH18].
Pyloromyotomy	Fredet–Ramstedt operation: longitudinal division of the pyloric muscle in HPS [CH19].

Term	Definition
RICE	Rest, Ice, Compression, Elevation — bleeding-joint first aid in hemophilia [CH15].
Retractions	Inward pulling of the chest wall with breathing (subcostal, intercostal, suprasternal) [CH12, CH16].
Surfactant	Surface-active agent that lowers alveolar surface tension; deficient in RDS [CH12].
Tachypnea	Abnormally fast respiratory rate [CH5, CH12].
Target cells	Abnormally flat RBCs with central hemoglobin dot — seen in thalassemia [CH13].
Transition zone	The segment between normal and aganglionic bowel in Hirschsprung disease [CH18].
Turgor (skin)	Skin elasticity used in dehydration assessment (tenting = reduced turgor) [CH17, App. B].
Vastus lateralis	The preferred pediatric IM injection site (middle third of the anterolateral thigh) [CH8, 20.13].

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END OF APPENDIX D

Glossary of Key Terms (English–English)